

EEE491 ENGINEERING PROJECT REPORT

FPGA-Based MFCC Spoken Digit Recognition

A fixed-point audio feature-extraction and template-matching system for the Basys 3

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Target device: Digilent Basys 3 / xc7a35tcpg236-1

Project scope: RTL, verification, FPGA integration, and home-fabricated microphone/ADC PCBs

Design language: VHDL

Tool flow: Vivado 2025.2, XSim, MATLAB

Design baseline: implemented and routed Basys 3 system

Repository: <https://github.com/Borek-32/EEE491-Spoken-Digit-Recognizer-BASYS3>

Final report - 20 August 2026

The report follows the complete path from signal acquisition to fixed-point recognition and physical prototyping.

Abstract

I built a fixed-point spoken-digit recognizer for the Digilent Basys 3. The design records 16,384 ADC samples, processes 63 overlapping frames, and extracts eight DCT coefficients using Hann windowing, a 512-point FFT, 32 MEL filters, and fixed-point log compression. Record mode writes a selected 16-frame feature sequence into writable reference RAM; compare mode searches the templates with SAD/L1 distance. A seed of 30 default templates supports comparison before speaker-specific slots are recorded. XSim and integer models check the processing stages, and the routed design meets its internal 100 MHz constraint. The 30-utterance development set separates 30/30 in offline leave-one-out analysis. I also designed and fabricated the microphone front-end PCB, integrated it through Pmod JC, and programmed the full design. Live spoken-digit comparison, runtime reference recording, slot readback and recall, corrected ADC receiver integration, and endpoint checks are complete. The PCB DRC/ERC findings have been resolved. The offline 30/30 figure remains the result of its stated analysis; it is separate from the completed live tests.

Keywords: FPGA; MFCC; fixed-point DSP; spoken digit recognition; VHDL; UART debugging.

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1 Introduction and Project Objective

This project asked me to implement spoken-digit recognition on an FPGA, including acquisition, frame control, feature extraction, template comparison, board controls, display, and debugging. I also built the microphone front end. The report follows the signal through the hardware and explains the main design choices, bugs, verification results, and remaining limitations.

The aim is not precision spectral measurement. It is to retain enough time-varying spectral information to distinguish digits, select a useful 16-frame interval, record it under a digit/trial label, and compare later utterances against the writable references. Runtime enrollment matters because voices, speaking style, microphone placement, and gain vary between users. The digital interfaces still have exact contracts: every WINDOW sample must reach the FFT once, FFT bin k must be stored at address k , and MEL must multiply $P[k]$ by $H[m,k]$.

1.1 Project overview

The complete system combines the microphone front end, utterance acquisition, fixed-point feature extraction, and runtime template handling. The first diagram shows that end-to-end path; the second expands only the digital signal-processing chain.

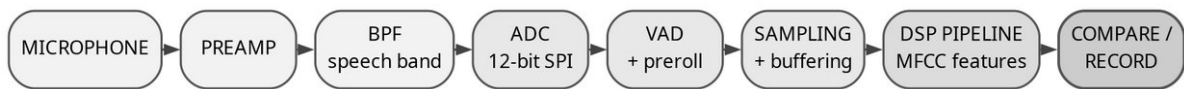


Figure 1. End-to-end project overview from microphone input to runtime enrollment or digit recognition.



Figure 2. Expanded digital feature path from the stored utterance to comparison or reference recording.

After voice detection, centered samples are assembled into a fixed 16,384-word utterance buffer. Sixty-three overlapping frames pass through Hann windowing, FFT/power, MEL, log-MEL, and DCT. COMP uses MEL energy to align a 16-frame DCT sequence, then either overwrites the selected reference mode slot in record mode or evaluates SAD/L1 distance against the stored templates in compare mode.

Table 1. Project objectives, implemented responses, and the evidence that supports them.

Objective	Implemented response	Evidence used
Acquire a repeatable utterance	12-bit SPI, 32-conversion average, preroll, VAD, fixed 16,384-sample RAM	ADC waveform and programmed-board captures
Extract speech features	63 frames; WINDOW, FFT, MEL, LOGMEL, DCT	Raw XSim, integer checkers, and UART stage plots
Enroll speaker-specific templates	Writable COE-initialized reference RAM; digit/trial-selected 16-frame overwrite	Focused COMP simulation records digit 2/trial 1 and recalls digit 2 afterward
Recognize 0-9	30 references; 16 frames x 8 coefficients; SAD/L1	COMP checker and 30-record leave-one-out analysis
Remain observable	Marker-framed 1 Mbaud UART export of all major memories	1,362-word packet check and live captures
Operate as physical hardware	Basys 3 controls, display, external ADC, custom analog PCB	Routed reports, UI verification, and fabrication evidence

2 Design Boundary and Evidence Strategy

Verification combines RTL inspection, XSim timing waveforms, integer-model comparisons, programmed-board UART captures, routed Vivado reports, and PCB evidence. Each source answers a different question, so simulation, numerical checking, implementation reports, and physical-board results are kept separate.

A waveform checks transaction order, an integer comparison checks fixed-point words, and a programmed-board capture checks memories populated by acquired data. Labeled live recognition and runtime reference recording, readback, and recall have also been completed. The offline development-set analysis retains its own test definition and result.

Unless stated otherwise, the explanatory diagrams were drawn from the implemented RTL states, ports, widths, and address equations.

Table 2. Evidence types and the engineering question assigned to each one.

Evidence family	Question answered	Interpretation rule
RTL and constraints	What is implemented?	Used for algorithms, states, widths, addressing, and the record/flush control paths
XSim waveform	Did the hardware transaction occur in the intended order?	Read for handshakes, latency, addressing, and writes; not treated as a numerical accuracy claim alone
MATLAB checker	Do saved words match the intended fixed-point model?	Used for exact word comparisons and explained implementation/model differences
UART capture	Did a programmed FPGA populate the complete signal chain?	Used for continuity, frequency localization, stage-memory population, framing, and result serialization
Routed reports	Does the integrated design fit and close timing?	Used for implementation feasibility, not as a substitute for functional explanation
PCB evidence	Was the analog path designed and physically produced?	Simulation and fabrication are identified separately; no unrecorded measurement is implied

3 Hardware Platform and User Operation

The target is the Digilent Basys 3 with the XC7A35T device and a 100 MHz board clock [3]. The FPGA receives external ADC data, stores an utterance, runs the feature chain, and either records a labeled speaker-specific reference or compares the utterance with the writable reference memory. A COE seed supplies 30 repeatable startup templates; runtime enrollment overwrites any selected digit/trial slot. Switches and buttons select the mode, label, trial, debug path, and reference-memory maintenance, while LEDs and the four-digit display show progress and the final result.

Table 3. Board controls and their role in a complete record or compare experiment.

Control or interface	Implemented behavior
SW0	High selects runtime enrollment; low selects compare mode. The synchronized start edge captures the selection for the complete 63-frame run.
SW1 / LD1	Enables the per-frame UART debugger; LD1 mirrors the debug-enable state.
SW6-SW15	One-hot reference digit selection from 0 through 9 for record mode.
BTNU	Starts one acquisition through edge detection; holding the button does not retrigger continuously.
BTNL / BTNR	Select trial 0, 1, or 2 through synchronized, debounced inputs.
BTND	Requests volatile reference-RAM flush while the complete controller is ready; requests during an active run are gated off.
BTNC	Resets the integrated system.
Seven-segment display	Record view: r d digit trial. Compare view: C d d result after a valid decision.
ADC pins	Registered chip select, MISO, and SCLK connect the external 12-bit converter.

This interface is not the central technical contribution, but it makes speaker adaptation possible without resynthesizing or regenerating the COE. At the synchronized start edge the top level captures record/compare mode, digit, and trial into run registers; later switch or button changes therefore cannot relabel an utterance or change its operation midway through the 63-frame transaction. The debug path is optional because full-memory UART export is intentionally slower than the normal enrollment or recognition path.

4 System Architecture and Data Contracts

The recognizer is divided into RAM-separated stages. Each module owns one operation, lowers ready while working, writes a complete local result, and raises ready when the next stage may safely read it. The memories absorb unequal BRAM, multiplier, FFT-IP, and UART latencies while keeping the fixed-point path observable.

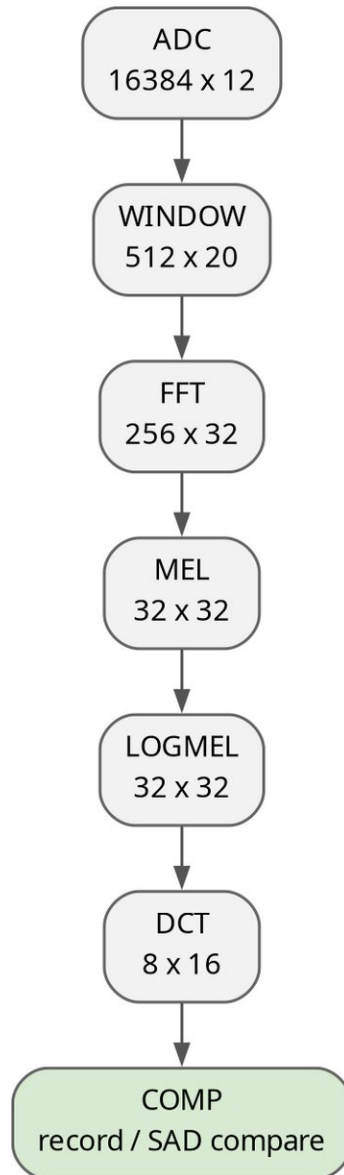


Figure 3. Processing pipeline and stored data dimensions.

The data contract becomes progressively smaller: 16,384 signed samples become 63 overlapping frames, each frame becomes 256 one-sided power values, then 32 perceptual bands, and finally eight signed DCT coefficients. The reduction is purposeful: COMP does not need analyzer-grade spectral detail; it needs a compact sequence that retains the speech envelope and can be compared repeatedly with modest hardware.

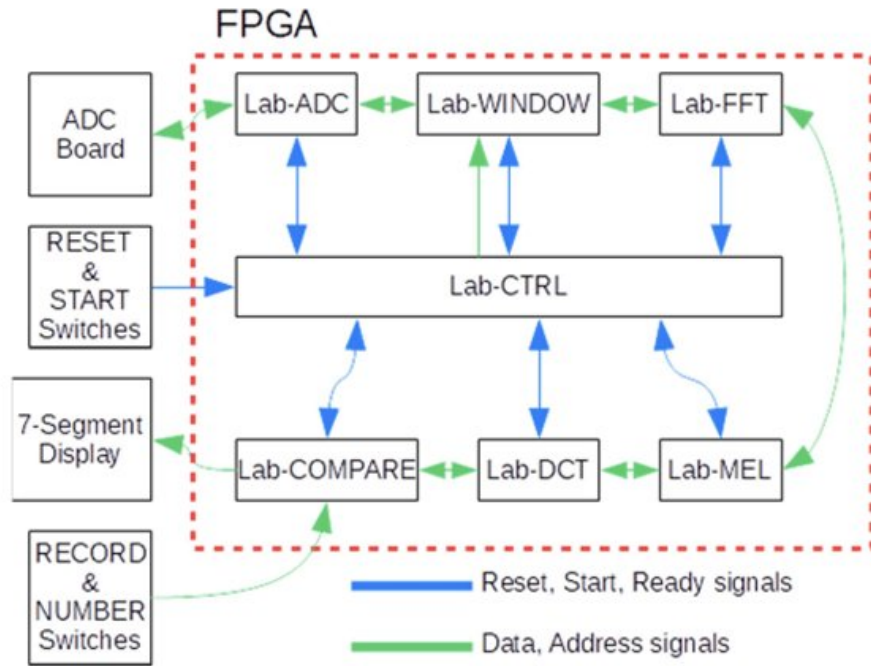


Figure 4. Original laboratory milestone architecture retained from the earlier report.

Historical design-stage diagram extracted from the earlier report.

This earlier diagram shows the modular starting point: an ADC board, separately developed WINDOW/FFT/MEL/DCT/COMPARE laboratories, one controller, and distinct control versus data/address paths. The final design adds LOGMEL, UART DEBUG, runtime record/flush behavior, and revised board interfaces.

Table 4. Implemented data widths and memory boundaries.

Stage	Principal operation	Stored interface
ADC	SPI receive, average, center, preroll, VAD, zero pad	16,384 x signed 12-bit
WINDOW	512 signed samples x 8-bit Hann coefficient	512 x signed 20-bit
FFT	512-point fixed-point FFT and magnitude squared	256 x unsigned 32-bit, bins 0-255
MEL	32 overlapping filters over 256 power bins	32 x unsigned 32-bit
LOGMEL	Piecewise interpolated log2	32 x unsigned 32-bit, 20 fraction bits
DCT	Non-DC DCT-II orders C1-C8	8 x signed 16-bit
COMP	Energy-aligned runtime enrollment or 30-reference SAD/L1	Digit/result flags or one volatile 16-word reference-slot write
DEBUG	Marker-framed readout of all stage memories	1,362 32-bit words per frame

```
frame_base(f) = 256 f, f = 0...62; frame = samples[frame_base ... frame_base + 511]
```

The 512-sample length with a 256-sample hop produces 63 frames and 50% overlap. Frame 62 begins at address 15,872 and ends exactly at 16,383. Reusing one arithmetic chain across those frames trades parallel throughput for resource efficiency and clarity. A push-button spoken-digit recognizer does not need to replicate 63 complete feature pipelines; it needs deterministic processing of every frame.

5 CTRL: Pipeline Orchestration

CTRL schedules the processing pipeline. Its ten-state main FSM is idle, ADC, WINDOW, FFT, MEL, LOGMEL, DCT, COMP, DEBUG, and done. One ADC acquisition occurs at the start of a run. WINDOW through COMP then repeat for all 63 frames, and DEBUG is inserted after COMP only when enabled. This organization ensures that the comparator receives the MEL energy and DCT vector from the same frame before the base address advances.

Every module transaction uses a three-phase start/ready contract. CTRL issues a one-clock start pulse, waits for ready to fall so that a stale idle-high level cannot be mistaken for completion, and then waits for ready to rise. A stage's pipeline depth can change without modifying CTRL, provided that its busy handshake remains unchanged.

The digit switches and trial buttons are also part of the control problem. Trial inputs pass through two flip-flops and a 500,000-clock debounce interval, approximately 5 ms at 100 MHz, and wrap through 0, 1, and 2. CTRL's UI selection registers may still update while idle or during ADC capture, but top_module copies mode, digit, and trial into dedicated run registers at the synchronized start edge. COMP uses those run registers, so later switch or button changes affect only the next run.

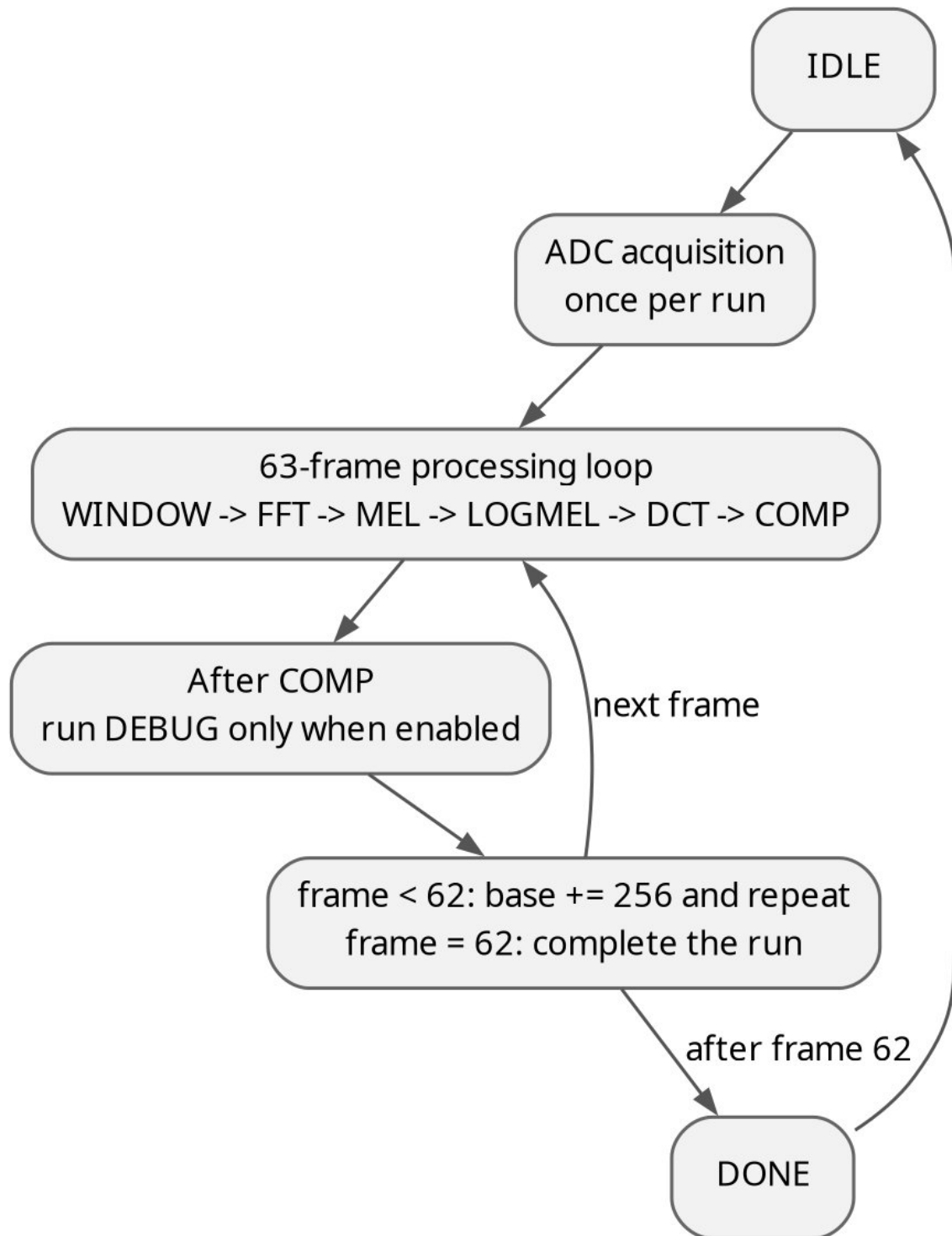
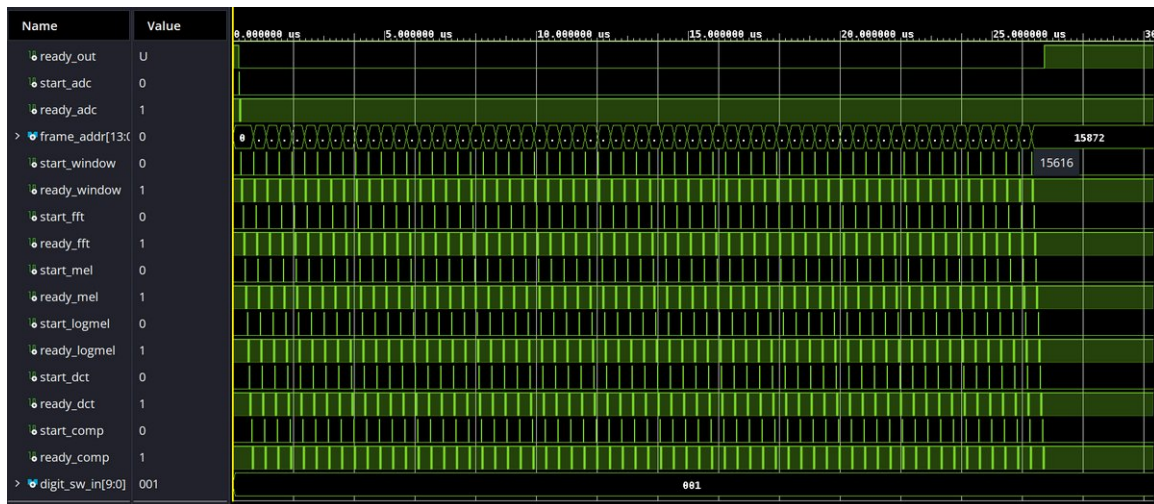


Figure 5. CTRL state progression, 63-frame loop, and optional DEBUG branch.

The diagram explains the key resource decision: acquisition occurs once, while one shared feature chain is reused across 63 overlapped frames. The optional debugger follows comparison and does not change the numerical route. Advancing only after a ready-low/ready-high transaction prevents one stage from reading a memory before the previous stage has finished writing it.

Table 5. Event encoding used by the CTRL numerical checker.

Checker event ID	Controller event
0	ADC
1	WINDOW
2	FFT
3	MEL
4	LOGMEL
5	DCT
6	COMP
7	DEBUG when enabled

**Figure 6. CTRL XSim waveform across the complete repeated frame sequence.**

Behavioral simulation capture from the indexed controller testbench.

The dense start and ready activity is the expected repeated pipeline, not unexplained switching. The frame-address bus advances in 256-sample steps and reaches 15,872 before ready_out returns high. Each start pulse is followed by a modeled busy interval, showing that CTRL responds to completed transactions rather than emitting an uncontrolled sequence of stage pulses.

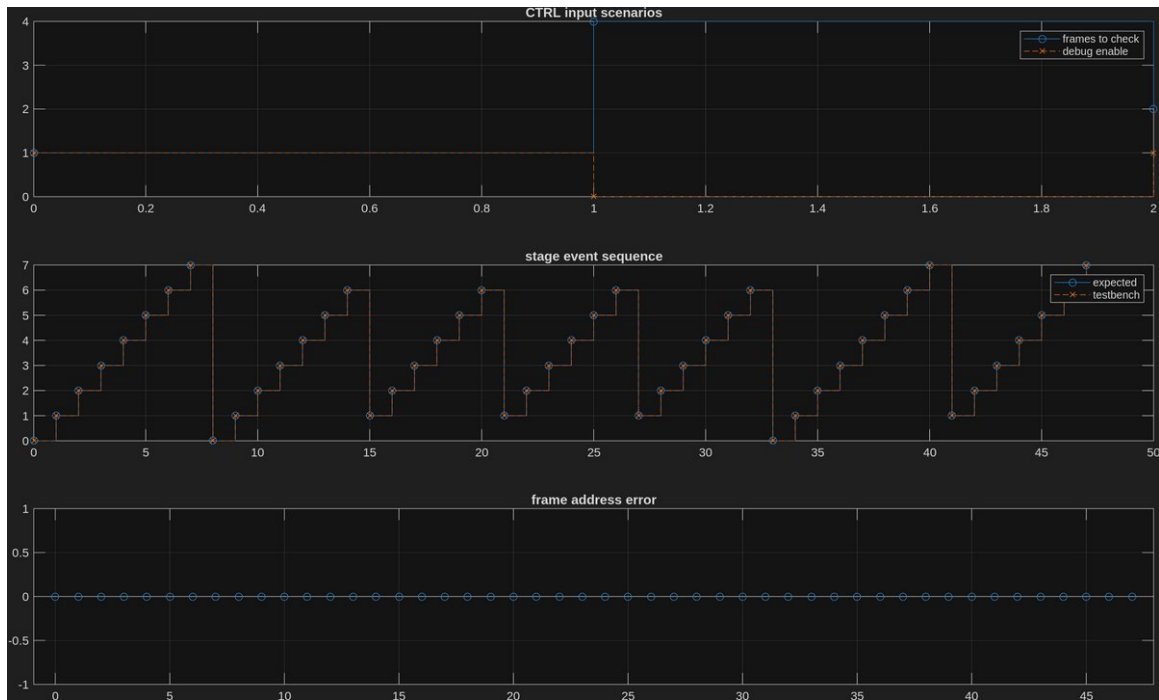


Figure 7. CTRL checker for stage order, debug branching, and frame addressing.

Current integer checker export from the controller testbench outputs.

The checker covers one frame with debugging enabled, four frames with it disabled, and two frames with it enabled. Expected and observed event IDs coincide in all 48 rows, and the frame-address error is zero. This checks both branches: DEBUG appears only after COMP when selected, and bypassing it does not disturb the remaining sequence.

The controller bench substitutes short ready pulses for the real processing blocks. It checks ordering, overlap, and debug branching; the following module chapters check the arithmetic and memory contents produced inside those transactions.

6 ADC: Acquisition, Averaging, and Voice Detection

ADC is more than a serial receiver: it accepts 12-bit words MSB first, averages 32 conversions, removes the unsigned midpoint, preserves 2,048 samples of preroll, detects the utterance boundaries, and guarantees a 16,384-word signed record for the downstream frame schedule.

The state machine remains in the 100 MHz system domain and advances on a one-cycle enable every three clocks. The resulting controller tick is 33.333 MHz and SCLK is approximately 16.667 MHz because the receive and decode phases alternate the clock. I originally pursued a separate 36 MHz generated-clock solution. Although its behavioral simulation was convincing, implementation around the ADC and memory boundary became placement- and clock-domain-sensitive, and programmed-board behavior could diverge even during a forced-low serial-input diagnosis. I tried explicit synchronization and timing exceptions, IOB placement, KEEP and DONT_TOUCH attributes, and localized placement before choosing the single-domain enable.

The final tick is approximately 7.4%, conventionally rounded to about 8%, below 36 MHz. I accepted that difference because repeatable speech acquisition and comfortable implementation margin matter more than reproducing a nominal tick exactly. The ADC-rate decision is a documented timing tradeoff; it is distinct from FFT sample-order and address correctness, which are enforced as hard regression properties.

Including the 32 conversions and the acquisition-state overhead, one stored audio sample consumes 2,017 ADC-enable ticks. With the implemented 33.333 MHz enable, the effective stored-sample rate is approximately 16.526 kSa/s. A 512-sample frame therefore spans about 30.981 ms, the 256-sample hop about 15.491 ms, and a full 16,384-sample record about 0.991 s. The separate FFT unit test uses an exact synthetic rate of 16 kSa/s and is labeled accordingly.

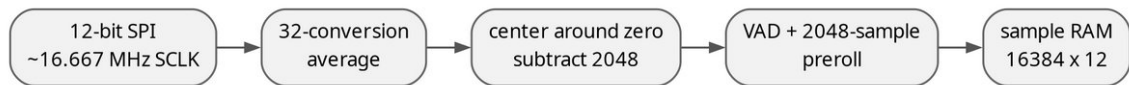
```
average[m] = floor(sum(raw[32m ... 32m+31]) / 32); signed_sample[m] = average[m] - 2048
```

The centered value remains signed 13-bit; its lower 12 bits are stored as two's-complement audio, and a separate 13-bit magnitude feeds detection. Before speech begins, averaged samples continuously overwrite a 2,048-word circular buffer. Once that buffer has wrapped, the active onset logic sums three consecutive magnitude results and compares the sum with 1,000. The mapped `n_on` value is not used by the implementation, so onset is not described as a ten-sample test.

After onset, I reserve the first 2,048 locations of the main RAM and write new audio after that prefix. At completion I copy the circular preroll into addresses 0 through 2,047 in chronological order. This fallback protects the first consonant or attack that occurred before the threshold crossing. After at least 4,096 post-preroll samples, 1,024 consecutive magnitude results below 80 terminate active recording; the rest of the RAM is zero padded. If speech continues, capture reaches the end naturally. Both paths deliver exactly the record length expected by CTRL.

Table 6. ADC parameters and the design reason for each one.

Acquisition parameter	Active value	Reason
Raw words per stored sample	32	Suppress isolated converter noise before feature extraction
Effective stored-sample rate	Approximately 16.526 kSa/s	2,017 ADC-enable ticks per averaged sample
Preroll	2,048 samples	Recover the speech onset that precedes the threshold crossing
Onset decision	Three-result magnitude sum > 1,000	Short level-sensitive trigger after preroll becomes valid
Minimum active record	4,096 samples after preroll	Prevent an early quiet interval from ending the word
End decision	1,024 consecutive magnitudes < 80	Detect a sustained low-level tail
Main sample RAM	16,384 x 12-bit	Provide the fixed 63-frame address space

**Figure 8. ADC processing path from SPI conversion to the fixed recording RAM.**

Averaging, midpoint removal, preroll, and end detection are part of the architecture because faults at acquisition would contaminate every later feature. The fixed output length is equally important: it allows the controller to generate all frame addresses without negotiating a variable-sized buffer.

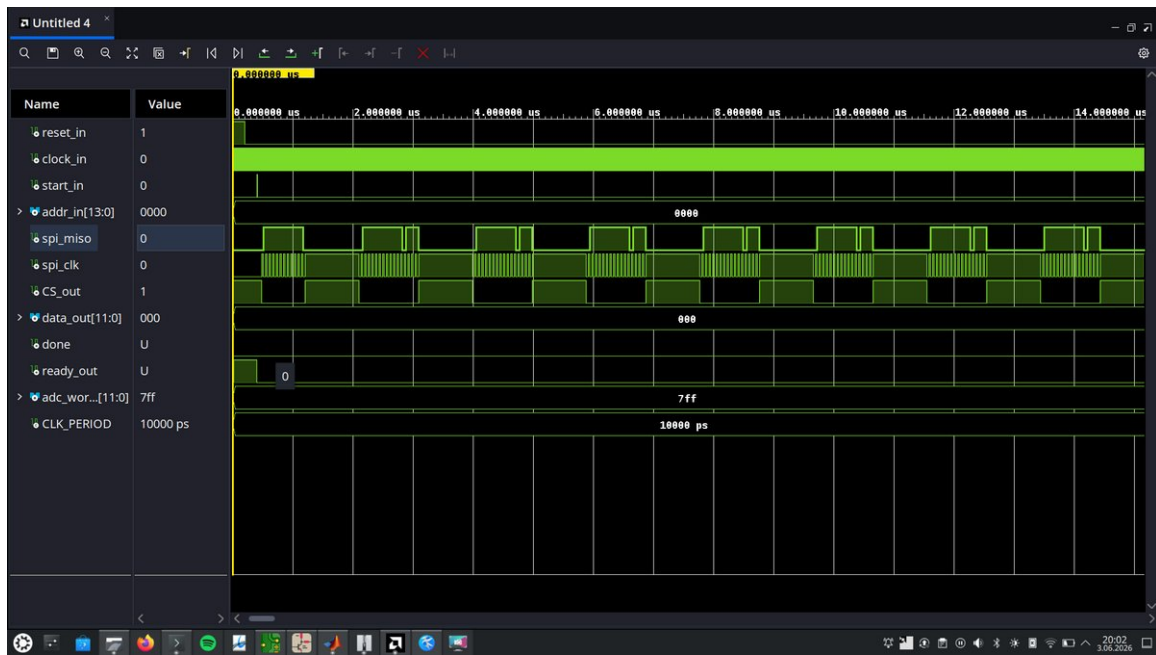


Figure 9. ADC XSim waveform showing the actual 12-bit serial transaction and stored-data interface.

Behavioral simulation capture from the indexed ADC testbench.

The waveform shows active-low chip-select intervals, SCLK bursts, programmed MISO activity, ready/busy behavior, and decoded sample storage. It checks polarity, bit order, transaction boundaries, and the separation between clock movement and data sampling. The testbench uses shortened/bypassed detection parameters so that a regression can reach RAM readback quickly; it does not tune the microphone or exercise the full runtime VAD thresholds.

Module simulation checks deterministic SPI capture and storage. The later programmed-board ADC plots and retained spoken-input records show physical input passing through the development acquisition path.

7 WINDOW: Frame Extraction and Hann Multiplication

WINDOW extracts one 512-sample region of the ADC RAM and tapers its endpoints before the FFT. An FFT treats a finite frame as one period of a repeating waveform; if the two ends do not meet smoothly, the artificial discontinuity spreads energy across unrelated bins. The Hann taper reduces that leakage while retaining the frame's dominant speech structure.

$$y_f[n] = x[\text{frame_base}(f) + n] h_q[n], \quad n = 0 \dots 511$$

The ADC sample is signed 12-bit and the quantized Hann coefficient is unsigned 8-bit. Their signed-by-unsigned multiplier produces a signed 20-bit product, which I retain without dividing by 255 in this stage. Carrying the coefficient scale forward avoids an unnecessary divider and keeps the convention identical in the FFT input and numerical checker.

The state sequence is idle, readr, four wait states, writer, write_ram, flush_last, and done. The waits align the synchronous ADC memory, coefficient ROM, and three-stage multiplier. The first writer pass primes the pipeline without writing; later passes commit addresses 0 through 510, and flush_last supplies the 512th write at address 511. A simulation-only contract checks that addresses 0 through 511 are written once and in order, and the final RAM vector matches all 512 Hann reference words. Because both Hann endpoints are zero, this test still cannot distinguish a stale or swapped final sample. A nonzero-endpoint diagnostic vector would close that specific gap. One multiplier is reused across the frame. At 100 MHz the operation takes about 3,585 clocks, or 35.85 microseconds, while CTRL depends only on the ready handshake.

The 50% overlap is not hidden inside WINDOW. CTRL changes the frame base by 256 and WINDOW always performs the same local sweep from 0 through 511. This separation keeps the arithmetic simple, makes the memory latency visible, and allows exactly the same window datapath to serve every frame.

Table 7. WINDOW numeric interface and resource choice.

Item	Implemented choice	Consequence
Input	Signed 12-bit ADC sample	Preserves positive and negative centered audio
Coefficient	Unsigned 8-bit Hann ROM	Compact coefficient storage with a deterministic scale
Product	Signed 20-bit	Retains the complete multiply result for FFT input
Frame	512 samples	Matches the configured FFT transform length
Resource use	One pipelined multiplier	Trades a short sequential latency for low duplication

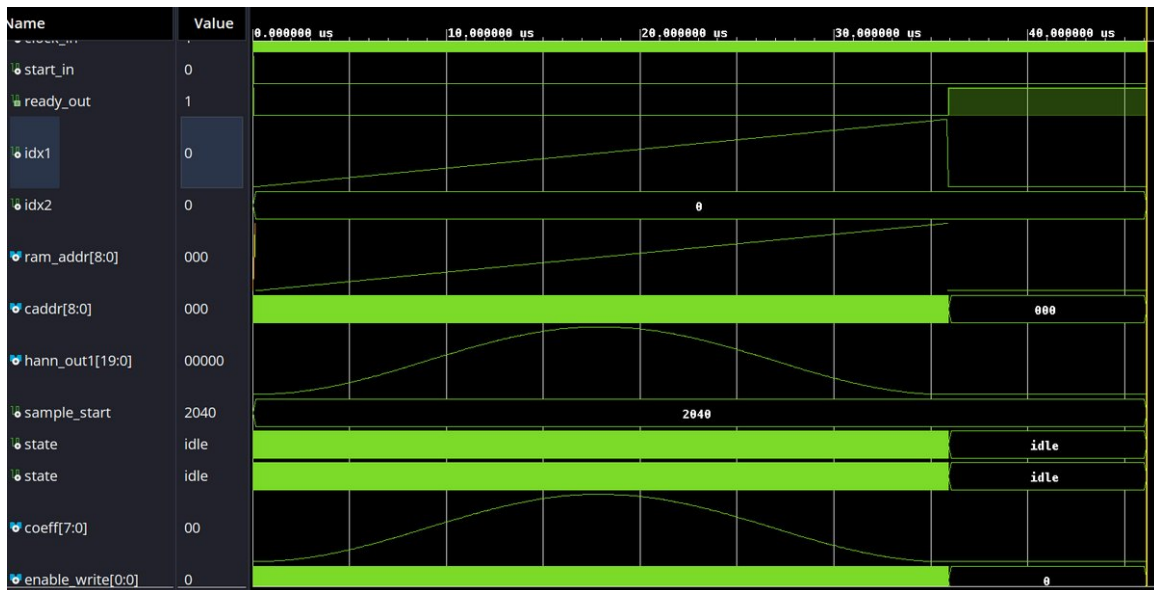


Figure 10. WINDOW XSim waveform for a complete 512-sample transaction.

Behavioral simulation capture from the indexed WINDOW testbench.

The local index, ADC address, coefficient address, and output write address traverse the frame while `ready` remains low. The Hann coefficient rises toward the center and falls at the far end, and the 20-bit product follows that envelope. The current regression adds an explicit 0-through-511 write-address assertion because the zero Hann endpoints alone could mask an endpoint association error.

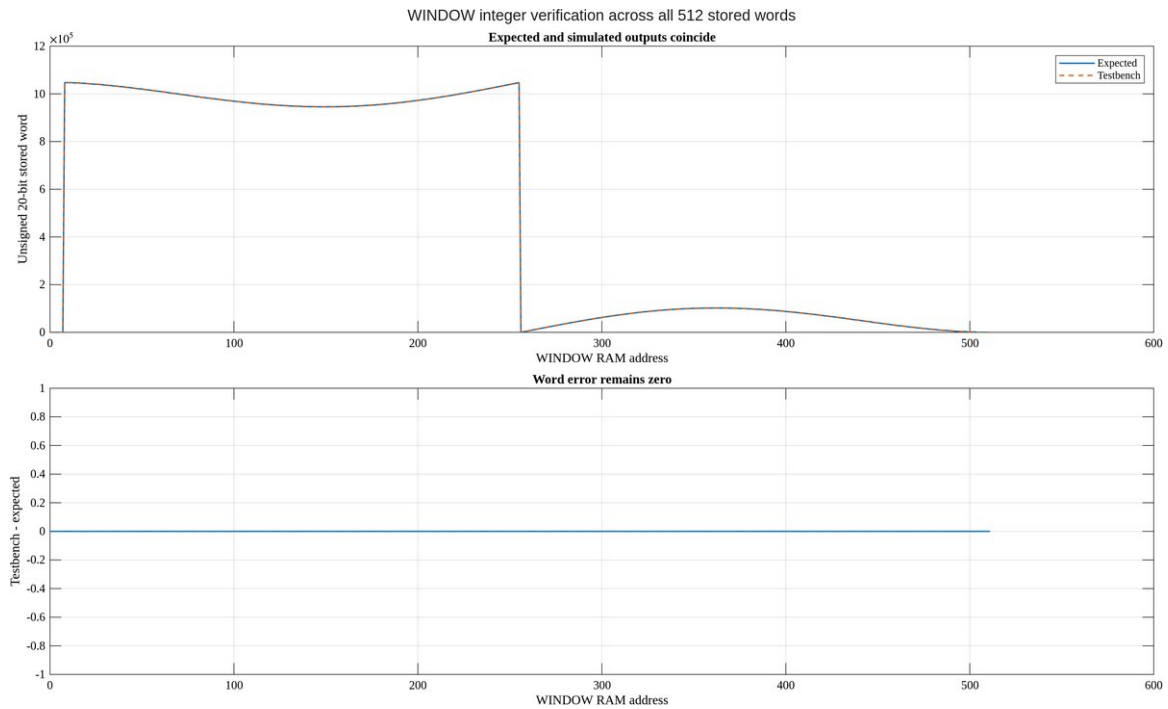


Figure 11. Bit-true WINDOW comparison across all 512 output words.

Light-theme checker rendered from the retained WINDOW expected and testbench output tables.

The checker independently multiplies the deterministic input by the retained 8-bit coefficient table. Expected and simulated curves coincide and the word-error trace remains zero. Where a viewer displays a negative 20-bit product as unsigned, it can appear close to 2^{20} ; that is the ordinary two's-complement representation consumed correctly as signed data by FFT, not an arithmetic discontinuity.

This deterministic vector checks coefficient addressing, signed storage, multiplier latency, and final RAM contents. The programmed-board window maps later in the report show the same operation applied across 63 frames from physical input.

8 FFT: AXI4-Stream Transform and Power Spectrum

FFT converts each 512-sample tapered frame into a one-sided power spectrum. This was my first AXI4-Stream IP integration. It was substantially simpler than my previous DMA experience because the recognizer needs only one configuration transaction, a real-valued input burst, tlast on the 512th accepted sample, valid complex outputs, and event monitoring. I left optional transport and transform features unused instead of carrying interface complexity that the application would never consume.

The AMD/Xilinx FFT core is configured as a 512-point fixed-point forward transform with a 20-bit input, 30-bit real and imaginary outputs, natural output order, unscaled arithmetic with truncation, Radix-2 Lite burst architecture, and a 100 MHz target [1], [2]. Each 20-bit WINDOW value is sign-extended into the real field of the 48-bit stream word and the imaginary field is set to zero. WINDOW RAM is synchronous, so the wrapper uses registered prefetch and holds each returned word until accepted. A one-word skid register captures an in-flight BRAM

response when TREADY deasserts, ensuring samples 0 through 511 transfer exactly once, with TLAST on accepted sample 511.

$$P[k] = \text{floor}((\text{Re}\{X[k]\}^2 + \text{Im}\{X[k]\}^2) / 2^{29}), \quad k = 0 \dots 255$$

Two signed 30-by-30 multipliers create 60-bit squares. I add them in 61 bits and store sum bits 60 down to 29 as an unsigned 32-bit magnitude-squared value. A square root would consume resources without changing which spectral region carries the most energy. Because the input is real, I retain bins 0 through 255: this includes DC and excludes the Nyquist bin at 256. The conjugate upper half contains no additional information for the MEL stage.

The magnitude multipliers have six registered stages. Together with the preceding real/imaginary wrapper latch, the power result for bin k is available when the output-event index reaches $k+7$. The wrapper therefore derives `POWER_RESULT_DELAY` as $6+1$ and writes bins 0 through 255 to identically numbered RAM addresses. The wrapper states are idle, set_config, load_input, get_output, and done. Connected event pins expose missing or unexpected tlast and channel-halt conditions, while completion is controlled by counted valid outputs.

8.1 Correcting FFT bin-address alignment in a synthetic 16 kSa/s unit test

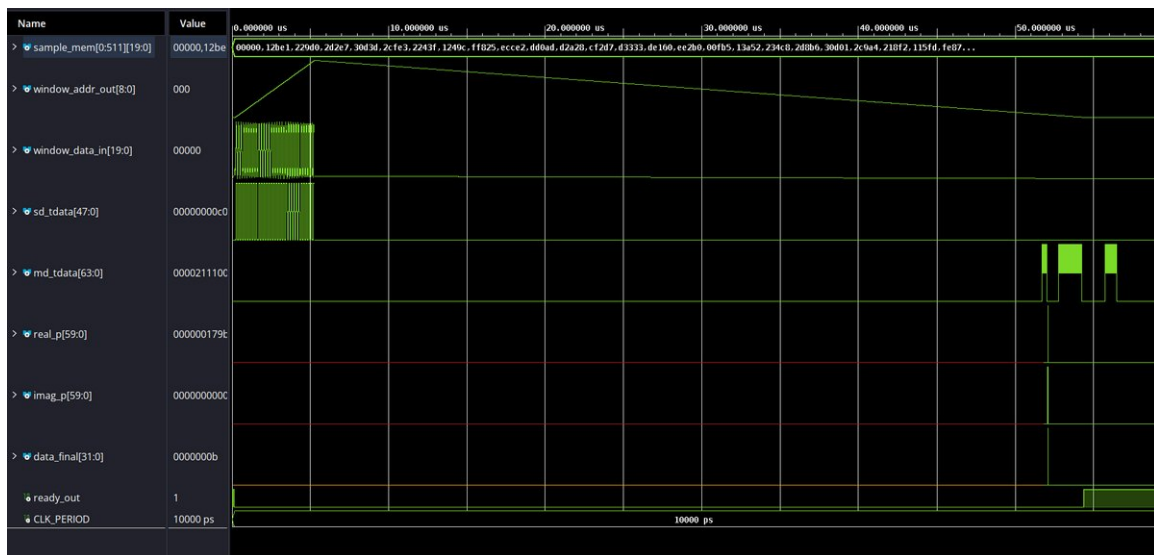
The FFT unit test uses an exact 16 kSa/s synthetic input, so a 1 kHz tone belongs at bin 32. The archived run placed the integer-reference peak at bin 32 but the DUT peak at RAM address 31. Although the magnitudes were close, the address was part of the wrapper contract and therefore incorrect. Cycle tracing showed that the old code waited eight output events even though the implemented path is one wrapper latch plus six multiplier stages. It dropped bin 0, stored bins 1 through 255 one address early, and duplicated bin 255 at the final address.

The first fix changed `POWER_RESULT_DELAY` to `MULTIPLIER_LATENCY + 1`, mapping bins 0 through 255 to addresses 0 through 255. A second review found that TREADY backpressure could still discard a pending synchronous WINDOW-RAM response, so I added registered prefetch and a one-word skid buffer. The wrapper now transfers samples 0 through 511 exactly once.

In the final synthetic 1 kHz test, the reference and RTL both peak at bin/address 32. The integer ideal is 4,882,812 and the DUT is 4,882,767; the maximum integer-ideal difference is 45, with one differing row out of 256. The RTL matches the AMD XFFT v9.1 bit-accurate C model in all 256 outputs.

Table 8. Corrected FFT addressing and numerical verification result.

FFT observation	Value	Engineering meaning
Integer ideal	Bin 32; 4,882,812	Independent integer reference
RTL DUT	Address 32; 4,882,767	Semantic FFT bin and RAM address agree
Integer-ideal comparison	Maximum difference 45; 1/256 rows differs	Passes the uniform 64-count bound
AMD XFFT v9.1 comparison	Exact 256/256; maximum difference 0	Bit-accurate vendor-model parity
Retained-output gates	Peak bin/address checked; bins/addresses 0..255 retained	One-sided power-spectrum properties used by the project

**Figure 12. Historical pre-fix FFT XSim waveform showing AXI input, transform latency, pipelined power calculation, and RAM writes.**

Behavioral simulation capture retained as defect-discovery context; it predates the synchronous-memory skid correction.

This view records the earlier transaction shape but is not used as current sample-order proof. Current retained-output evidence consists of the post-fix peak-address check, the independent integer comparison, and exact AMD bit-accurate agreement for all 256 stored one-sided power bins.

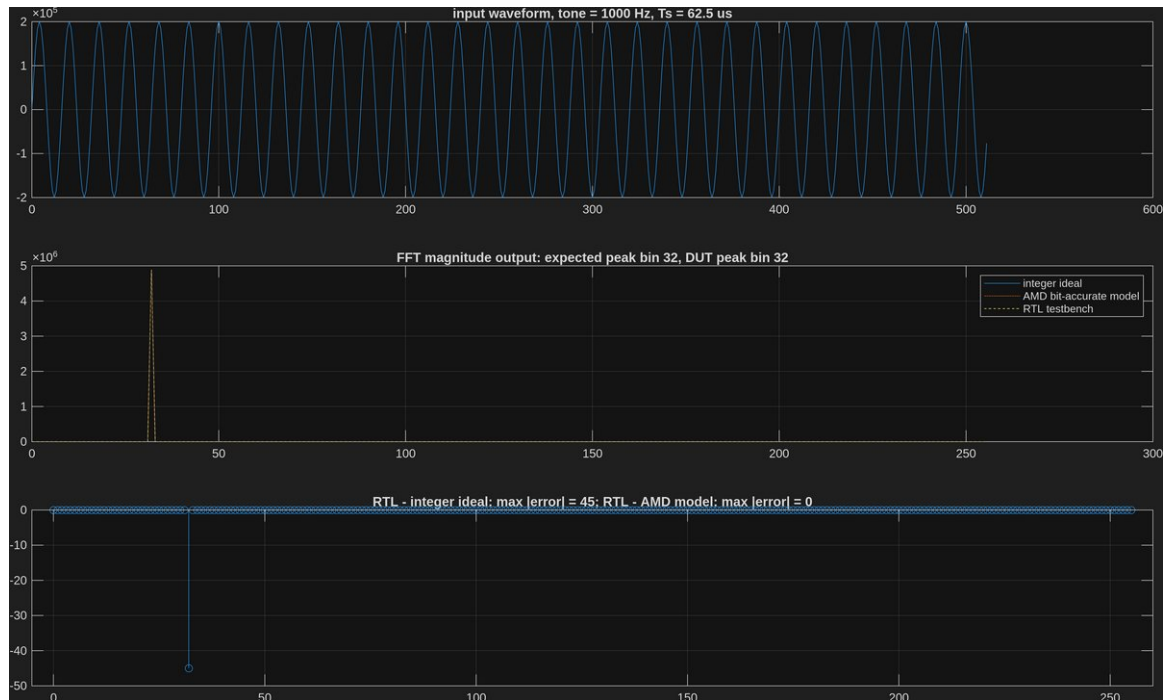


Figure 13. Synthetic 16 kSa/s FFT unit test: corrected 1 kHz reference bin 32 and DUT address 32.

Post-fix XSim output checked directly by the strengthened MATLAB verifier.

All three spectra peak at bin/address 32. RTL differs from the independent integer ideal by at most 45 counts in one row and matches the AMD bit-accurate model exactly in all 256 rows.

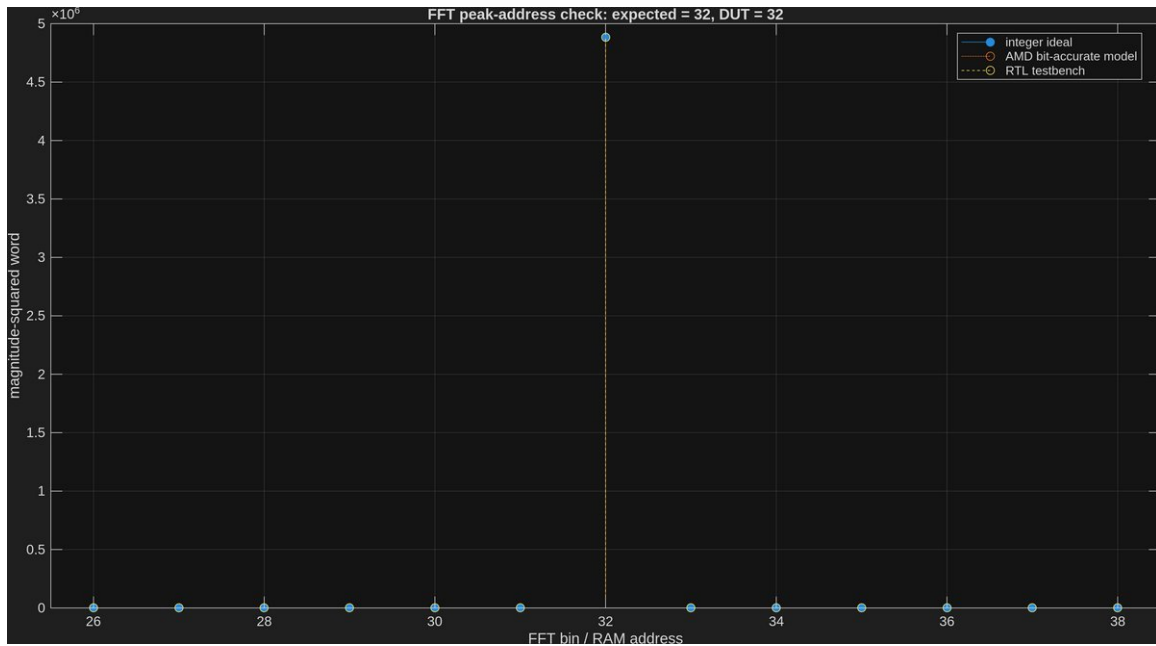


Figure 14. Synthetic 16 kSa/s unit-test close-up of exact bin-to-RAM alignment after the RTL correction.

Post-fix MATLAB address assertion rendered around bins 26 through 38.

Both markers coincide at bin/address 32. This close-up makes the corrected interface property visible rather than relying on similar peak magnitudes or downstream MEL behavior.

The single-tone bench isolates AXI transfer, transform output, power arithmetic, and storage. Address correctness is enforced locally; cross-frequency hardware captures and real spoken-input maps then provide separate application-level spectral evidence.

9 MEL: Perceptual Filter-Bank Accumulation

MEL reduces the 256 correctly addressed linear-frequency power values to 32 overlapping band energies. The filter bank converts the linear-bin spectrum into a perceptual representation that emphasizes the broad spectral envelope used in speech features [4].

9.1 Addressed memories and fixed-point datapath

$$E_f[m] = \sum(P_f[k] H[m,k], k=0\dots255); \quad MEL_f[m] = (E_f[m] \gg 16) \bmod 2^{32}, \quad m=0\dots31$$

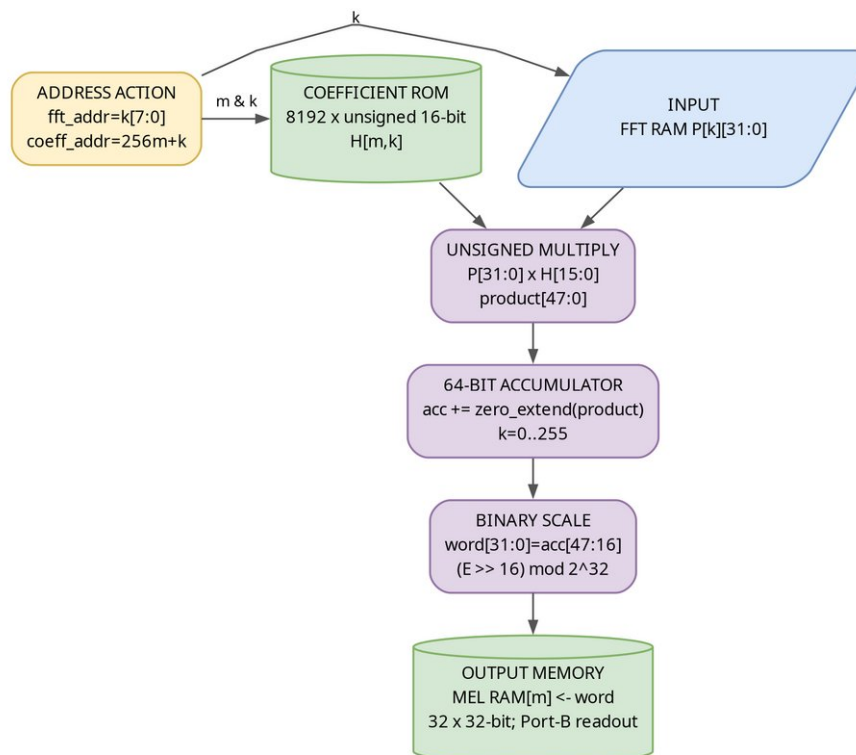


Figure 15. MEL addressed-memory and fixed-point arithmetic datapath.

The same filter index m and bin index k acquire FFT power $P[k]$ at address k and coefficient $H[m,k]$ at address $256m+k$. Their registered outputs feed one unsigned 32-by-16 multiplier and a 64-bit accumulator; bits 47 down to 16 are written as the 32-bit band result.

The coefficient memory contains $32 \times 256 = 8,192$ unsigned 16-bit words. Concatenating the five-bit filter number with the eight-bit FFT-bin number directly creates address $256m+k$. One unsigned 32-by-16 multiplier produces a 48-bit product, which I accumulate in 64 bits. After the final bin, accumulator bits 47 down to 16 become the unsigned 32-bit band value, removing the coefficient's 16 fractional bits.

The implemented read schedule presents $FFT[k]$ while the coefficient address presents $256m+k$. The address is held until the top-level mux and synchronous FFT/coefficient memories have returned stable words; latency is handled by the counter, not by changing the requested mathematical index. This removes the former residual $k+1$ compensation that

remained after the FFT RAM itself was corrected to store bin k at address k . One multiplier and accumulator are reused across all filters, requiring 8,192 multiply-accumulate iterations per frame and avoiding 32 duplicated filter datapaths.

9.2 Serialized address, accumulate, and write control

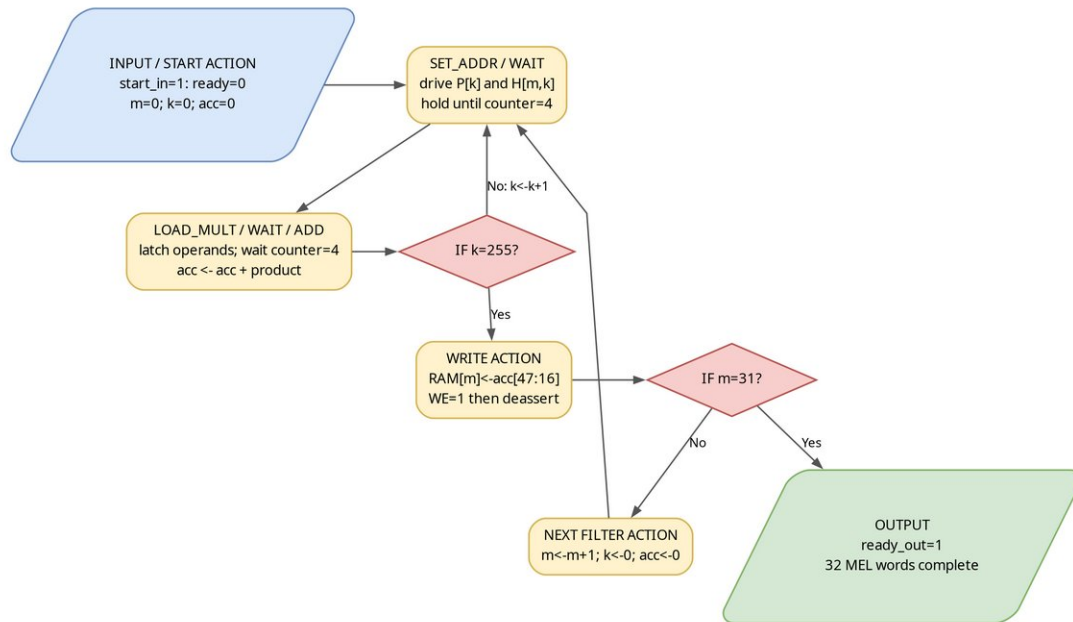


Figure 16. MEL nested-loop control for 256 bins and 32 filters.

Start clears m and k . Each address/wait sequence acquires one synchronized FFT/coefficient pair; $k=255$ writes the completed band, and $m=31$ ends the transaction.

The states idle, start_filter, set_addr, load_mult, add, write_ram, write_wait, and done wait until the address-path and multiplier counters reach four. These waits are not arbitrary idle time. They prevent an address that looks correct in a waveform from being paired with data that still belongs to an earlier request.

9.3 Numerical and timing evidence



Figure 17. Retained MEL XSim waveform showing the repeated address, MAC, and 32-write control structure.

Development-stage behavioral capture retained from before the final FFT-bin/address identity correction.

The dense activity represents the 256-bin traversal repeated for every filter, and the state/latency structure remains applicable. This historical capture predates removal of the residual $k+1$ FFT request. Figure 15 shows the current address mapping, and Figure 18 checks numerically that $H[m,k]$ consumes $P[k]$.

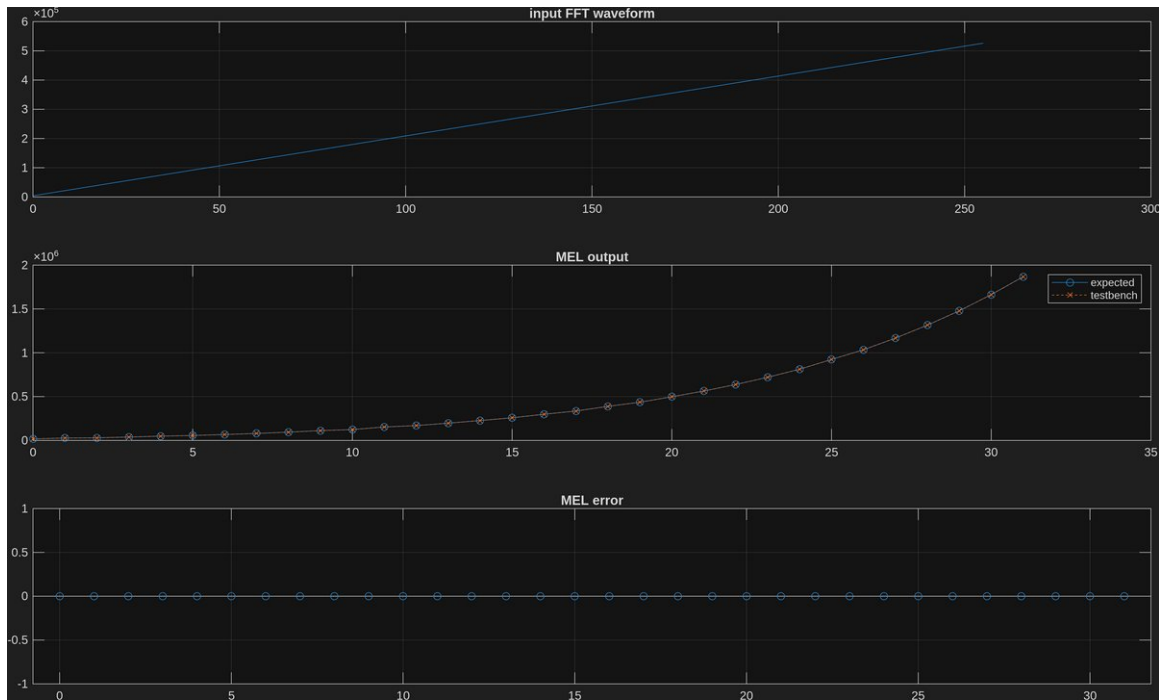


Figure 18. Exact 32-band MEL comparison after enforcing FFT-bin/address identity.

Post-correction integer checker export from the MEL testbench output.

The stimulus begins at 4,096 and increases by 2,048 at every FFT address, so pairing coefficient $H[m,k]$ with any FFT word other than $P[k]$ changes the result. Expected and simulated outputs overlap from band 0 through band 31 and the error is zero. This checks the ROM address, FFT address, multiplication, accumulation, scaling, and write path together. This ramp also checks the fixed-point filter bank and complements the later programmed-board frequency sweep.

10 LOGMEL: Piecewise Logarithmic Compression

Raw MEL energies can span several orders of magnitude. Feeding them directly to DCT would let a few large bands dominate the feature vector, so LOGMEL approximates $\log_2$ in fixed-point hardware. The logarithm compresses level while preserving the contour used by the later DCT [4], [5], without a floating-point unit or long iterative algorithm.

10.1 Binary normalization and interpolation datapath

$$x' = \max(x, 1); \quad e = \text{floor}(\log_2(x')); \quad L = (e \ll 20) + \text{base}[a] + ((\text{slope}[a] \text{ residual}) \gg 8)$$

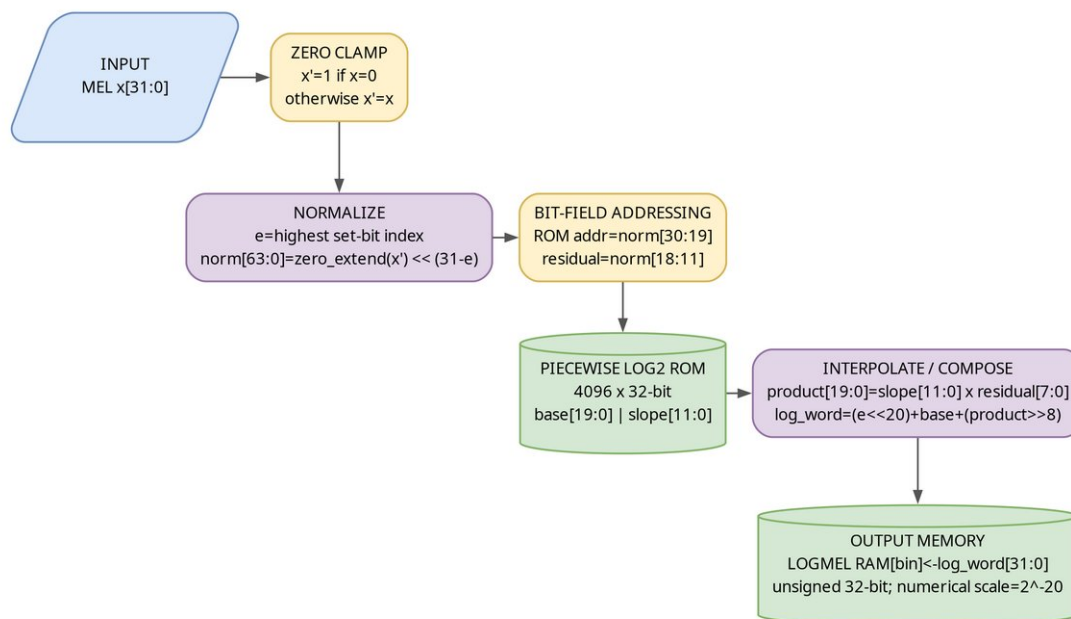


Figure 19. LOGMEL binary normalization, ROM lookup, and interpolation datapath.

The leading-one search converts the logarithm into an exponent plus normalized mantissa. Bits 30 down to 19 form the 12-bit ROM address, bits 18 down to 11 form the interpolation residual, and the packed base/slope word drives a 12-by-8 multiply. The exponent and interpolated fraction share one unsigned 32-bit word with scale 2^{20} .

After clamping zero to one, I find the highest set bit e and shift the word until its leading one occupies bit 31. Normalized bits 30 down to 19 select one of 4,096 ROM segments, and bits 18 down to 11 form an eight-bit interpolation residual. Each 32-bit ROM word packs a 20-bit base and 12-bit slope. A 12-by-8 multiplier produces the correction, and adding the exponent in a 2^{20} scale yields the 32-bit result. Zero and unity consequently map deterministically to zero.

10.2 Per-band address, wait, and write control

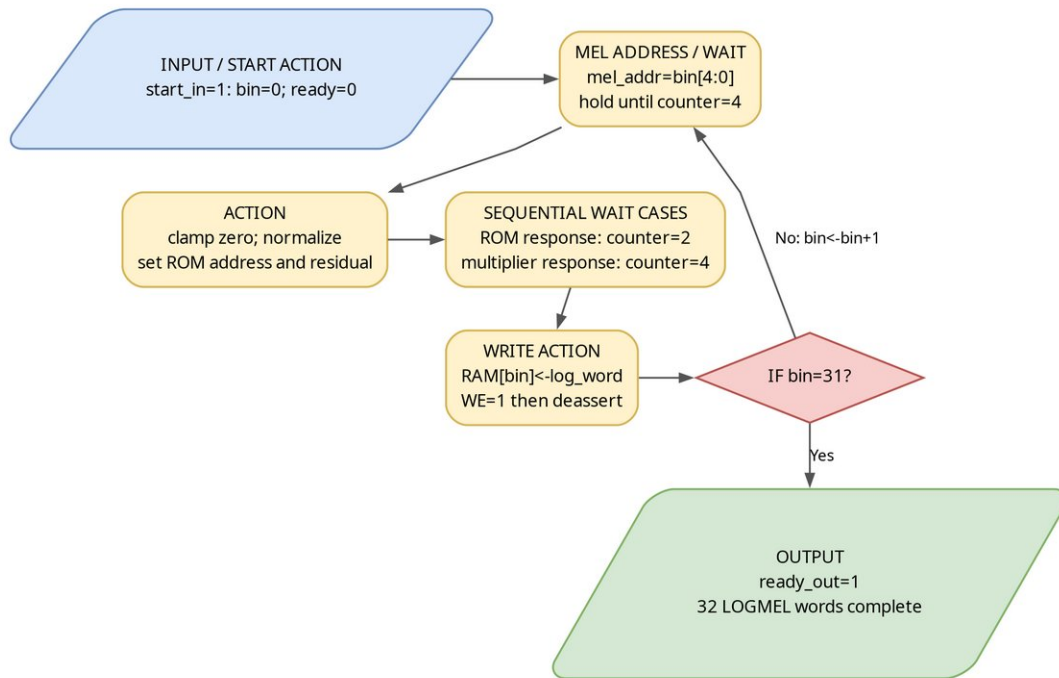


Figure 20. LOGMEL 32-band sequential control and synchronous-memory waits.

Start initializes `bin` once. Each iteration requests `MEL[bin]`, waits four counts, forms the ROM address, waits two counts for the ROM and four for the multiplier, and writes `LOGMEL[bin]`. `bin=31` raises `ready`; otherwise `bin` increments.

The sequential states are `idle`, `set_mel_addr`, `wait_mel`, `set_rom_addr`, `wait_rom`, `load_mult`, `wait_mult`, `write_ram`, `write_wait`, and `done`. For each of 32 bands I allow four counts for the MEL memory, two for the logarithm ROM, and four for the multiplier. One small ROM/multiplier path is reused for every band, giving a fixed and observable order from address 0 through 31.

10.3 Boundary behavior and numerical evidence

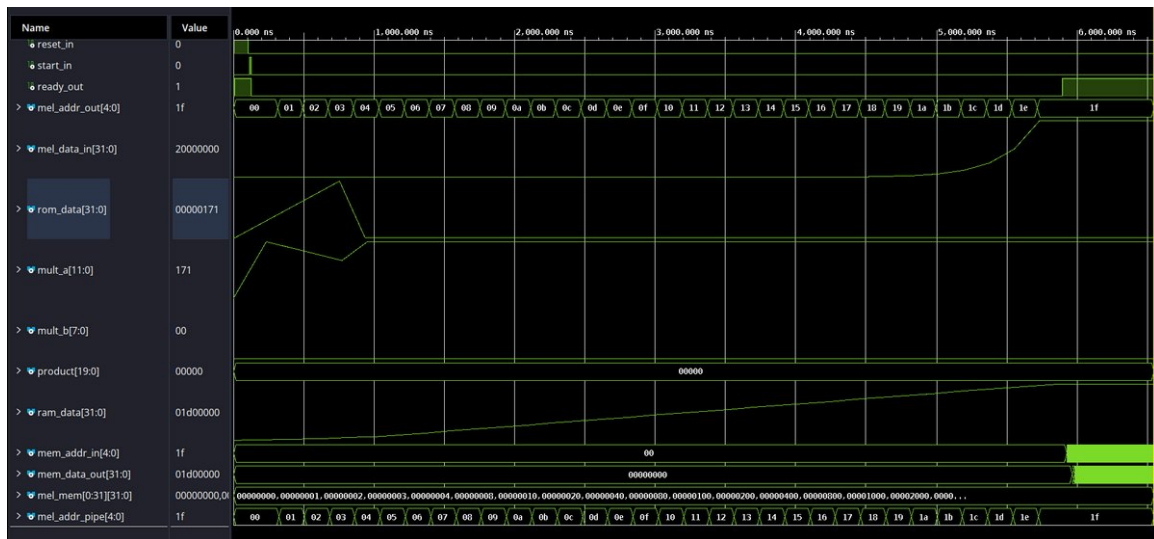


Figure 21. LOGMEL XSim waveform showing MEL reads, normalization, ROM interpolation, and output writes.

Behavioral simulation capture from the indexed LOGMEL testbench.

The MEL address steps across all 32 entries while the exponent, ROM data, slope/residual product, and final word change after their scheduled delays. ready remains low until the last write. The timing view shows an explicit piecewise datapath with controlled synchronous latency.

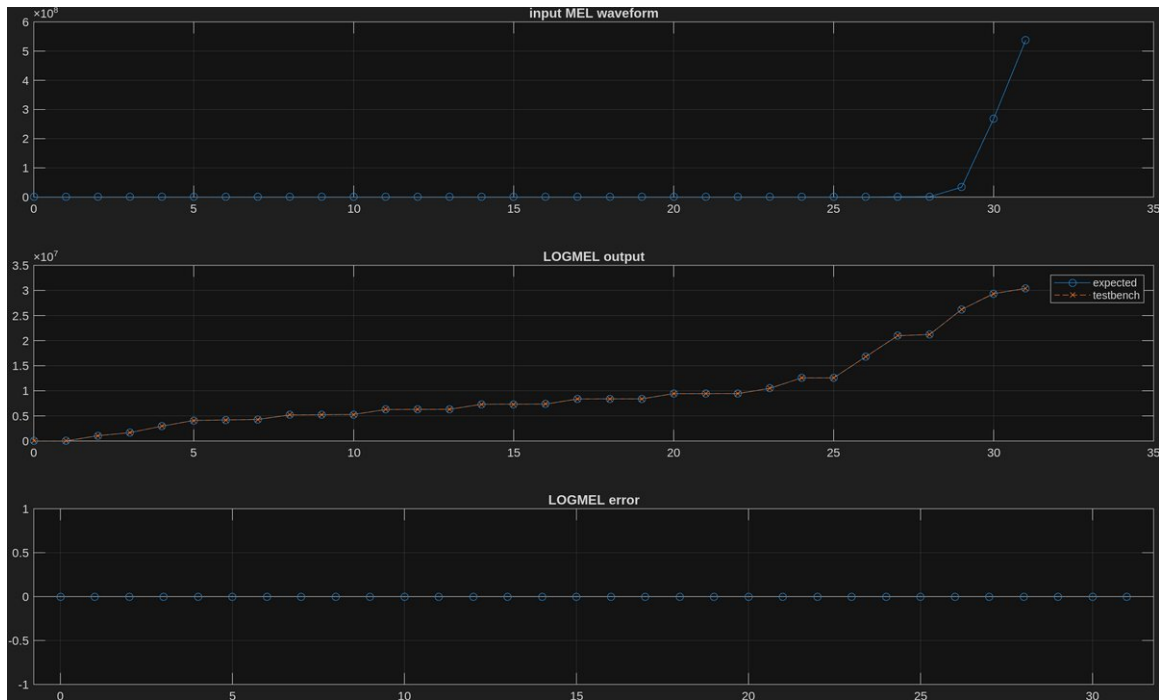


Figure 22. Boundary-focused LOGMEL comparison across zero, powers of two, and neighboring values.

Current integer checker export from the LOGMEL testbench output.

The 32 inputs include zero, unity, powers of two, and values immediately above or below exponent boundaries. Expected and simulated words coincide and the error is zero. Exact powers confirm the 20-fraction-bit scale directly, while adjacent values exercise the ROM address and interpolation residual. The visible compression of the high-dynamic-range input is the intended feature, not a loss of the spectral contour.

The checker covers the implemented approximation and boundary handling, not equivalence to an unrestricted floating-point logarithm. Reference features use the same quantization, so COMP compares like with like.

11 DCT: Eight-Coefficient Feature Compression

DCT converts each 32-value log-MEL contour into eight signed coefficients [4], [5]. The cosine ROM implements non-DC DCT-II orders C1 through C8; hardware index 0 represents C1, not C0. MEL energy is retained separately for speech-onset selection, while the non-DC coefficients describe spectral shape for template distance.

$$A_q = \sum(L[m] Q[q,m], m=0\dots31); \quad C_q = \text{ASR}(A_q + 2^{29} \text{ if } A_q \geq 0 \text{ else } A_q - 2^{29}, 30)$$

Each LOGMEL word is unsigned 32-bit and each cosine coefficient is signed 16-bit, approximately Q1.15. Their product is signed 48-bit and the 32 terms accumulate in signed 64-bit precision. The RTL adds 2^{29} for a non-negative accumulator and subtracts 2^{29} for a negative accumulator before the arithmetic right shift by 30, then retains signed 16 bits. This is a deterministic sign-dependent quantization convention reproduced by the checker; the negative side is biased away from zero, so it is not symmetric round-to-nearest.

The cosine address is $32(q-1)+m$. The LOGMEL request is issued as $(m+31)$ modulo 32 to compensate the registered upstream read path; this is a memory-scheduling alignment, not a rotation of the reported feature vector. The states idle, set_addr, load_mult, add, write_ram, write_wait, next_dct, and done perform 32 multiply-accumulate iterations per coefficient and repeat for all eight orders using one datapath.

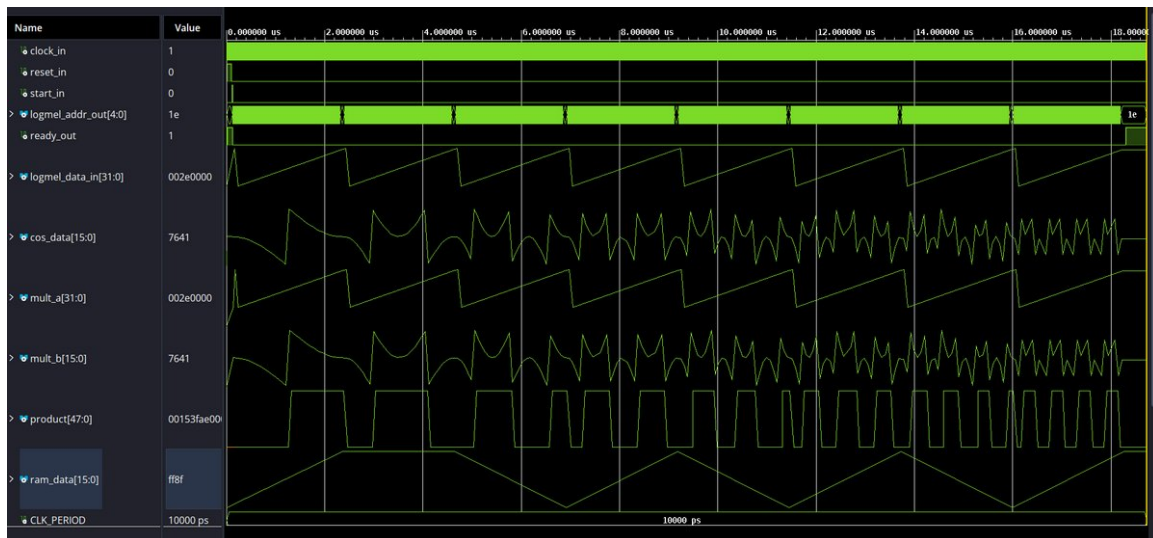


Figure 23. DCT XSim waveform across eight 32-sample cosine sweeps.

Behavioral simulation capture from the indexed DCT testbench.

The LOGMEL input contour is revisited for every transform order while the cosine sequence oscillates progressively faster. Product and accumulator values change polarity, and the output RAM contains both positive and negative coefficients. The repeated pattern explains how one arithmetic path produces all eight features before ready returns high.

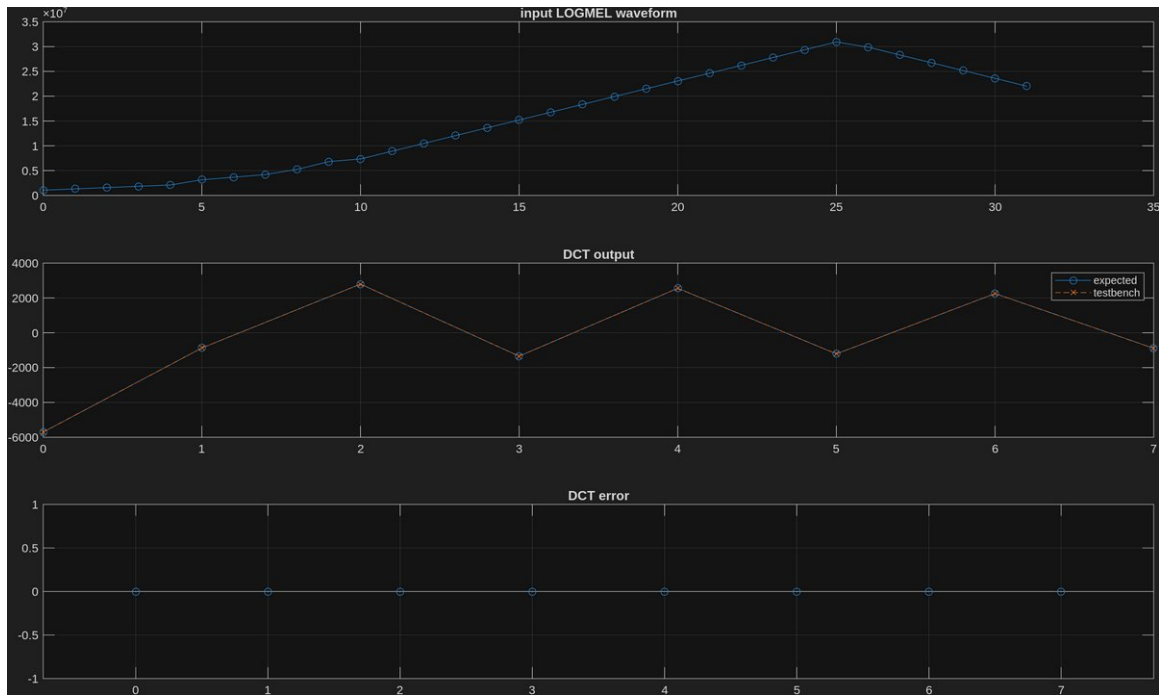


Figure 24. Exact fixed-point DCT comparison for the eight C1-to-C8 outputs.

Current integer checker export from the DCT testbench output.

The tested contour produces [-5702, -873, 2794, -1342, 2555, -1211, 2250, -891]. Expected and simulated traces overlap at every coefficient and the error is zero. Alternating positive and negative results simultaneously exercise cosine addressing, signed multiplication, wide accumulation, sign-dependent rounding, 16-bit conversion, and RAM ordering.

The deterministic contour checks the complete fixed-point DCT datapath. Spoken-input and programmed-board DCT maps later show the same eight-word interface populated by acquired signals.

12 COMP: Runtime Enrollment, Active-Window Selection, and Template Matching

COMP handles both runtime enrollment and nearest-template classification. In record mode, one processed utterance is reduced to an energy-aligned interval of sixteen 128-bit DCT feature words and written into the selected digit/trial slot. This is the main adaptation path because it replaces a startup example with a speaker-specific template without resynthesis. In compare mode, the same writable RAM supplies the templates for SAD/L1 distance.

12.1 Frame feature acquisition and binary packing

COMP receives one completed feature frame at a time. It addresses all 32 MEL bands and accumulates them into one unsigned 64-bit energy value. It then addresses the eight DCT outputs and packs the signed 16-bit C1-through-C8 words into one 128-bit frame. A 64-word utterance RAM stores the 63 captured feature frames.

```
energy[f] = sum(MEL_f[m], m=0...31)
```

```
U[f][127:0] = C1[15:0] | C2[15:0] | ... | C8[15:0]
```

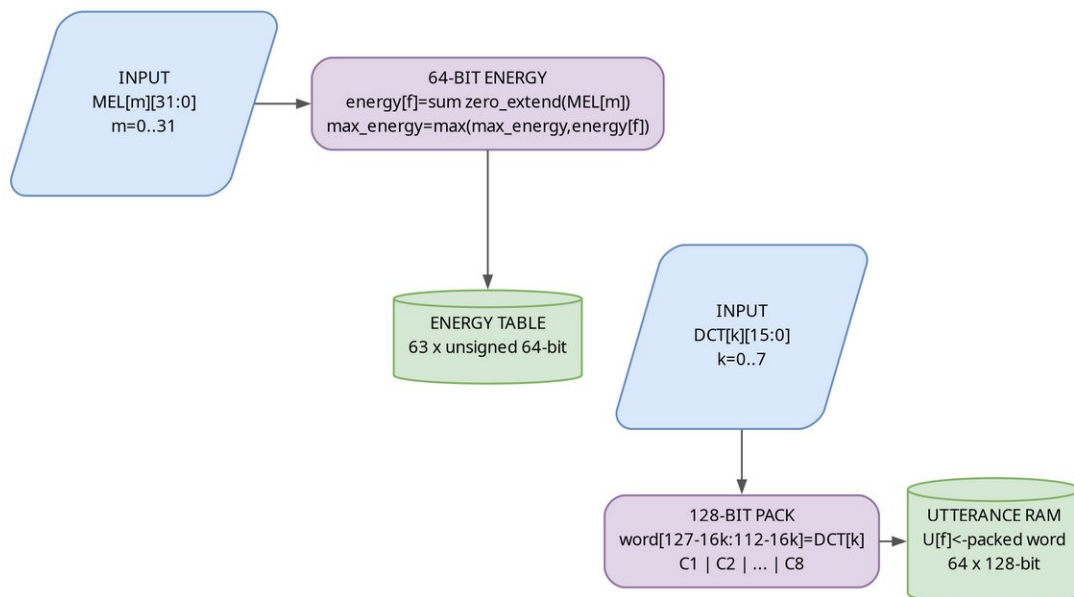


Figure 25. COMP per-frame MEL-energy accumulation and DCT feature packing.

Thirty-two addressed unsigned MEL words form one 64-bit frame-energy sum. Eight signed 16-bit DCT words are packed in C1-through-C8 order into one 128-bit utterance-RAM word.

The two serialized reads preserve different information: energy provides speech timing while the signed DCT word provides spectral shape. The resulting energy[f] and U[f] entries share the same frame number, so later enrollment and comparison use the timing decision without changing the feature-word layout.

12.2 Active-window threshold and start-address control

$\text{threshold} = \text{floor}(\text{max}(\text{energy}) / 8)$; $\text{active_start} = 0$ if no crossing, else $\text{min}(\text{first } f \text{ with } \text{energy}[f] > \text{threshold}, 47)$

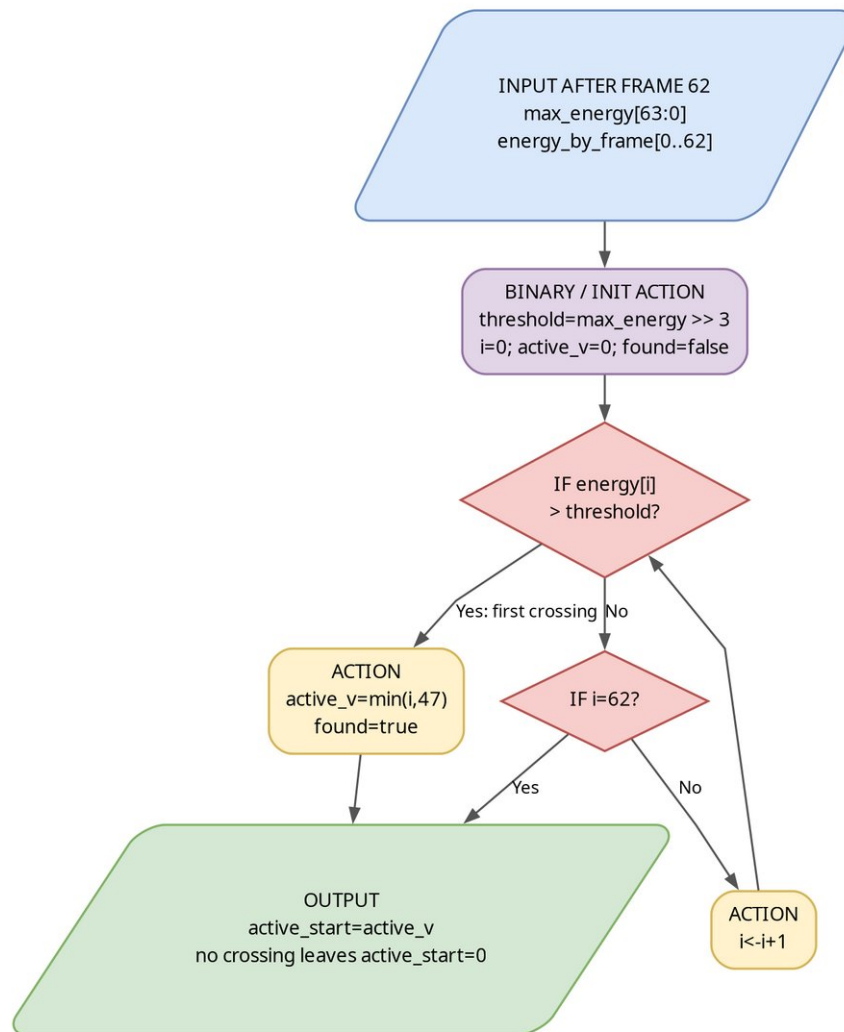


Figure 26. COMP final-frame activity threshold scan and legal 16-frame start selection.

After frame 62, $\text{threshold} = \text{max_energy} \gg 3$ implements division by eight. The scan selects the first energy strictly above threshold, retains zero when no crossing exists, and clamps a crossing after frame 47 to the final legal 16-frame start. Record and compare modes therefore receive the same active_start value.

The activity threshold is one eighth of the maximum frame energy. The first frame strictly above that threshold selects the speech start, and any later value is ignored. If the first crossing occurs after the last complete 16-frame interval, the start is clamped to 47; if no frame crosses the threshold, the initialized start value remains zero. Enrollment and comparison therefore consume the same $U[\text{active_start}]$ through $U[\text{active_start}+15]$ interval.

12.3 Writable reference-RAM address map and configuration-time seed

The implemented reference store is a writable 512-by-128 simple dual-port RAM. FPGA configuration initializes addresses 0 through 479 with 30 default templates: ten digits, three trials per digit, and 16 frames per trial. Addresses 480 through 511 are zero-filled spare locations. The initialization image is included for immediate comparison, deterministic simulation, and convenient demonstration; it does not turn the store into a ROM and it is not a substitute for runtime enrollment.

reference = $3d+t$; slot_base = $16(3d+t) = 48d+16t$; address = slot_base+i

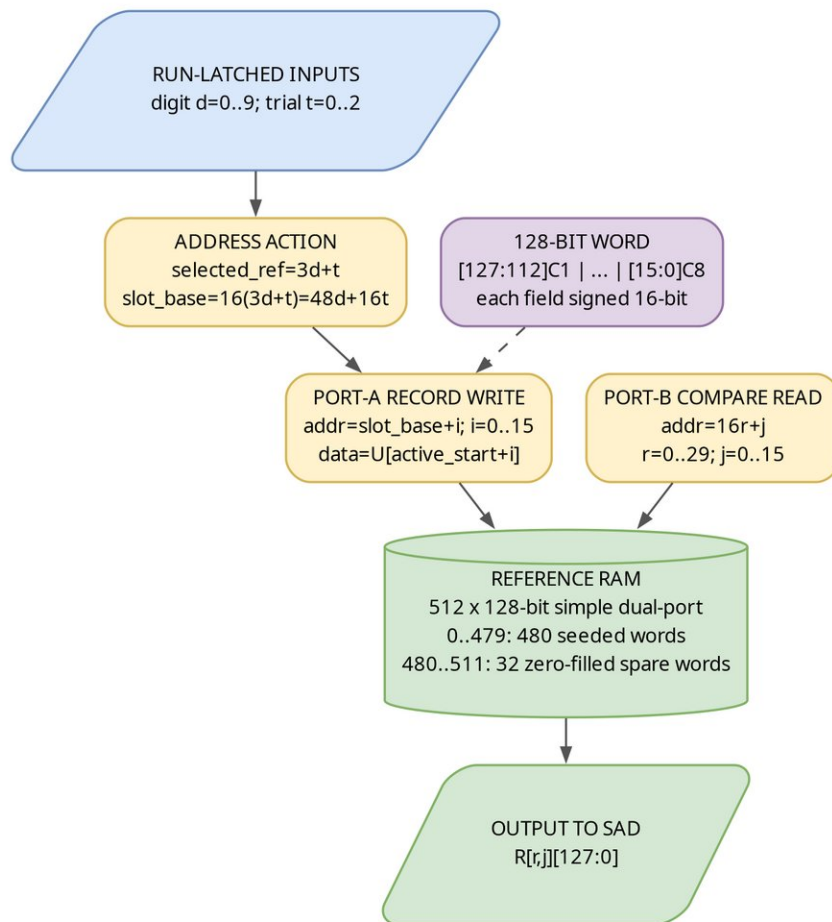


Figure 27. Writable COMP reference-RAM dimensions, address mapping, and 128-bit word format.

The 512-by-128 simple dual-port RAM holds 30 populated 16-frame slots at addresses 0 through 479 and leaves 32 spare zero-filled words. Record address $16(3d+t)+i$ and compare address $16r+j$ use the same C1-through-C8 word layout.

The COE was generated from saved development recordings and gives the FPGA a repeatable template set at configuration. Runtime recording remains the normal adaptation path and overwrites whichever digit/trial slot the user selects. Because the saved recordings predate the final FFT/MEL address corrections, the seed is not used as a current live-accuracy result.

12.4 Runtime speaker-specific enrollment

At the synchronized top-level start edge, record/compare mode, selected digit, and selected trial are copied into run registers and remain stable for the full acquisition and 63-frame feature sequence. In record mode, the final active_start drives a sixteen-word copy from utterance RAM into the selected reference slot. Every recorded word uses the same C1-through-C8 packing as the COE, so comparison does not need a separate format for default and user-enrolled templates.

```
R[3d+t, i] <- U[active_start+i], i=0...15
```

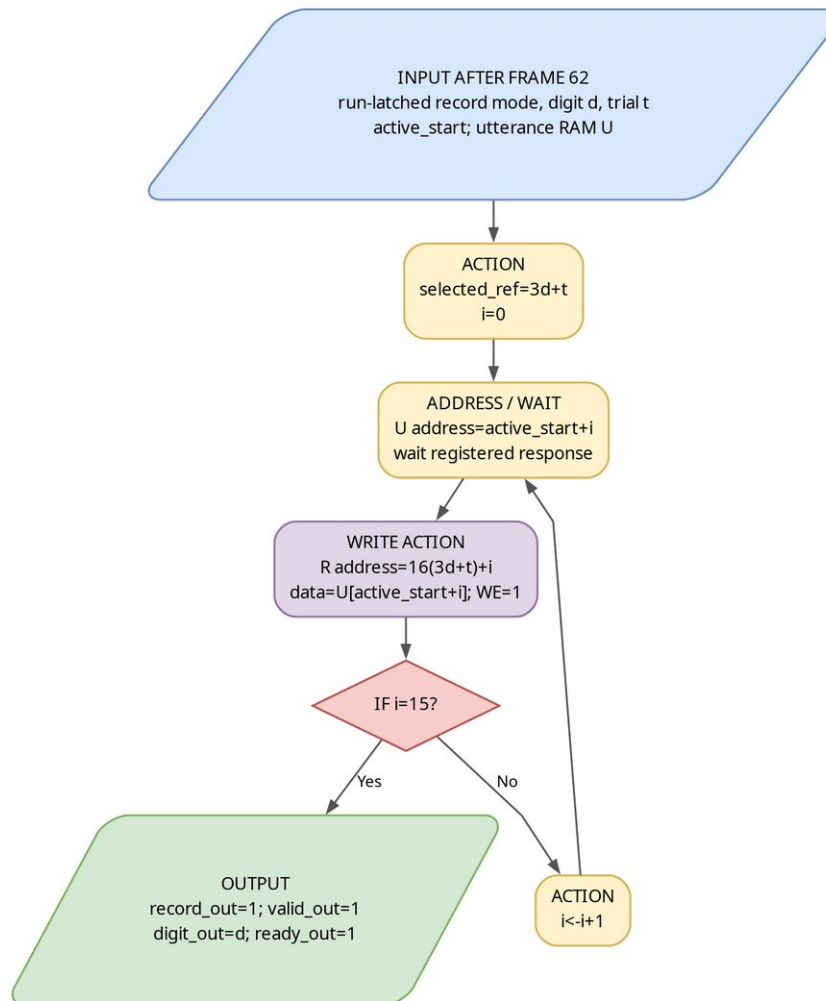


Figure 28. Runtime enrollment state flow and exact utterance-to-reference address translation.

One shared BRAM read path retrieves $U[\text{active_start} + i]$, and one reference-RAM write pulse stores that 128-bit word at $16(3d+t) + i$. The $i=15$ decision terminates after exactly sixteen writes, asserts record_out and valid_out , returns the enrolled digit, and makes COMP ready for the next transaction. No COE regeneration or bitstream rebuild occurs during enrollment.

The update is volatile but persists through an ordinary control reset because that reset does not rewrite BRAM. Reconfiguring the FPGA restores the COE-seeded startup collection. This

behavior supports two useful workflows: immediate comparison using the repeatable default set, or speaker adaptation by replacing any selected digit/trial slot at runtime.

12.5 Reference-memory lifecycle and maintenance ownership

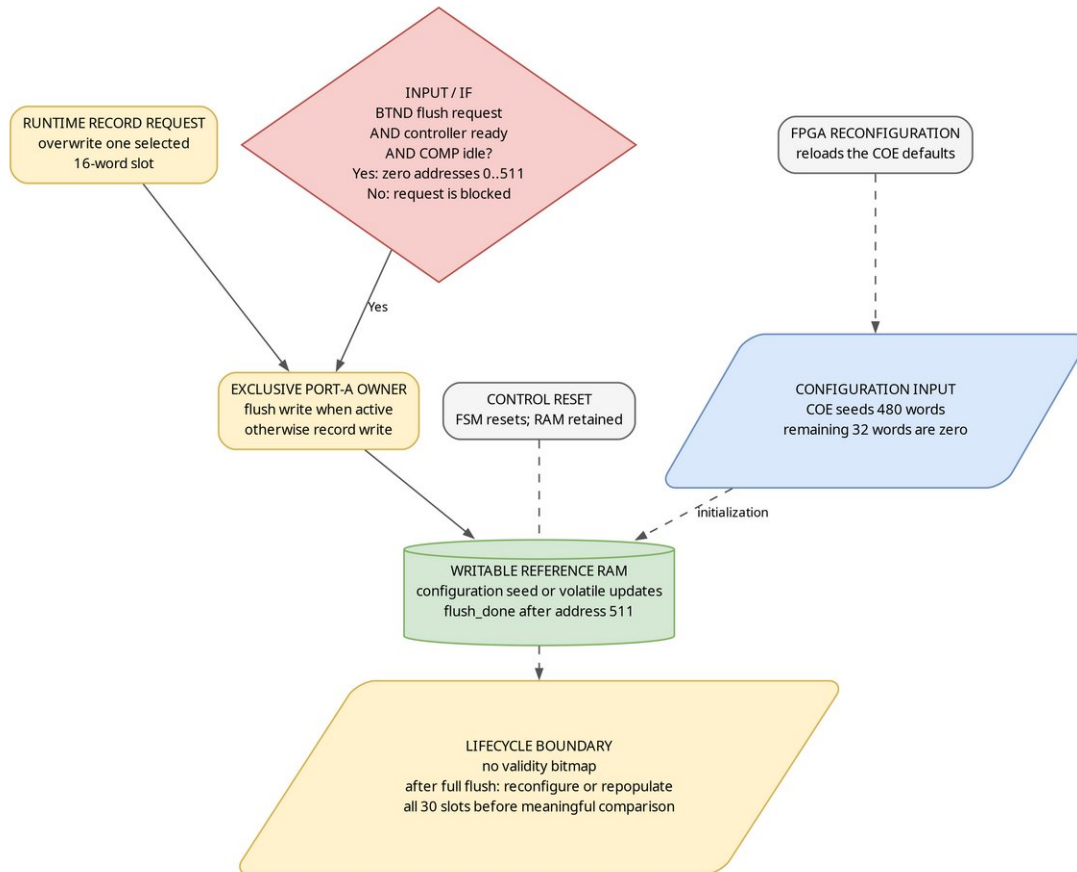


Figure 29. COMP reference-memory initialization, runtime ownership, reset, flush, and reconfiguration lifecycle.

The COE seeds the RAM only at FPGA configuration. Runtime enrollment overwrites one selected slot, control reset preserves those words, reconfiguration restores the seed, and a gated maintenance flush writes zero to all 512 addresses. The current design has no validity bitmap, so the lifecycle explicitly states that a complete flush requires all 30 slots to be repopulated or the FPGA to be reconfigured before meaningful comparison.

Runtime recording and flush share reference-RAM port A through an explicit owner selection rather than bitwise-OR address composition. `top_module` first gates BTND with the complete CTRL ready state, and COMP then synchronizes the request and admits it only while its own FSM is idle at the run boundary. A press during ADC, feature extraction, comparison, or sixteen-word enrollment therefore cannot steal the reference-RAM write port. Flush writes zero to all 512 locations and then asserts `flush_done`.

Flush means erase, not restore defaults. The current classifier has no per-slot validity bitmap, so zero-filled locations still participate in SAD. Meaningful comparison after a complete flush consequently requires reconfiguration to reload the COE or repopulation of all 30 reference

slots. Runtime overwrite itself does not require flushing: any chosen digit/trial slot can be replaced directly.

12.6 Signed-coefficient SAD/L1 datapath

$$D(r,w) = \sum(\text{abs}(U[w+j,k] - R[r,j,k]), j=0\dots15, k=0\dots7)$$

Starting at received frame 15, each newly available 16-frame window is compared with all 30 references. Utterance address $w+j$ and reference address $16r+j$ acquire the two 128-bit frame words. For each of their eight coefficients, both signed 16-bit values are extended to 17 bits before subtraction; the 17-bit magnitude is accumulated into unsigned 64-bit pair, reference, and best-distance registers.

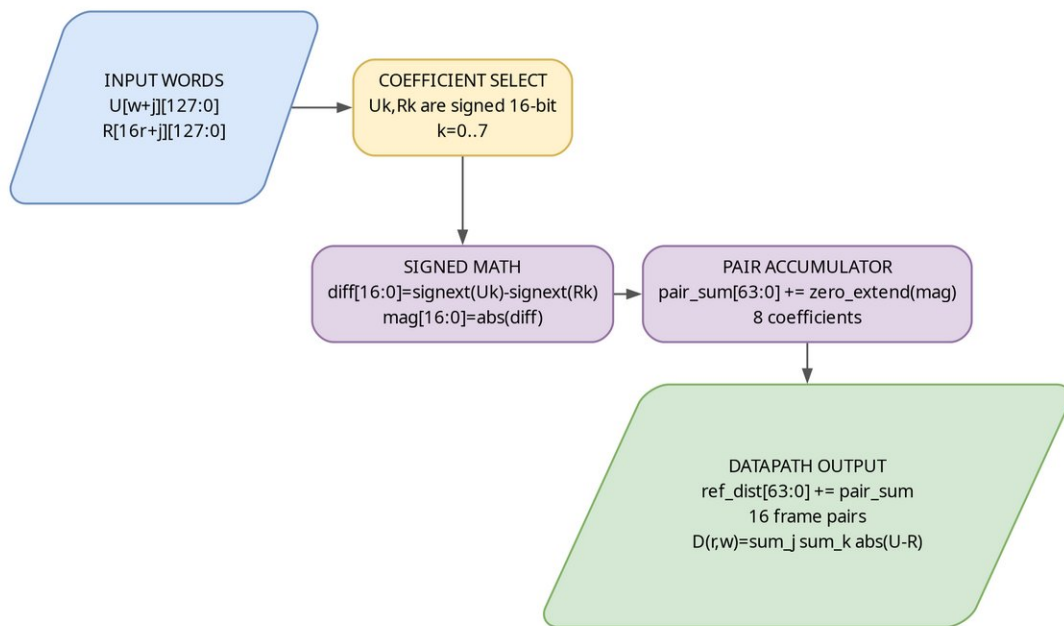


Figure 30. COMP addressed signed-coefficient absolute-difference datapath.

Utterance address $w+j$ and reference address $16r+j$ acquire two packed 128-bit words. Coefficient field k from each word is sign-extended to 17 bits before subtraction; the magnitude accumulates across eight fields and then sixteen frame pairs in 64-bit registers.

12.7 Nested comparison control and final digit selection

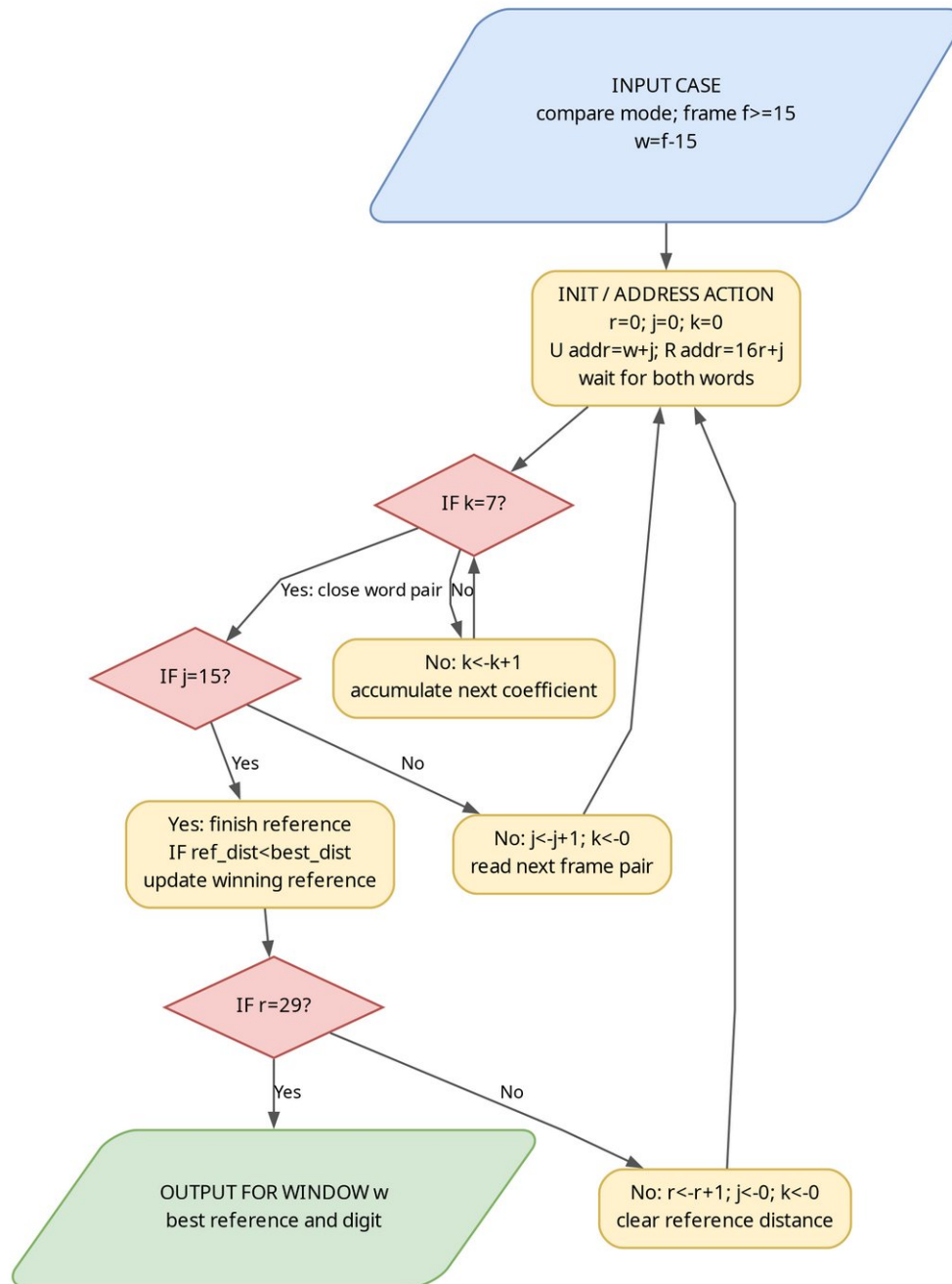


Figure 31. COMP coefficient, frame-pair, and reference-loop control for one window.

$k=7$ closes one 128-bit word pair, $j=15$ closes one reference distance, and $r=29$ yields the winning reference for window w .

The workload is 48 window starts $\times$ 30 references $\times$ 16 frames $\times$ 8 coefficients = 184,320 absolute differences for one utterance. Reusing one comparison path avoids 30 parallel engines. Registered-memory wait states ensure that the address equations shown in the diagram acquire the intended utterance and reference words before arithmetic begins.

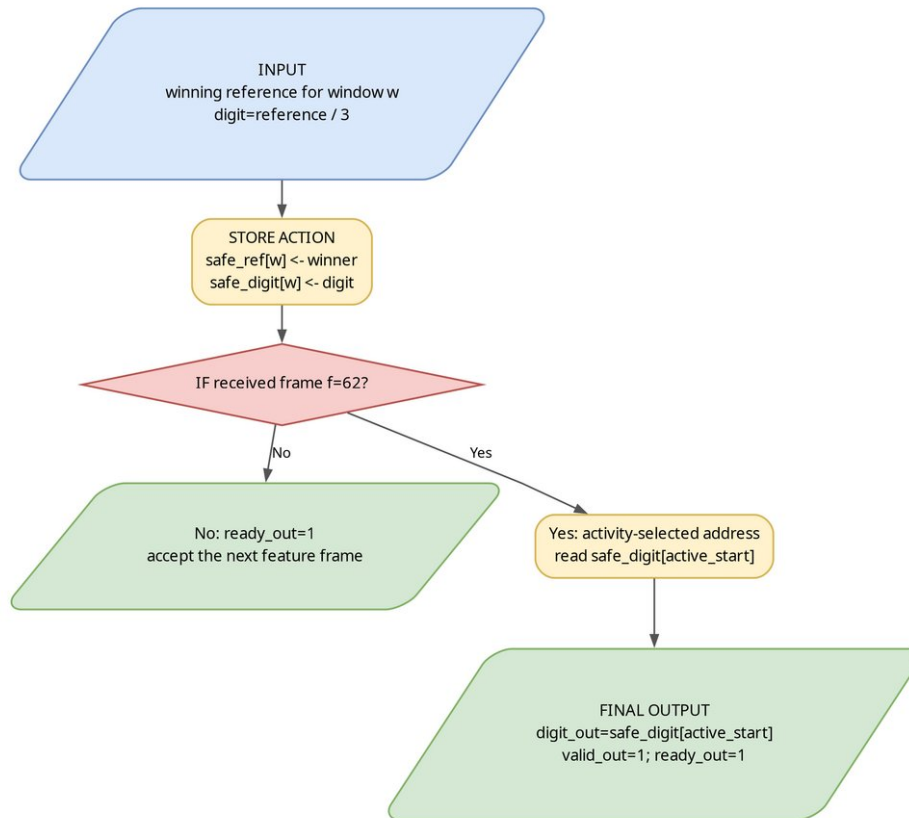


Figure 32. COMP per-window winner storage and final activity-aligned digit selection.

Frames 15 through 61 store a winner and return ready for the next feature frame. Frame 62 uses `active_start` to select the stored window digit and raises `valid_out`.

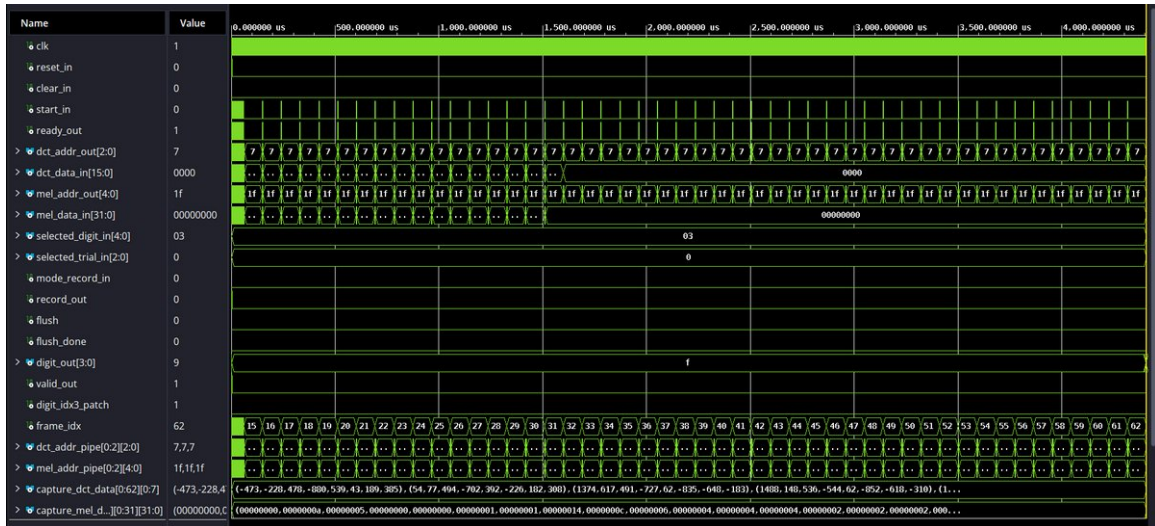


Figure 33. COMP XSim waveform across the 63-frame compare-mode sequence.

Behavioral simulation capture from the indexed COMP testbench.

The waveform shows repeated frame transactions, MEL and DCT memory reads, window/reference progress, compare mode held active, and the final valid digit 9. The selected digit/trial input is irrelevant in compare mode, so the result is produced by the stored SAD calculation rather than copied from a label port.

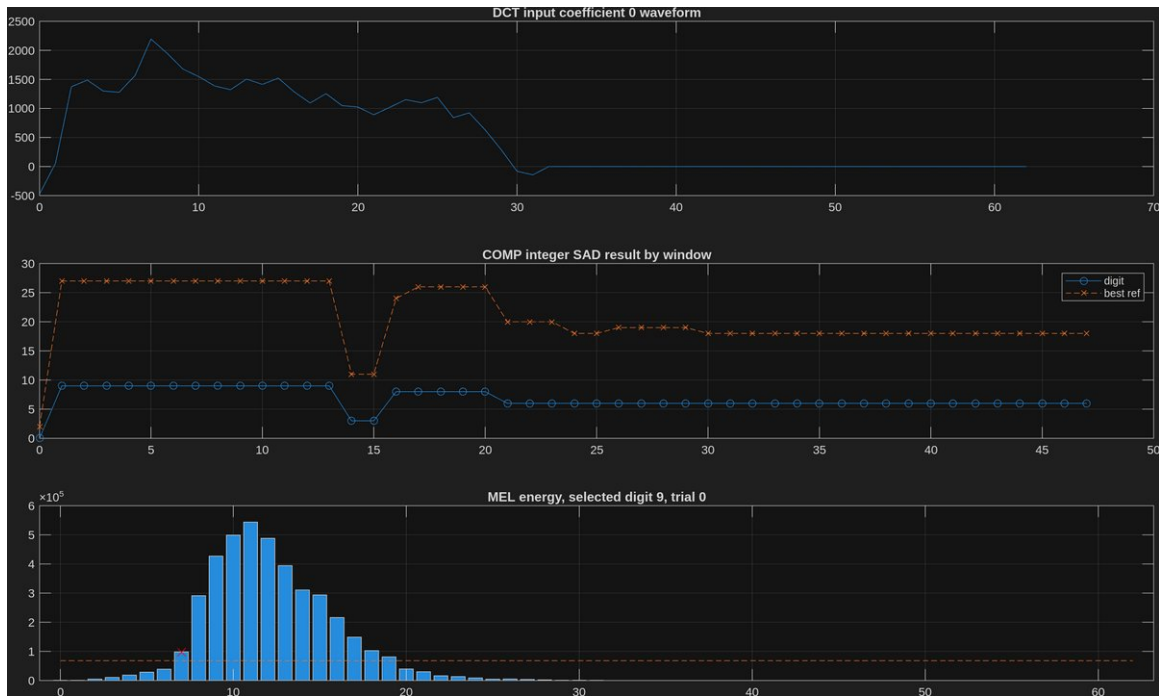


Figure 34. COMP checker showing energy onset, best reference by window, and the digit-9 result.

Current integer checker export from the COMP testbench output.

The maximum MEL energy is 543,737, making the integer threshold 67,967. The first crossing is frame 7 with energy 98,039. At that start, reference 27 has the winning SAD of 27,338 and maps to digit 9. Other starts may prefer other references; the final result follows the stated energy-aligned rule rather than a fixed or globally forced digit.

12.8 Focused record-and-recall verification

Table 9. Focused COMP record-and-recall simulation result.

Focused COMP transaction	Observed result	Property exercised
Compare the retained capture against the configuration seed	valid=1; digit=9; record=0	Baseline compare path and seeded-RAM read
Enroll the same active 16-frame window into digit 2, trial 1	valid=1; digit=2; record=1	Functional enrollment completion for reference 7, mapped to addresses 112..127
Clear only run state and compare the unchanged capture again	valid=1; digit=2; record=0	Runtime word persistence and subsequent readback

For digit 2/trial 1, $\text{selected_ref} = 3(2) + 1 = 7$ and the implemented slot formula maps its sixteen words to addresses 112 through 127. The two-line test output is an assertion-backed functional

summary of enrollment completion and later recall; it is not an address-by-address dump of all sixteen write cycles.

The COMP bench first classifies the saved sequence as digit 9, records its selected 16-frame window into digit 2/trial 1, and then classifies the same sequence as digit 2. This checks a write followed by readback at the COMP boundary without the impractical one-second ADC stimulus. It does not exercise microphone acquisition or measure live recognition accuracy. Likewise, a stationary tone that returns digit 3 checks completed comparison and result serialization, not recognition of a spoken label.

13 DEBUG: UART Observability

DEBUG acts as an internal instrument for the programmed FPGA. When enabled after COMP, it temporarily owns the read addresses of the ADC, WINDOW, FFT, MEL, LOGMEL, and DCT memories and serializes a complete frame. This makes it possible to locate the first stage with an unexpected value instead of diagnosing the recognizer from one output digit.

The packet is self-delimiting. A frame marker is followed by the frame base address and one packed COMP-status word. Markers AABBC01 through AABBC06 then introduce 512 ADC samples, 512 WINDOW samples, 256 FFT values, 32 MEL values, 32 LOGMEL values, and eight DCT coefficients. AA5503CC terminates the packet. Including the frame/status words, seven section markers, and end marker gives 1,362 32-bit words.



Figure 35. DEBUG UART packet order and section lengths.

The marker structure turns one serial byte stream into a deterministic snapshot of all major stage memories. The same address convention is preserved from the FPGA through the PC plot, so I can compare simulation, checker output, and programmed-board data at a defined module boundary rather than guessing from a final digit alone.

Table 10. DEBUG packet fields and their interpretation.

Packet item	32-bit marker or length	Meaning
Frame	AABBC00 + address + COMP info	Identifies the 256-sample frame base and current mode/result flags
ADC	AABBC01 + 512 words	Signed 12-bit source samples, extended into 32-bit transport words
WINDOW	AABBC02 + 512 words	Signed 20-bit Hann products
FFT	AABBC03 + 256 words	Unsigned 32-bit one-sided power values
MEL	AABBC04 + 32 words	Unsigned perceptual-band energies
LOGMEL	AABBC05 + 32 words	Unsigned fixed-point log2 values
DCT	AABBC06 + 8 words	Signed 16-bit C1-to-C8 coefficients
End	AA5503CC	Closes the frame packet

Each word is sent most-significant byte first; each UART byte follows the ordinary least-significant-bit-first 8N1 convention. Four bytes require 40 serial bits, so 1,362 words at 1 Mbaud require 54.48 ms per frame. Exporting all 63 frames takes approximately 3.43 seconds. That cost exists only with debugging enabled. In normal operation CTRL bypasses DEBUG and the recognition path is not burdened by full-memory serialization.

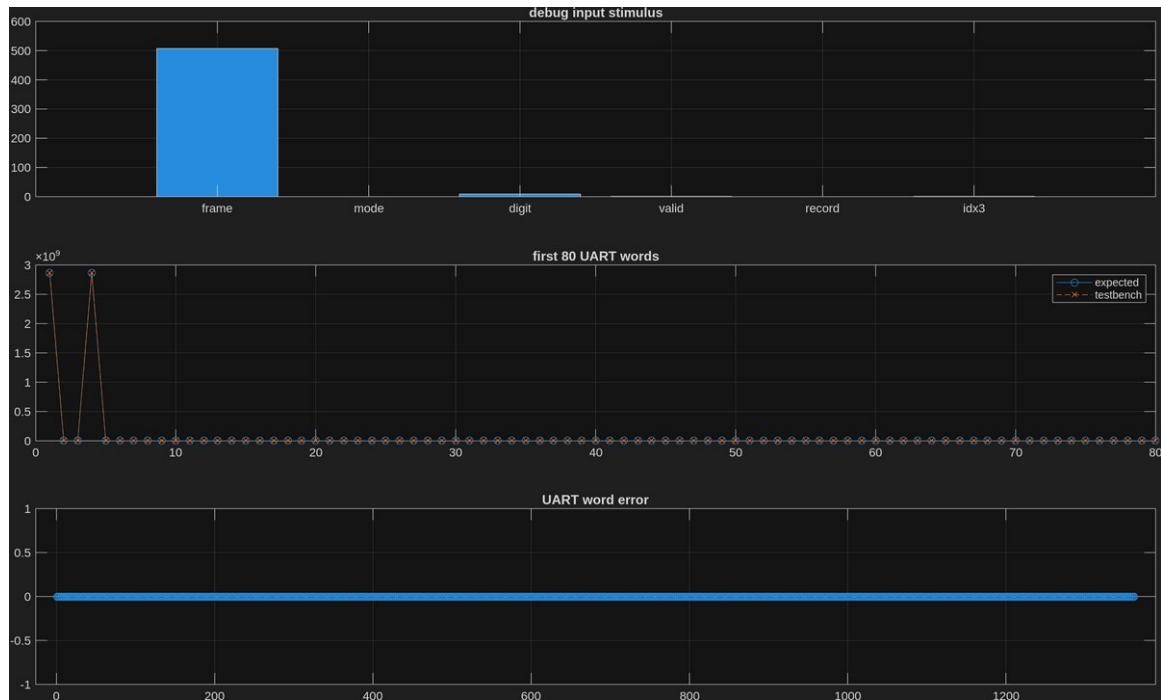


Figure 36. DEBUG packet comparison for the complete 1,362-word UART sequence.

Current checker export from the DEBUG testbench output.

The test stimulus deliberately uses the synthetic frame-address field 507 and a COMP status encoding for digit 9 so that serialization is checked with nontrivial values. Address 507 is not presented as an operational 256-aligned frame base. The marker/status words and following data overlap the expected trace, and the error remains zero through word 1,362. A raw hexadecimal-to-decimal comparison also gives 1,362 matches and zero mismatches, checking packet markers, section order, word extension, and complete-length serialization.

The DEBUG bench checks digital packet construction. The programmed-board sections later show a PC parser recovering stage arrays and a final result from the physical UART link.

14 Seven-Segment Display and Top-Level Integration

The display translates mode and result state into a compact user-facing view. In record mode I display r d digit trial. In compare mode I display C d d result after COMP asserts valid; before that event the result position remains zero. This makes it possible to conduct a basic board experiment without keeping the UART receiver connected.

The four anodes and seven segment lines are active low. Counter bits 16 down to 15 select one display position, so each phase lasts 327.68 microseconds at 100 MHz and a complete four-digit scan lasts 1.31072 ms. The decoder covers digits 0 through 9 and the letters r, C, and d. That refresh is fast enough for a steady visual result while requiring only one shared segment decoder.

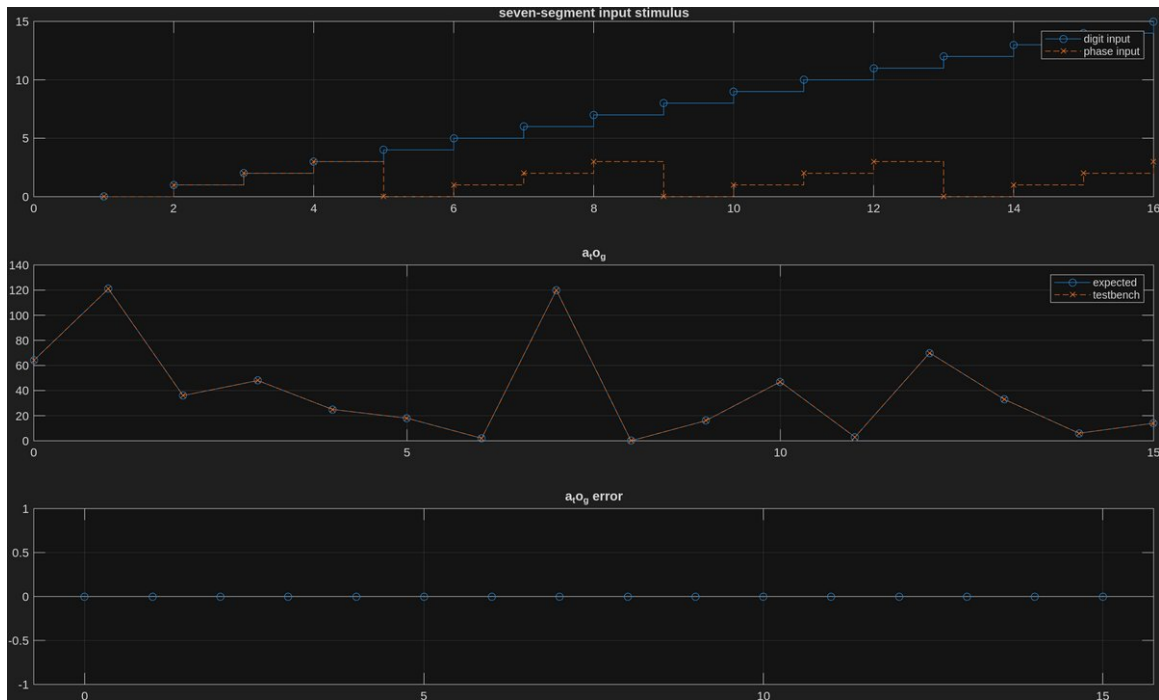


Figure 37. Seven-segment decoder comparison across numerical and letter symbols.

Current checker export from the display testbench output.

The stimulus traverses the supported digit/symbol and multiplexing-phase combinations. Expected and simulated active-low segment words overlap in all 16 cases and the error remains zero. This validates the symbol table and polarity used by the top-level display path, not merely one photographed result.

The top-level integration connects the stage memories, start/ready signals, controls, display, LEDs, ADC pins, and UART. A compact smoke test verifies simple integrated observables such as a switch reflected on a PMOD path and debug enable reflected on an LED. I do not treat that limited smoke test as proof of the complete recognizer; the controller, module, packet, and programmed-board evidence supply that coverage.

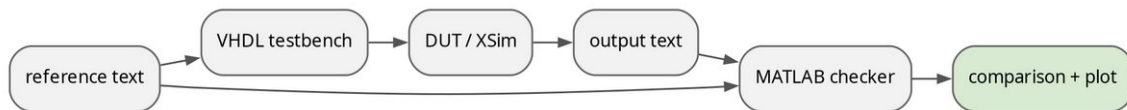
A full ADC-to-COMP behavioral run is intentionally absent. The implemented acquisition alone represents about 0.991 s of sampled time and requires many more simulator clock events than the safe module suite, so `tb_ADC` is excluded from the default regression. The focused COMP enrollment/recall bench bypasses that long acquisition while preserving the writable-reference transaction under test.

Table 11. Role and limit of the top-level integration checks.

Integration layer	What it checks	What it does not replace
Display checker	Active-low symbols and scan selection	Recognition arithmetic
Top-level smoke	Basic net connectivity and selected observables	Full ADC-to-COMP operation
CTRL checker	Ordered stage launches and frame addresses	Per-stage numeric correctness
Programmed-board UART	Integrated memories and serialized result	A multi-speaker recognition study

15 Verification Methodology and Module Results

Verification was developed alongside the RTL. Each major stage uses a VHDL testbench, a saved text interface, and a MATLAB checker. The testbench exercises the handshakes and synchronous-memory timing, the text output provides a stable boundary, and the checker recomputes or interprets the fixed-point result. This keeps the hardware timing context while making long memory outputs practical to inspect.

**Figure 38. Text-first XSim-to-MATLAB verification workflow.**

The same repeatable interface is used throughout the project: a deterministic stimulus enters the DUT, XSim writes addressable results, and an independent numerical routine reports both the expected/actual traces and their error. Raw waveforms remain beside the checker plots because agreement without timing context would not expose a misaligned BRAM or handshake.

Table 12. Module-level verification results and why each one matters.

Module	Direct retained result	Meaning in the system
ADC	Serial transactions and six explicit RAM readback values agree	Acquisition words reach deterministic storage
CTRL	48/48 expected event rows; zero frame-address error	Stage order, overlap, and debug branches are correct
WINDOW	512/512 words agree; write addresses 0..511 asserted	Final RAM contents and write-address count/order are checked
FFT	Peak bin/address 32; integer-ideal max error 45; AMD parity 256/256	Peak address and all 256 retained one-sided power words are checked against independent references
MEL	32/32 words agree	All filter/coefficient/address paths are exercised
LOGMEL	32/32 displayed differences are zero	Manual diagnostic review covers exponent, ROM, interpolation, and zero clamp
DCT	8/8 displayed differences are zero	Manual diagnostic review covers C1-C8 addressing, accumulation, and quantization
COMP	Energy start 7, reference 27, digit 9, SAD 27,338	The compare result follows the implemented onset rule
DEBUG	1,362/1,362 UART words agree	Every packet section is serialized in the intended order
Display	16/16 tested symbol/phase words agree	Active-low result presentation is correct

The automated checks and the diagnostic plots are distinguished deliberately. FFT, MEL, WINDOW, CTRL, COMP, DEBUG, and display checks contain assertion or mismatch gates at their stated boundaries. The retained LOGMEL and DCT figures were manually reviewed against generated expected vectors and show zero displayed differences; their current MATLAB scripts are diagnostic comparisons rather than automated numerical pass gates. An XSim PASS for those benches denotes bounded simulation completion, not an independent numerical verdict.

16 Programmed-Board Debugger Demonstration

I exercised the programmed FPGA with stationary inputs at 100 Hz, 500 Hz, 1 kHz, 2 kHz, and 4 kHz and exported the internal stage memories over the DEBUG channel. A tone is not a spoken digit, but it is a strong integration probe because its frequency is known and its expected movement through FFT and MEL is easy to interpret. The 1 kHz sequence below is

shown without skipping a component: ADC, Hann overlap-add view, WINDOW frame map, FFT, MEL, LOGMEL, and DCT.

These 3 June captures predate the later FFT and MEL address-identity corrections [7]. They are historical evidence of frequency-responsive end-to-end memory population; current $P[k]$ -to- $H[m,k]$ correctness comes from the post-correction module regressions.

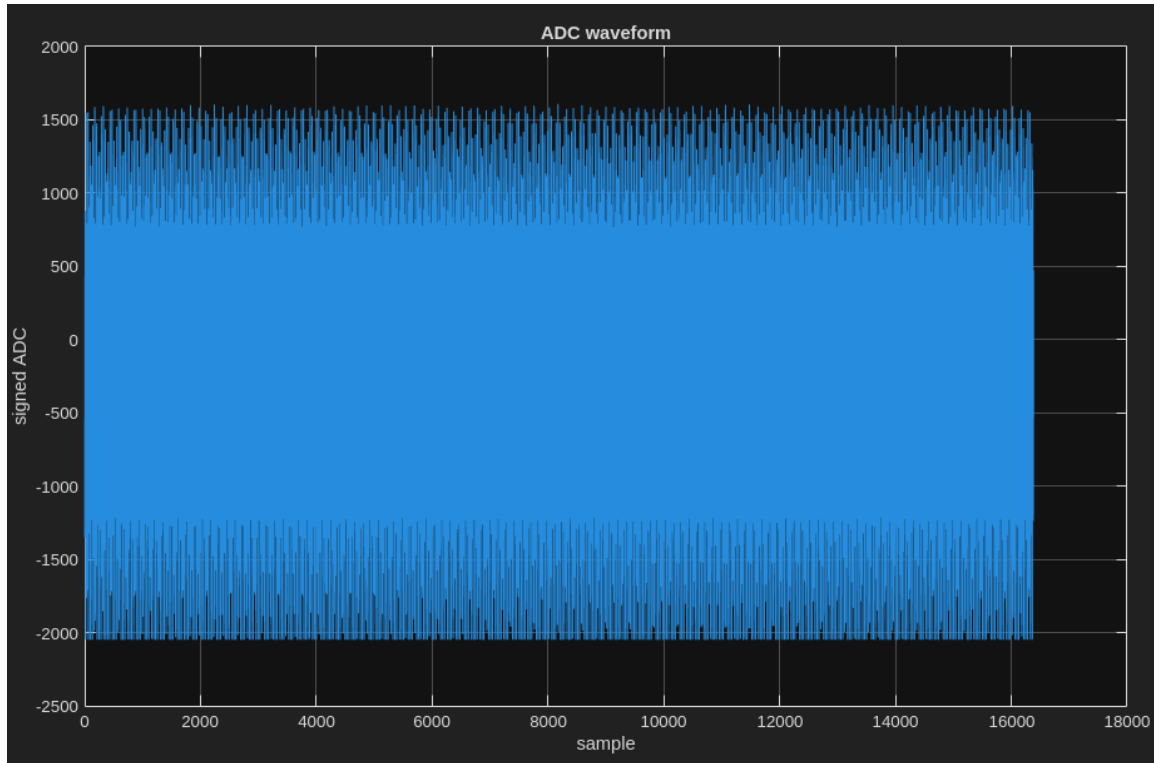


Figure 39. Programmed-board 1 kHz experiment: ADC capture.

UART capture from the programmed Basys 3, 3 June 2026.

The complete 16,384-sample record contains continuous centered tone activity. At page scale many cycles merge into a dense band; the relevant result is sustained signed acquisition, RAM storage, and successful UART reconstruction rather than a precision distortion measurement.

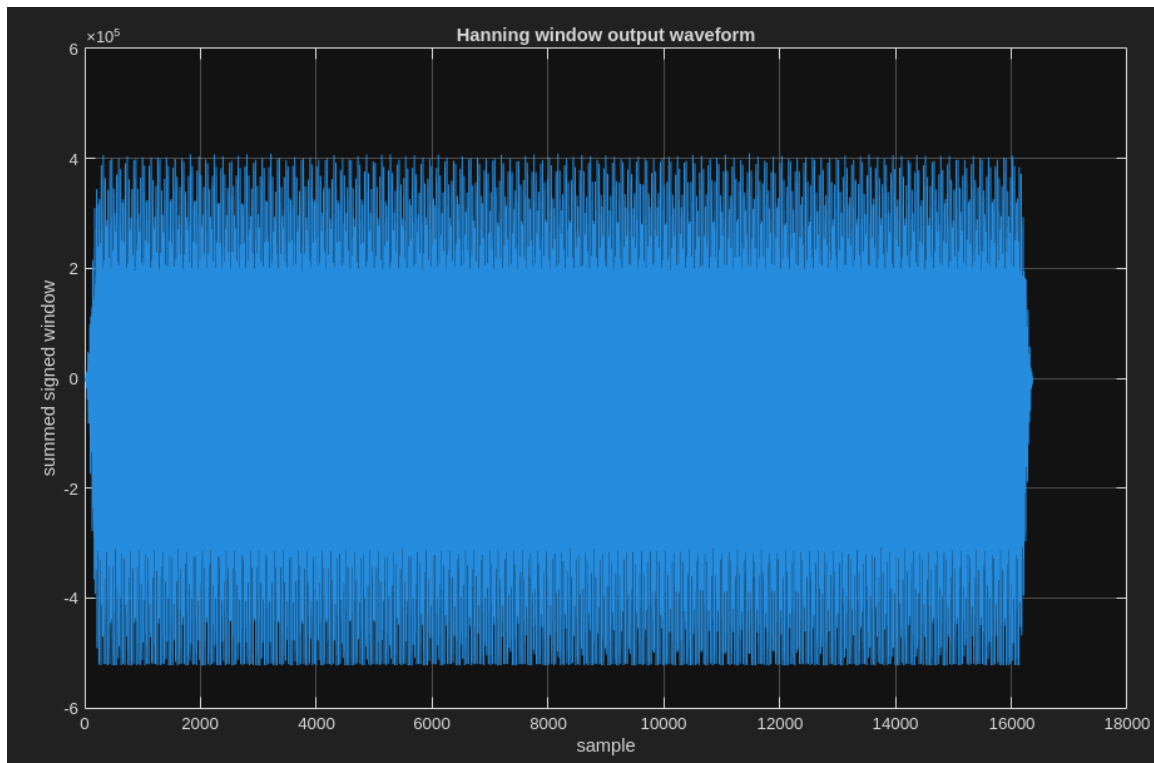


Figure 40. Programmed-board 1 kHz experiment: Hann-window overlap-add reconstruction.

UART capture from the programmed Basys 3, 3 June 2026.

This view overlays/reconstructs all 63 tapered frames, so it should not be read as one isolated coefficient curve. It shows continuous 256-sample-hop coverage and the intended overlap between adjacent frames.

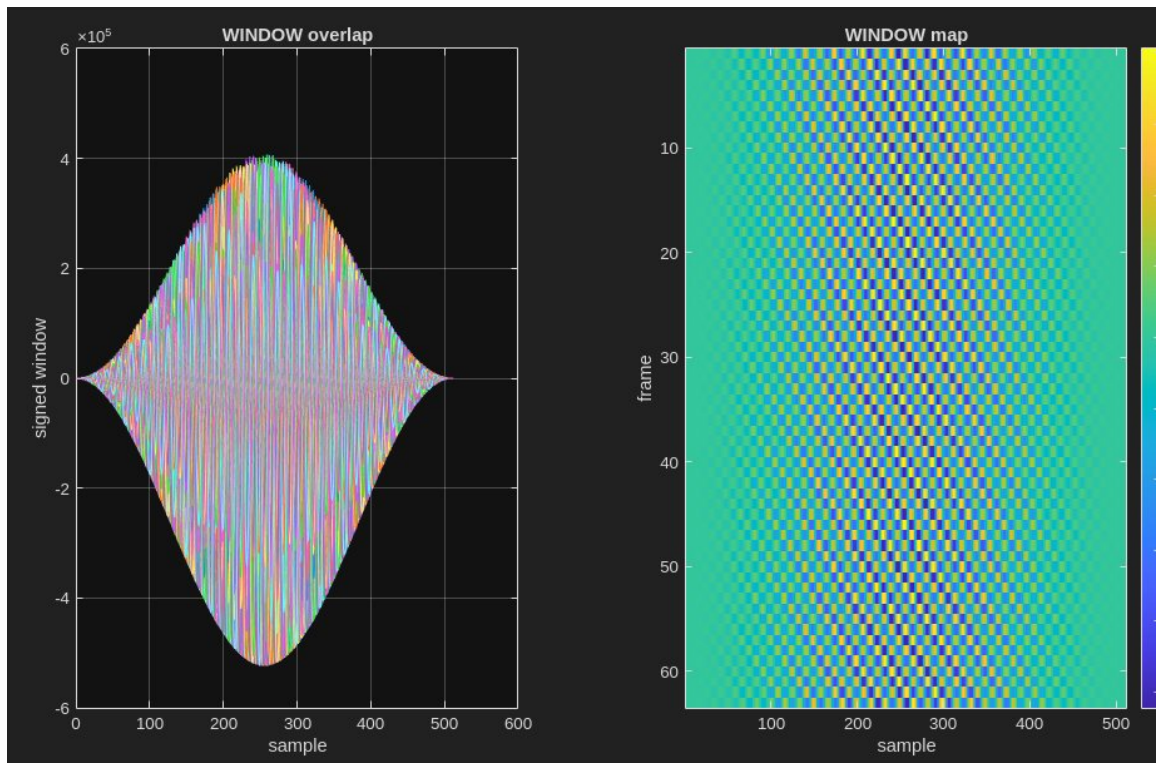


Figure 41. Programmed-board 1 kHz experiment: WINDOW output across all frames.

UART capture from the programmed Basys 3, 3 June 2026.

The frame-domain plot and map show repeated tapered 512-sample records rather than one unexplained trace. Stable structure across frames confirms that frame-base progression, sample reads, coefficient application, and UART frame assembly remain coherent on programmed hardware.

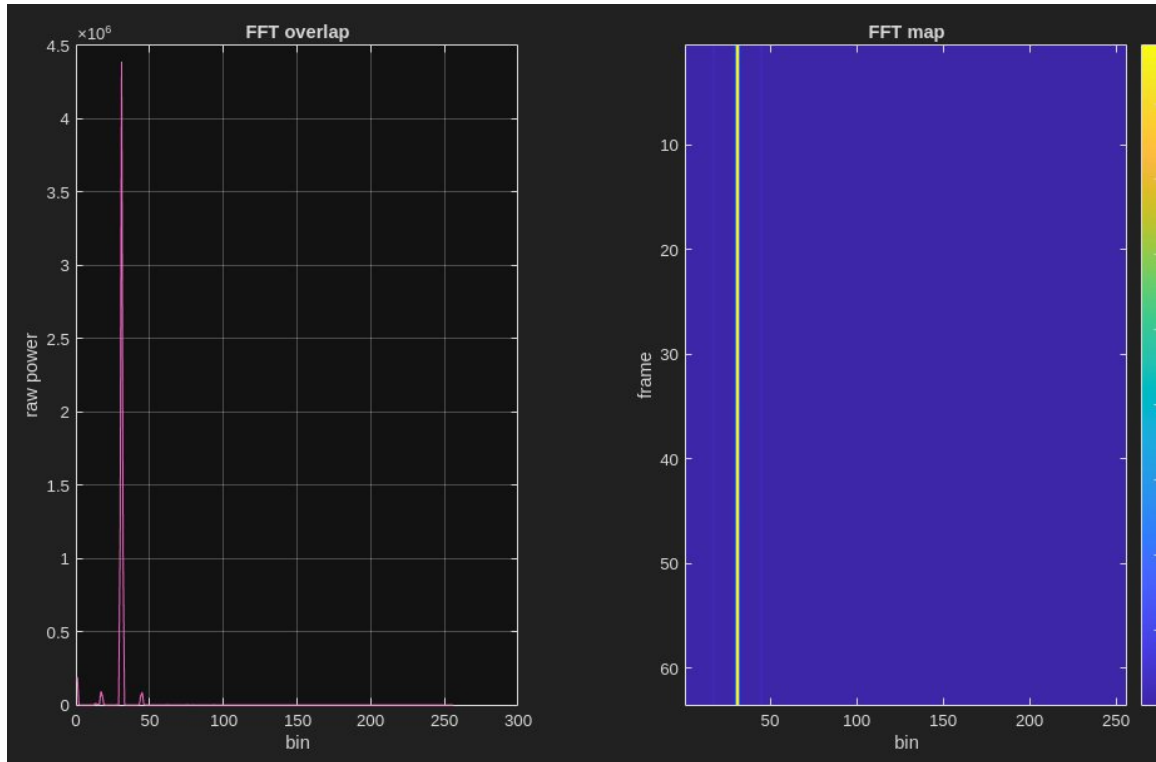


Figure 42. Programmed-board 1 kHz experiment: FFT power output.

UART capture from the programmed Basys 3, 3 June 2026.

The spectrum remains concentrated near the applied-tone region across the frame sequence. This historical capture shows frequency localization after physical acquisition through the development AXI/power pipeline.

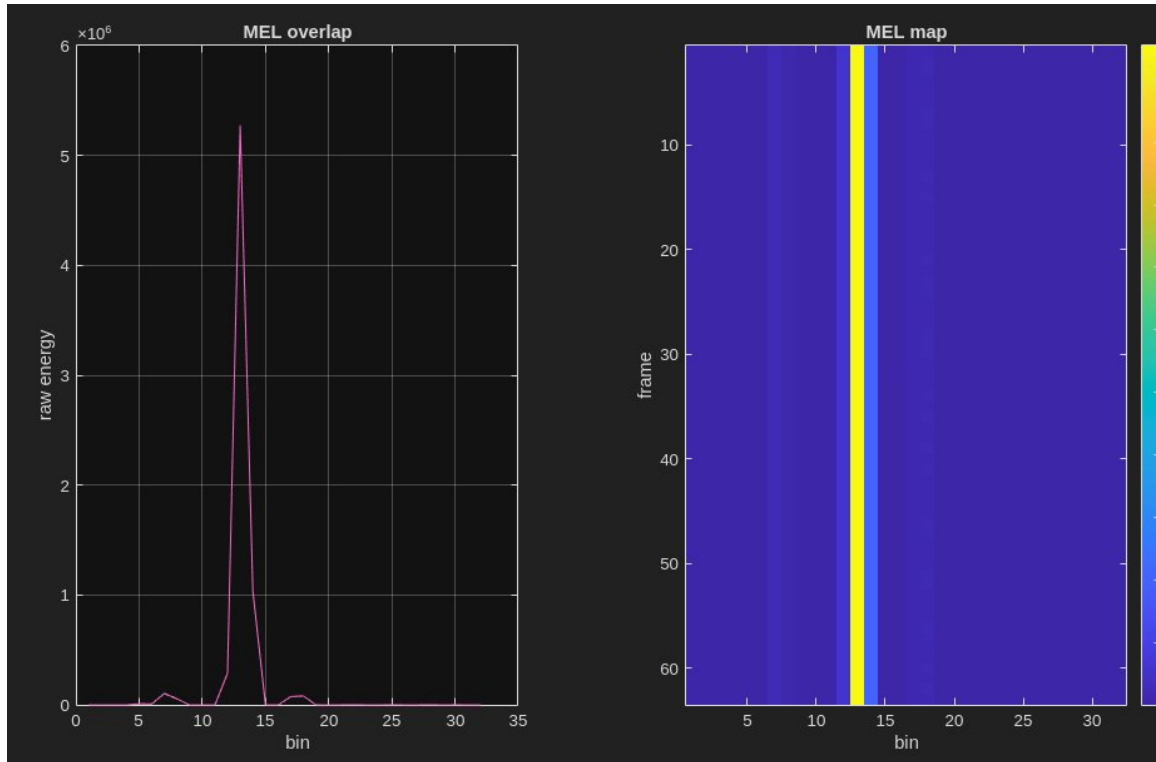


Figure 43. Programmed-board 1 kHz experiment: MEL filter-bank output.

UART capture from the programmed Basys 3, 3 June 2026.

MEL gathers the development-bitstream FFT values into overlapping perceptual bands. This historical application-level behavior is not used as evidence for the later corrected bin/address contract.

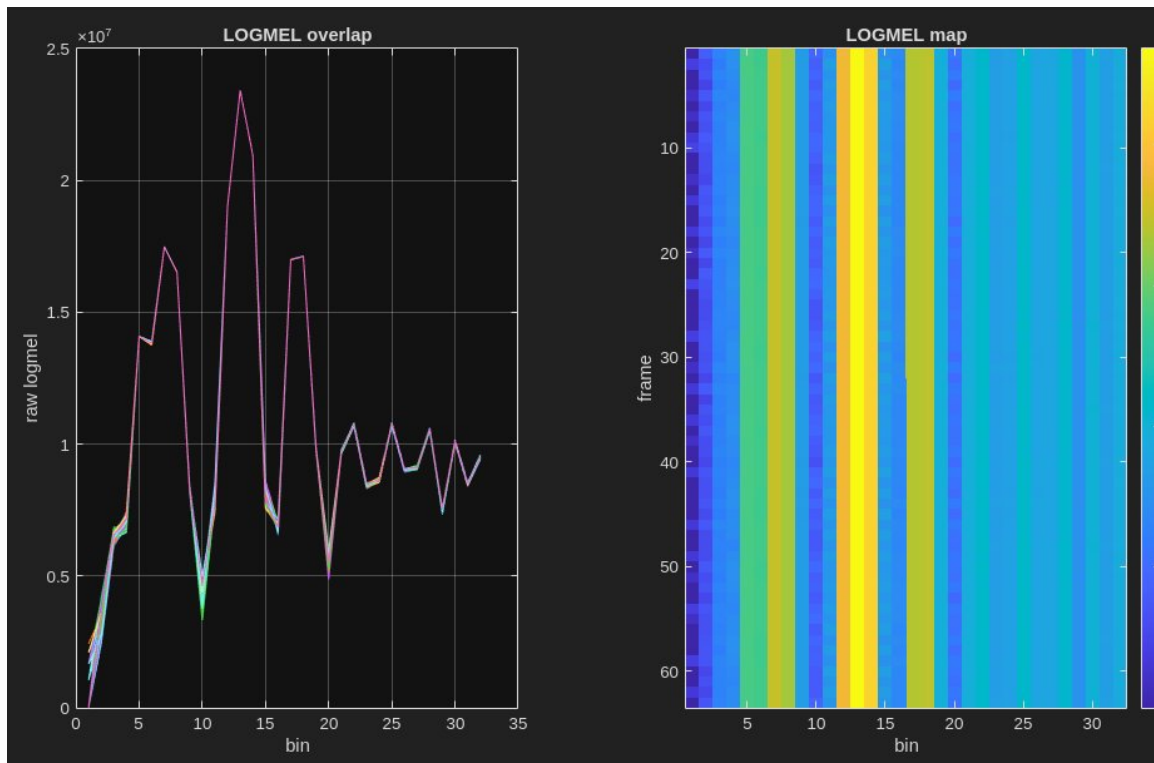


Figure 44. Programmed-board 1 kHz experiment: LOGMEL output.

UART capture from the programmed Basys 3, 3 June 2026.

The same band contour remains visible after logarithmic compression while the dynamic range is reduced. This is the intended preparation for DCT, not a separate classification result.

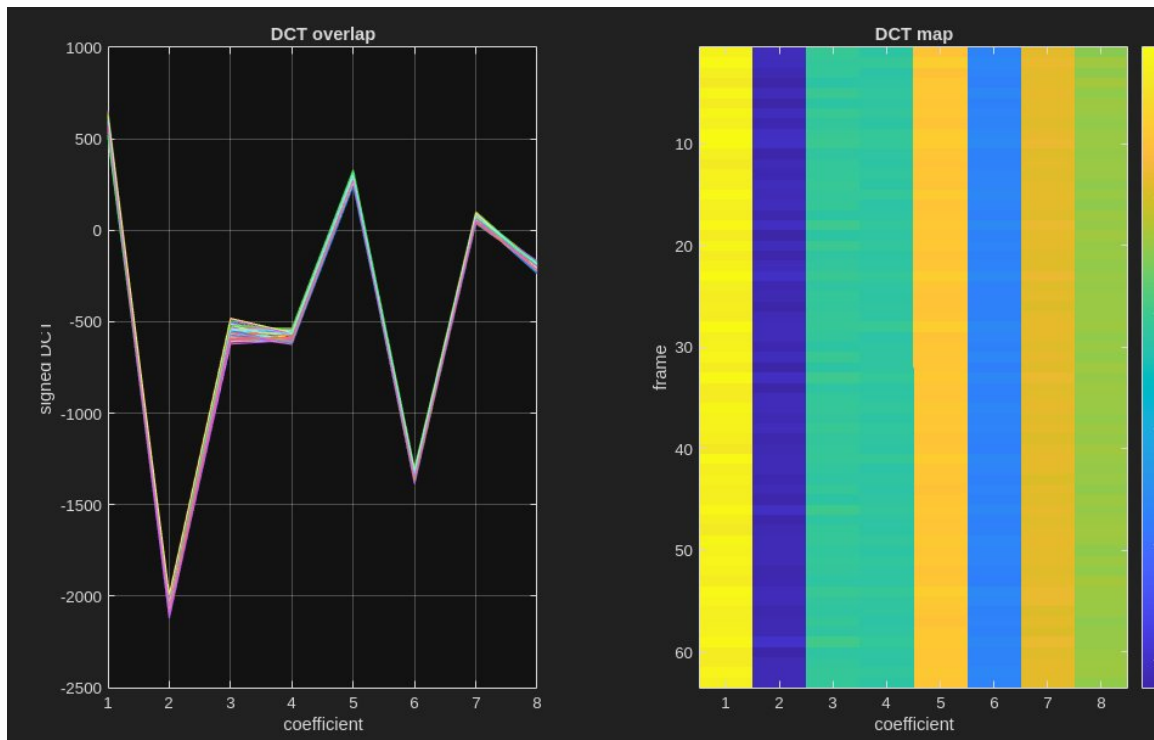


Figure 45. Programmed-board 1 kHz experiment: DCT C1-to-C8 output.

UART capture from the programmed Basys 3, 3 June 2026; a blank left margin was added for label readability without altering the captured pixels.

The final eight-coefficient pattern remains stable across the stationary capture. This shows that data progressed coherently through every fixed-point stage and reached the exact compact interface consumed by COMP.

16.1 Complete frame progression and serialized compare result

A retained 4 kHz capture contains 63 packets with frame bases 0, 256, 512, and so on through 15,872. Every packet is in compare mode. At the physical acquisition rate, 4 kHz corresponds to fractional FFT bin 123.924, whose nearest semantic bin is 124. The retained development bitstream stores the maximum at RAM address 123 for all 63 frames, consistent with its legacy one-address shift. The MATLAB plot labels MEL hardware index 31 as band 32. At the final frame COMP valid rises and digit code 3 is serialized. This checks execution of the comparison and result path, not recognition of a labeled spoken digit.

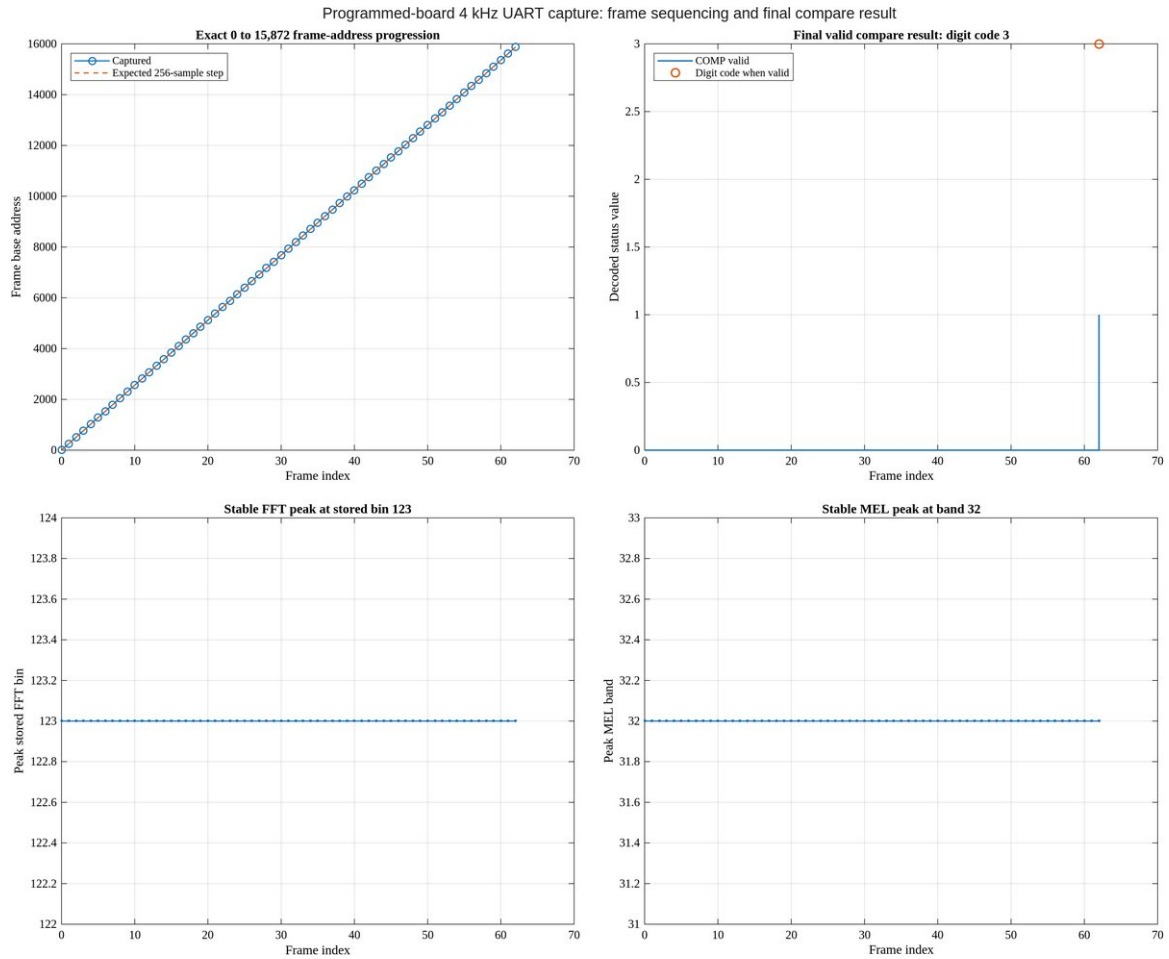


Figure 46. Retained 4 kHz UART run: exact frame addresses, stable spectral peaks, and the final valid COMP code.

Rendered from the saved programmed-board UART capture without altering its values.

The final packet asserts valid and carries digit code 3.

Table 13. Programmed-board tone evidence by frequency and retained stage.

Applied tone	Indexed stage views	Coverage note
100 Hz	ADC, Hann, WINDOW, FFT, MEL, LOGMEL, DCT	Complete seven-stage set
500 Hz	ADC, Hann, WINDOW, FFT, MEL, LOGMEL, DCT	Complete seven-stage set
1 kHz	ADC, Hann, WINDOW, FFT, MEL, LOGMEL, DCT	Complete seven-stage set shown above
2 kHz	ADC, Hann, WINDOW, FFT, MEL, LOGMEL	No retained DCT plot for this run
4 kHz	ADC, Hann, WINDOW, FFT, MEL, DCT	No retained LOGMEL plot for this run

Appendix B collects the remaining 26 retained tone plots. Across the sweep, the dominant FFT region moves upward as the applied frequency rises, and the MEL concentration moves into higher bands. These programmed-board captures complement the corrected bin-address regression with historical end-to-end observations.

17 Retained FPGA-Computed Spoken-Digit Outputs and Separability Results

The retained startup/test collection contains 30 labeled utterances acquired through the FPGA-UART workflow: digits 0 through 9 with three trials each [7]. These are not MATLAB-generated MFCC examples. For every recording, the programmed FPGA computed and exported the 16,384-sample ADC record and all 63 WINDOW, FFT, MEL, LOGMEL, and DCT frames; the plots in this section render those exact exported integer words. MATLAB was used to parse, store, and plot the UART words and, in Section 17.2, to apply the COMP-compatible leave-one-out distance analysis. The retained representations also supplied the 480 populated 128-bit words used to initialize the writable reference RAM before runtime speaker-specific enrollment.

17.1 One retained spoken-input path from samples to coefficients

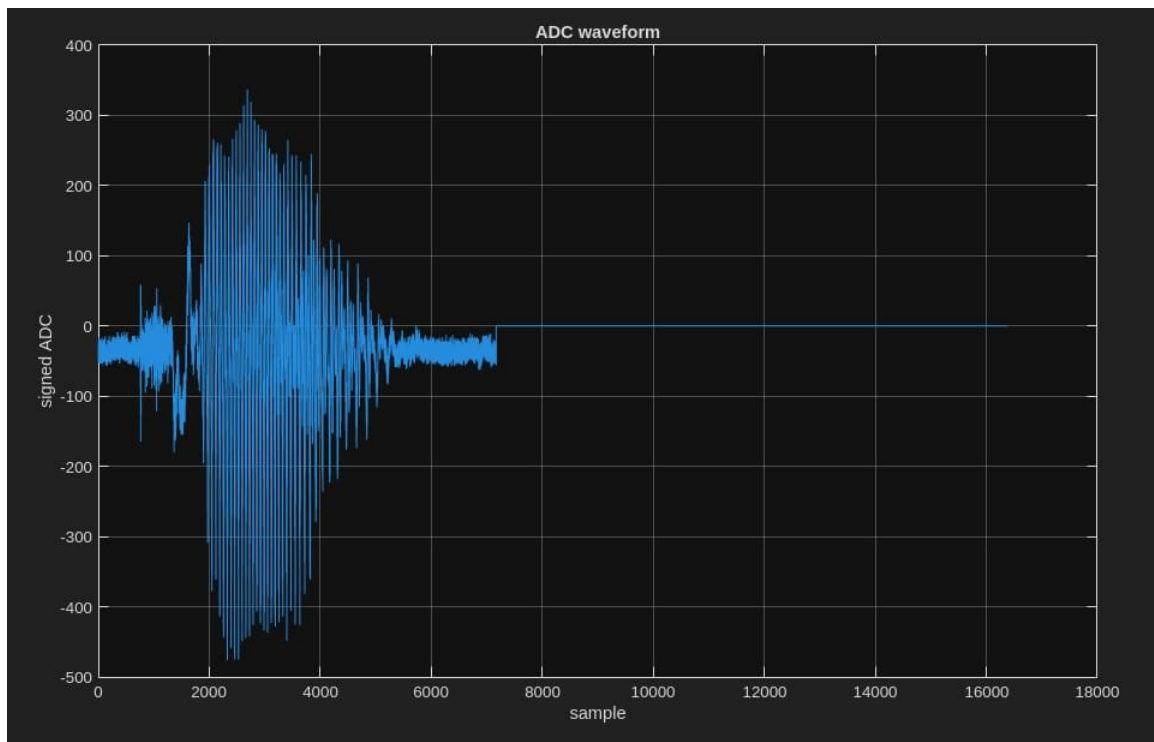


Figure 47. Retained spoken-input experiment: ADC waveform.

Distinct spoken-input figure retained in the project archive.

The recorded waveform has a time-localized speech envelope rather than the constant amplitude of a stationary tone. It is the physical signal described by the following maps.

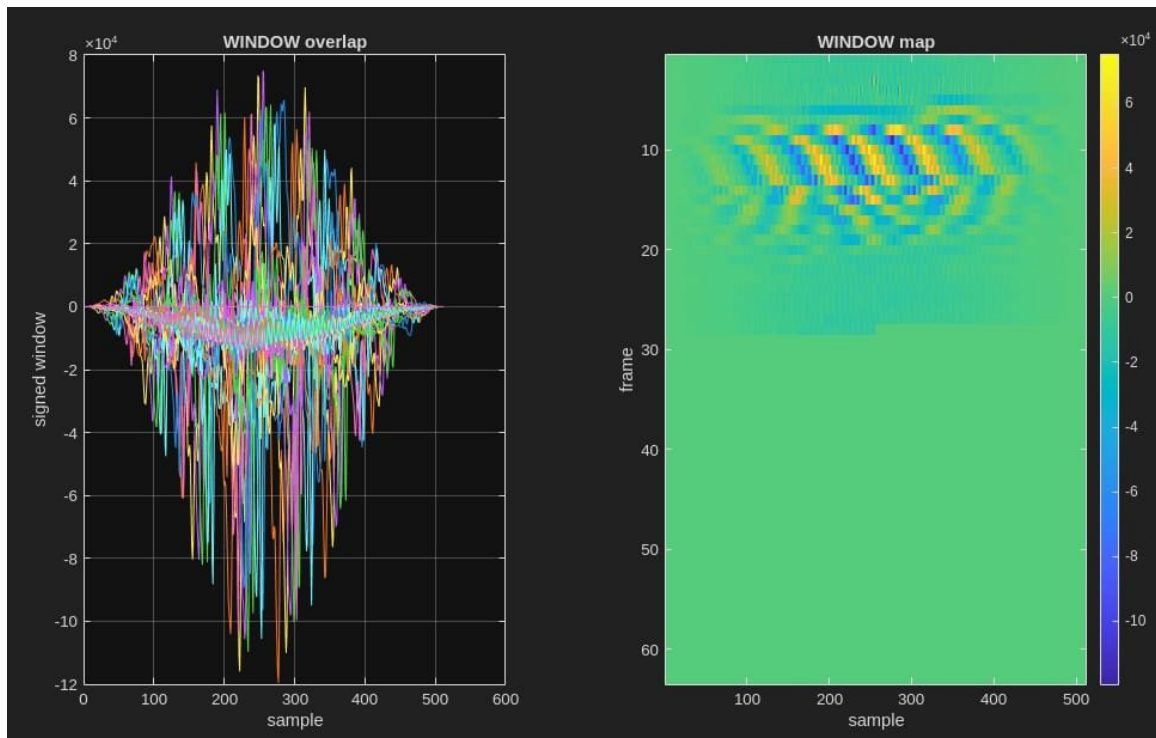


Figure 48. Retained spoken-input experiment: WINDOW frame sequence.

Distinct spoken-input figure retained in the project archive.

The same speech interval appears across overlapping tapered frames, preserving transitions that would otherwise fall on an arbitrary frame boundary.

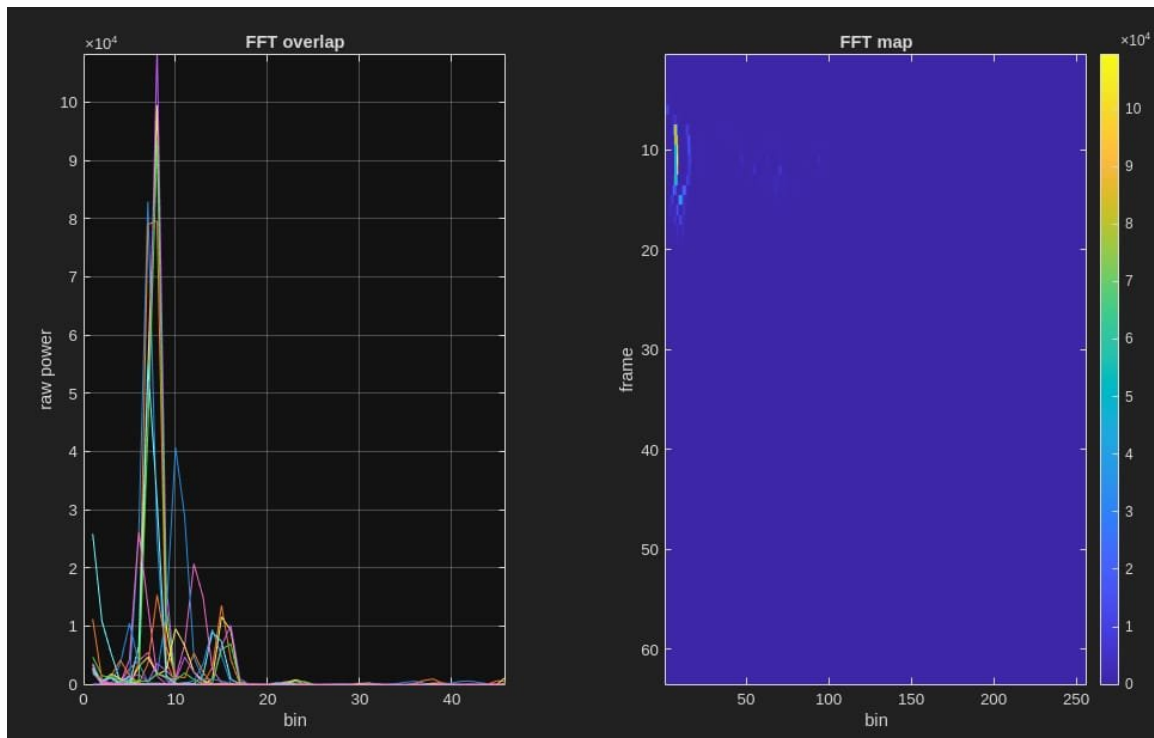


Figure 49. Retained spoken-input experiment: FFT frame map.

Distinct spoken-input figure retained in the project archive.

Time-varying spectral concentrations follow the utterance rather than staying at one tone bin. This is the frequency structure needed by the perceptual filter bank.

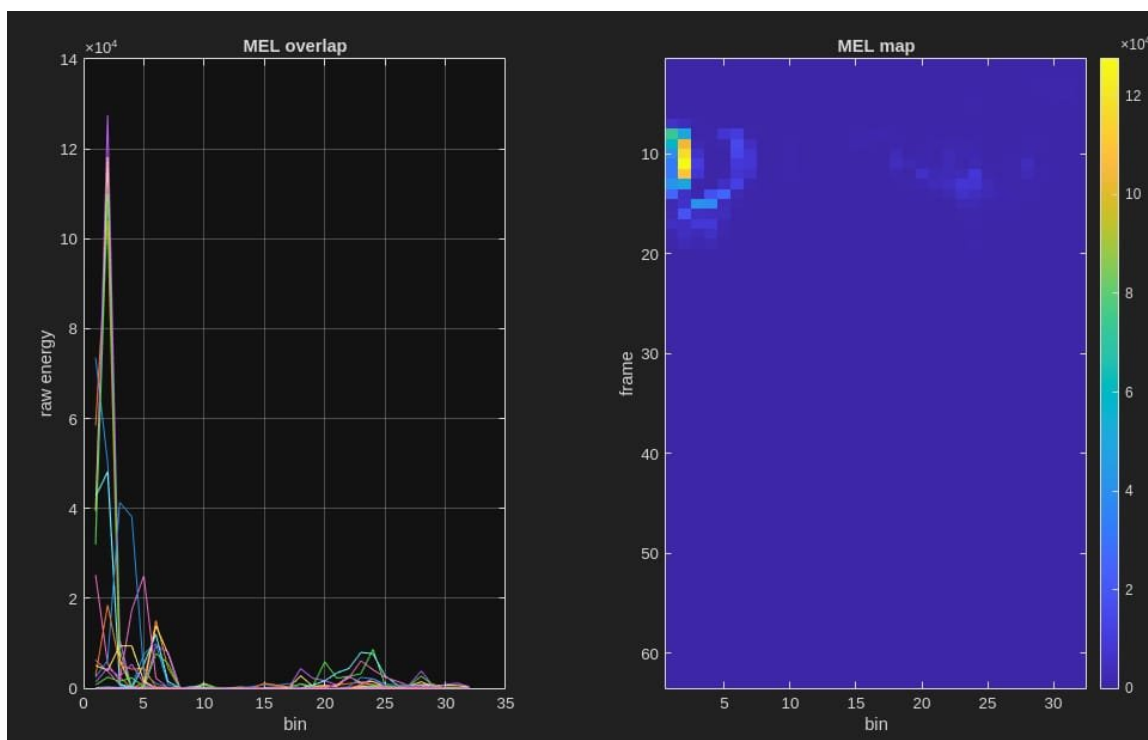


Figure 50. Retained spoken-input experiment: MEL frame map.

Distinct spoken-input figure retained in the project archive.

The 256-bin spectrum is reorganized into 32 broader bands while its time-varying contour remains visible.

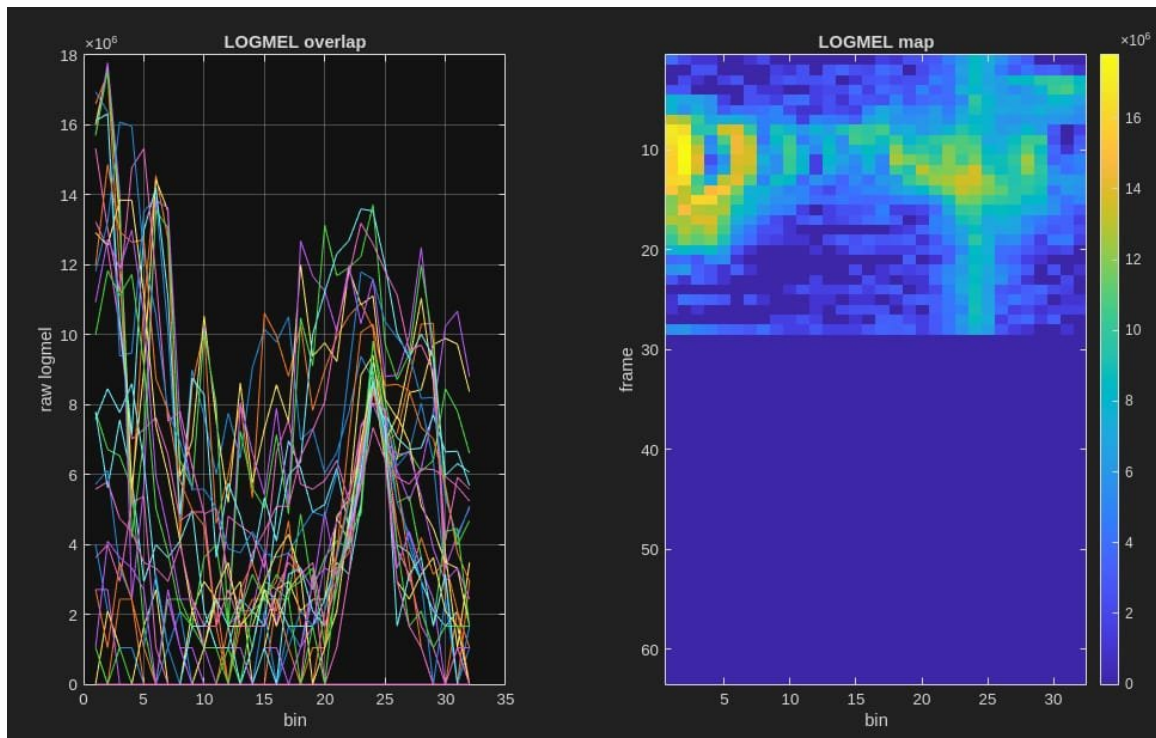


Figure 51. Retained spoken-input experiment: LOGMEL frame map.

Distinct spoken-input figure retained in the project archive.

Logarithmic compression reduces absolute dynamic range without erasing the band pattern.

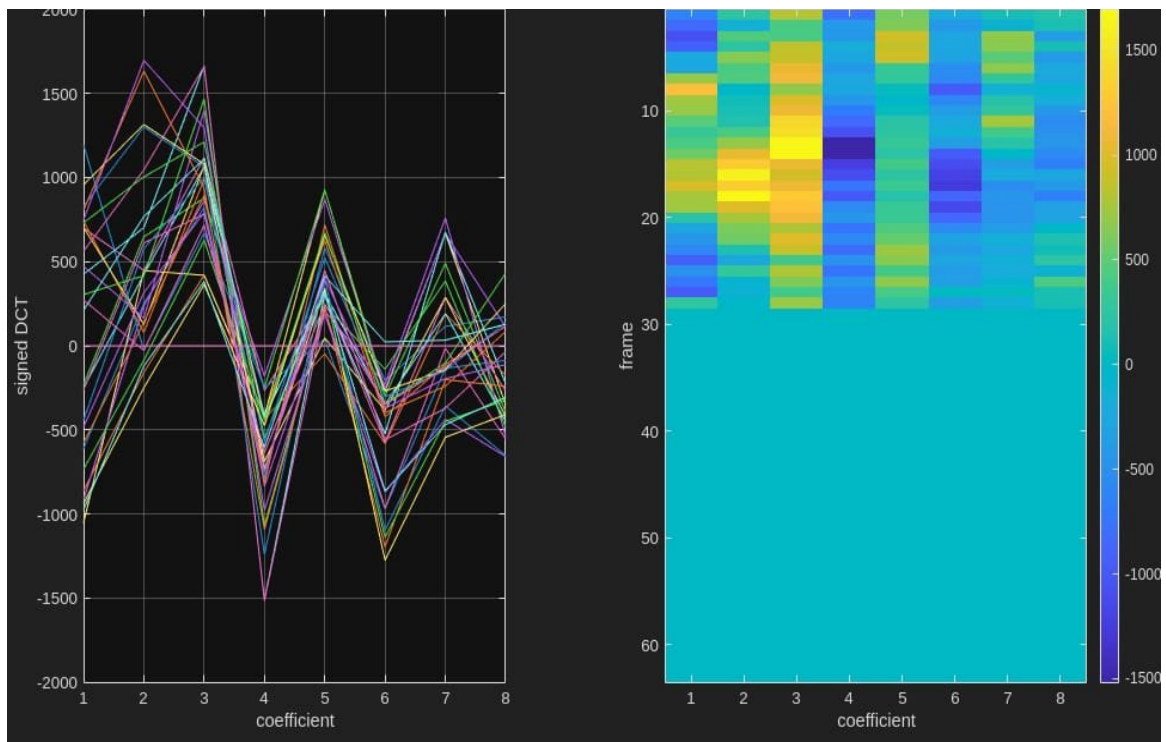


Figure 52. Retained spoken-input experiment: DCT frame map.

Distinct spoken-input figure retained in the project archive.

Eight signed non-DC coefficients retain a compact, changing signature that can be compared over the selected 16-frame interval.

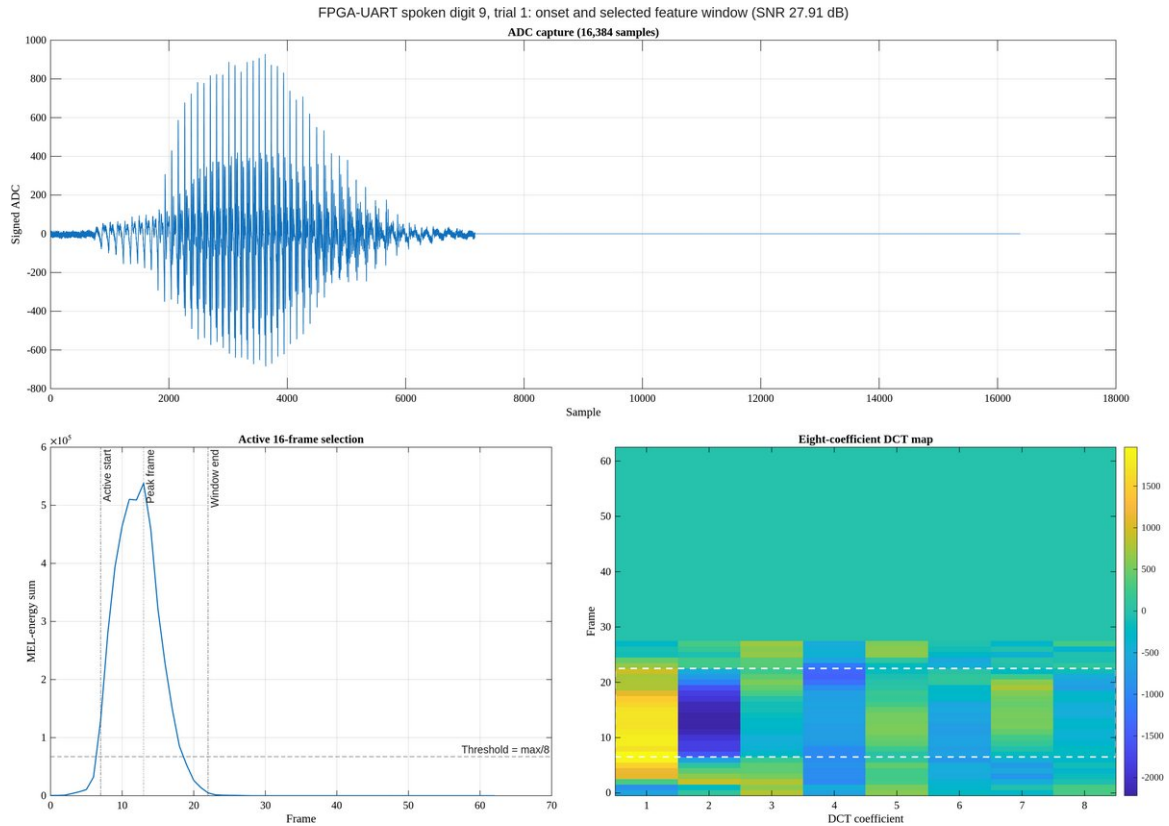


Figure 53. Saved digit-9 UART record: ADC envelope, MEL-energy onset, DCT map, and selected 16-frame window.

Rendered from the retained dataset record; all plotted arrays are saved FPGA-UART values.

The six preceding retained figures show the complete ADC-to-DCT representation. This focused view makes their timing relationship readable: the record has an estimated SNR of 27.91 dB, the first MEL-energy threshold crossing selects frame 7, peak MEL energy occurs at frame 13, and the dashed DCT-map rectangle marks the exact 16-frame feature window passed to SAD/L1 comparison. The saved record still contains 63 WINDOW, FFT, MEL, LOGMEL, and DCT frames; those counts are stated here rather than compressed into an unreadable plot annotation.

17.2 SAD/L1 decision and leave-one-out separation

Applying the implemented 16-frame by eight-coefficient SAD/L1 rule offline to the exact DCT words computed by the FPGA gave 30/30 leave-one-out decisions on this 30-record development set. Rebuilding the 480 populated reference words from the same retained FPGA values produced zero mismatches with the COE. The 30/30 figure is therefore an offline separability analysis of FPGA-computed features, not a retained log of 30 on-board COMP decisions. The focused COMP simulation in Section 12.5 separately exercises the RTL decision path. Because the recordings predate the FFT/MEL address fixes, this result is used to check the SAD/L1 rule and seed packing, not current live accuracy.

Table 14. Retained FPGA-UART startup/test collection under the COMP SAD/L1 distance convention.

Digit	Retained trials	Correct leave-one-out decisions
0	3	3 / 3
1	3	3 / 3
2	3	3 / 3
3	3	3 / 3
4	3	3 / 3
5	3	3 / 3
6	3	3 / 3
7	3	3 / 3
8	3	3 / 3
9	3	3 / 3
Total	30	30 / 30

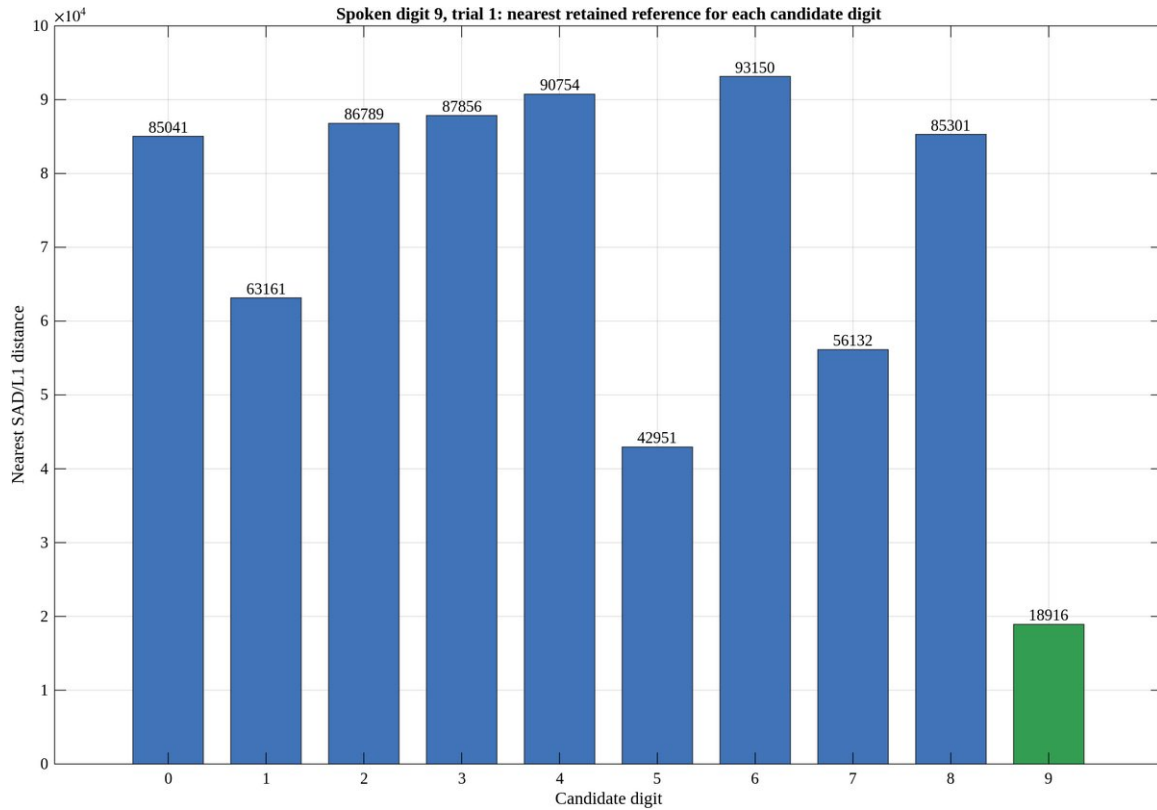


Figure 54. Digit 9, trial 1: nearest retained reference for each candidate digit.

Rendered from the retained C1-to-C8 feature records with the implemented 16-frame SAD/L1 metric.

The correct digit-9 distance is 18,916. The nearest wrong digit is 5 at 42,951, approximately 2.27 times farther away; the remaining digit-wise nearest distances range from 56,132 to 93,150. The plot therefore explains this individual decision rather than presenting the result as an unexplained label.

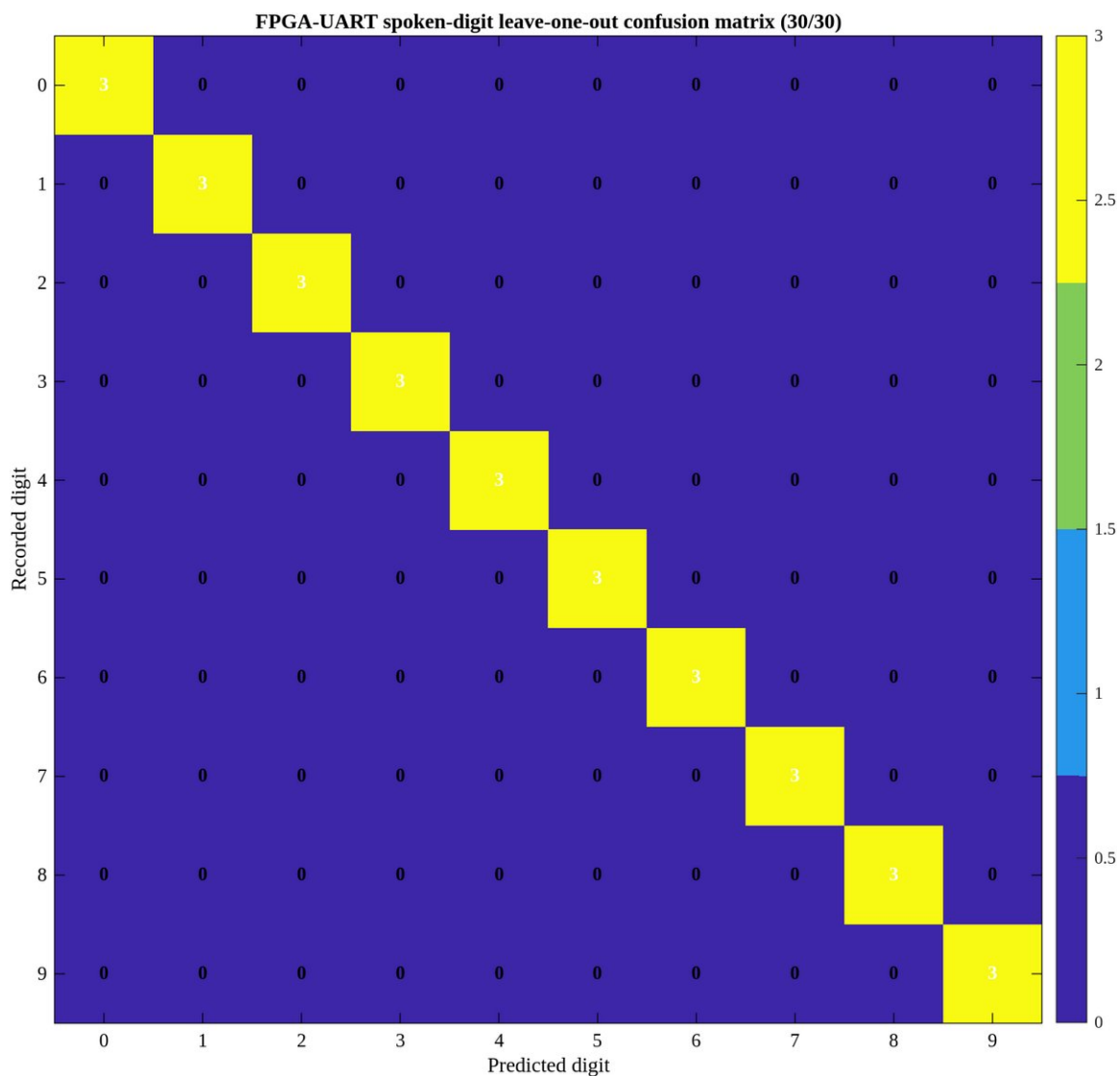


Figure 55. Leave-one-out confusion matrix for all 30 retained FPGA-UART utterances.

Rendered from three retained recordings per digit using the COMP-compatible SAD/L1 convention.

All 30 leave-one-out decisions lie on the diagonal for this development set.

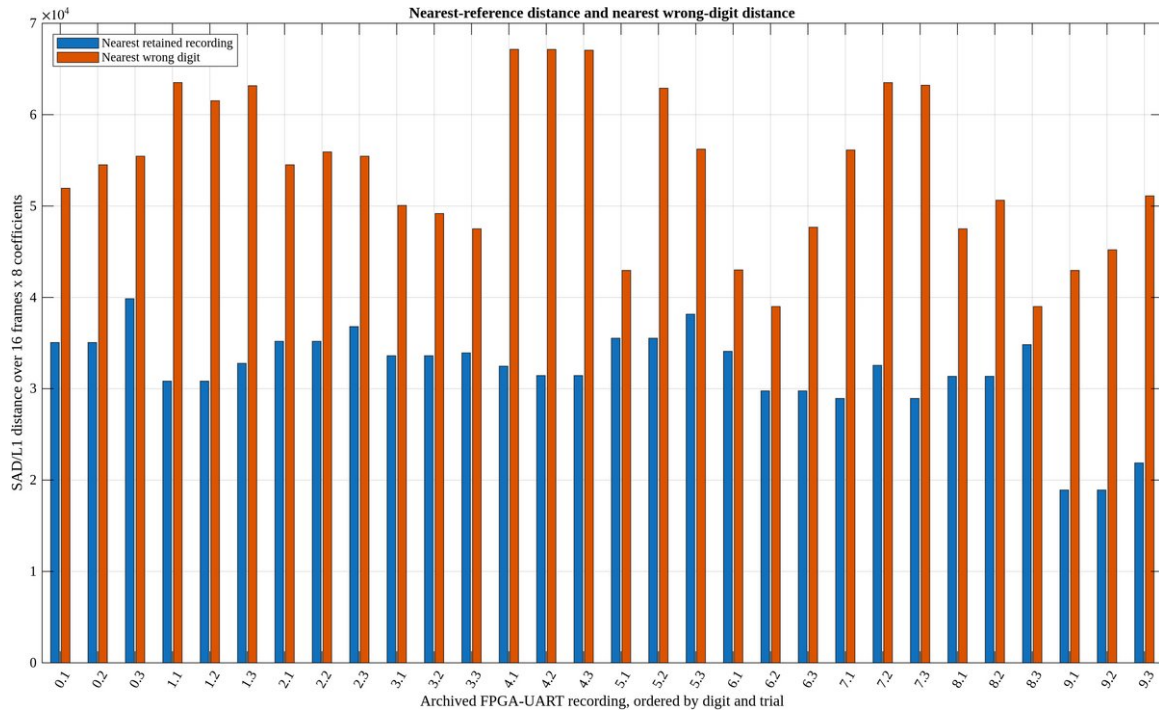


Figure 56. Nearest retained-recording distance versus nearest wrong-digit distance for all 30 queries.

Rendered from the retained FPGA-UART dataset using leave-one-out SAD/L1 comparison.

For every query, the nearest retained recording is closer than the nearest example carrying a different digit. Showing both distances prevents the perfect confusion matrix from becoming a context-free score: it exposes the separation margin for every digit/trial and identifies recordings that are relatively easier or harder under this small dataset.

18 Vivado Implementation and Clock Engineering

The current RTL was synthesized, placed, routed, and converted into a Basys 3 bitstream on 27 July 2026 [6]. It uses the 100 MHz board clock as a single processing domain; the ADC's 33.333 MHz transaction enable is derived inside that domain rather than generated as a second clock. The 3 June programmed-board captures belong to an earlier development bitstream and are used only as historical evidence of acquisition and memory population. Current post-correction simulation, current implementation reports, and historical board captures are identified separately.

18.1 The 36 MHz attempt and the final divide-by-three decision

During development I attempted a generated ADC-controller clock near the original 36 MHz objective. Behavioral simulation alone did not expose the practical difficulty. The generated clock placed control and memory activity across a more sensitive boundary, implementation iterations were unstable, and board behavior could differ from the simulated receiver even during a diagnostic in which the serial input was forced low. I added synchronizer/clock-path constraints, I/O registers, KEEP and DONT_TOUCH attributes, and a localized ADC placement region while investigating the problem.

I ultimately judged the exact nominal tick to be the wrong priority. Advancing the receiver every three 100 MHz clocks produces 33.333 MHz, approximately 7.4% below 36 MHz, while eliminating the separate clock-domain dependency. For a speech recognizer, that uniform difference is safer than retaining an inconsistent acquisition implementation. Only the retained final 100 MHz implementation results are reported. The current routed internal timing and the historical programmed-board behavior are separate results.

Table 15. Final timing result and its relationship to the ADC clock decision.

Timing item	27 July routed value	Interpretation
Primary clock	100.000 MHz	One synchronous processing domain
WNS / TNS	+0.630 ns / 0.000 ns	Positive setup margin and no setup-failing endpoints
WHS / THS	+0.014 ns / 0.000 ns	Positive hold margin and no hold-failing endpoints
Routing	12,825 / 12,825 routable nets; 0 errors	The complete design is fully routed
External I/O delays	Not constrained	No board-level SPI or other interface timing-closure claim
ADC controller enable	33.333 MHz	Single-domain divide-by-three transaction cadence
ADC SCLK	Approximately 16.667 MHz	Clock toggles over alternating receive/decode phases

18.2 Resource use in relation to the architecture

Table 16. Placed resource utilization of the integrated recognizer.

Resource	Used	Available	Utilization
Slice LUTs	5,604	20,800	26.94%
Slice registers	8,437	41,600	20.28%
Block RAM tiles	23.5	50	47.00%
DSPs	15	90	16.67%
Bonded I/O	39	106	36.79%
BUFGCTRL	1	32	3.13%

BRAM is the largest proportional use because the design stores every intermediate stage and 30 reference templates. Sequential MEL/DCT arithmetic and multiplier-free SAD keep DSP use to 15.

All constrained internal 100 MHz paths meet setup and hold timing. External SPI and board-interface input/output delays have not been constrained or characterized, so this is not a board-level interface timing-closure claim.

19 Analog Front End and PCB Development

This began as my first PCB layout and fabrication project. I designed the microphone bias, gain, and low-pass stages, laid out a single-sided through-hole board, and fabricated it with a photoresist process. Sections 19.1 through 19.4 retain that first-board history. Section 19.5 onward documents the later two-layer home-fabricated revision that integrates the ADC and direct Basys 3 connection.

19.1 Assignment constraints and fourth-order filter synthesis

The assigned front end had to do more than amplify a microphone. It had to use one supply, remain compatible with a 3.3 V ADC input, keep the 100 Hz-to-3 kHz band within one decibel, place the low-pass corner between 4 and 4.5 kHz, approach an 80 dB-per-decade high-frequency slope, provide adjustable gain, and fit on a single PCB smaller than 25 cm². I treated those limits as linked design constraints rather than choosing a schematic first and justifying it afterward.

Table 17. Assigned analog/PCB constraints and the evidence used to evaluate them.

Assigned constraint	Design response	Retained evidence
100 Hz-3 kHz within -1 to +1 dB	Two unequal-Q Sallen-Key sections preserve the speech-band shape	SPICE change is 0.395319 dB from 100 Hz to 3 kHz
Corner between 4 and 4.5 kHz	E12 search targets nearly equal section natural frequencies	Isolated cascade is -3.0008225 dB at 4.053 kHz
80 dB/decade stopband slope	Two second-order sections form a fourth-order response	SPICE falls 80.006246 dB from 10 to 100 kHz
Single supply and 3.3 V ADC interface	Buffered reference biases the microphone and filter path	Recalled bring-up reading: 1.632 V at a 9.16 V supply; no meter photograph was retained
Adjustable gain	10 kOhm potentiometer is included in the non-inverting feedback network	Live ADC-code UART/MATLAB tuning; zero exact rail hits in the later 30-utterance set
PCB area no greater than 25 cm ²	44 mm x 40 mm, single-sided, through-hole layout	17.6 cm ² artwork, 70.4% of the permitted area

A fourth-order Butterworth response requires two different second-order pole pairs. The target section Q values are approximately 0.5412 and 1.3066; using two identical $Q = 0.707$ sections would not recreate those poles. At the common natural frequency, identical 0.707 sections multiply to an amplitude of 0.5, or -6.02 dB. The selected unequal sections multiply to approximately 0.7058, or -3.03 dB, which is the intended fourth-order corner behavior.

selected $Q_1 \times Q_2 = 0.541307 \times 1.303840 = 0.70578 \rightarrow 20 \log_{10}(0.70578) = -3.03 \text{ dB}$

Table 16–5. Butterworth Coefficients

n	i	a_i	b_i	k_i = f_{ci} / f_c	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.4142	1.0000	1.000	0.71
3	1	1.0000	0.0000	1.000	—
	2	1.0000	1.0000	1.272	1.00
4	1	1.8478	1.0000	0.719	0.54
	2	0.7654	1.0000	1.390	1.31
5	1	1.0000	0.0000	1.000	—
	2	1.6180	1.0000	0.859	0.62
	3	0.6180	1.0000	1.448	1.62
6	1	1.9319	1.0000	0.676	0.52
	2	1.4142	1.0000	1.000	0.71
	3	0.5176	1.0000	1.479	1.93
7	1	1.0000	0.0000	1.000	—
	2	1.8019	1.0000	0.745	0.55
	3	1.2470	1.0000	1.117	0.80
	4	0.4450	1.0000	1.499	2.25

Figure 57. Fourth-order Butterworth section targets used before the E12 search.

Retained design-reference table from the earlier report.

The fourth-order row supplies two unequal section targets rather than one repeated damping value. I used those targets to evaluate realizable resistor/capacitor combinations. This is why the two physical filter sections intentionally have different component ratios even though their natural frequencies are nearly equal.

The earlier report preserved the ranked search output as vector text, so it was missed when only embedded pictures were extracted. The complete ranking is restored below. Candidate rows were restricted to less than one percent damping-factor error and sorted primarily by the separation between the two realized natural frequencies. Resistance values are shown in kOhm and capacitances in nF.

Table 18. Ranked E12 component candidates recovered from the earlier report.

Ran k	Delta fc (kHz)	f1 / f2 (kHz)	Damping error S1/S2 (%)	Section 1 R; C	Section 2 R; C
1	0.0190	4.050 / 4.069	0.0 / 0.2	12, 10; 3.9, 3.3	15, 15; 6.8, 1.0
2	0.0190	4.050 / 4.069	0.8 / 0.2	39, 22; 1.5, 1.2	15, 15; 6.8, 1.0
3	0.0190	4.050 / 4.069	0.8 / 0.2	39, 33; 1.2, 1.0	15, 15; 6.8, 1.0
4	0.0557	4.125 / 4.069	0.7 / 0.2	47, 12; 2.2, 1.2	15, 15; 6.8, 1.0
5	0.0643	4.005 / 4.069	0.1 / 0.2	39, 10; 2.7, 1.5	15, 15; 6.8, 1.0
6	0.0736	4.154 / 4.228	0.4 / 0.0	82, 68; 5.6, 4.7	12, 12; 8.2, 1.2
7	0.0856	4.154 / 4.069	0.4 / 0.2	82, 68; 5.6, 4.7	15, 15; 6.8, 1.0
8	0.1035	4.125 / 4.228	0.7 / 0.0	47, 12; 2.2, 1.2	12, 12; 8.2, 1.2
9	0.1254	4.353 / 4.228	0.1 / 0.0	15, 10; 3.3, 2.7	12, 12; 8.2, 1.2
10	0.1655	4.394 / 4.228	0.5 / 0.0	27, 10; 2.7, 1.8	12, 12; 8.2, 1.2
11	0.1655	4.394 / 4.228	0.8 / 0.0	27, 18; 1.8, 1.5	12, 12; 8.2, 1.2
12	0.1746	4.403 / 4.228	0.3 / 0.0	22, 15; 2.2, 1.8	12, 12; 8.2, 1.2
13	0.1782	4.050 / 4.228	0.0 / 0.0	12, 10; 3.9, 3.3	12, 12; 8.2, 1.2
14	0.1782	4.050 / 4.228	0.8 / 0.0	39, 22; 1.5, 1.2	12, 12; 8.2, 1.2
15	0.1782	4.050 / 4.228	0.8 / 0.0	39, 33; 1.2, 1.0	12, 12; 8.2, 1.2

I selected the first minimum-separation row: 12 kOhm, 10 kOhm, 3.9 nF, and 3.3 nF for section 1; 15 kOhm, 15 kOhm, 6.8 nF, and 1.0 nF for section 2. The calculated natural frequencies are 4049.862 and 4068.876 Hz, only 19.014 Hz apart. The realized Q values are 0.541307 and 1.303840, with damping-factor errors of approximately 0.0204% and 0.2088% against the

Butterworth targets. Two other rows tie the frequency separation, but their first-section damping error rounds to 0.8%; the retained records do not state whether stock availability or layout convenience also influenced the final tie-break.

19.2 SPICE verification of passband, cutoff, and roll-off

I first simulated the filter in isolation so that microphone and gain-stage variables could not hide its pole behavior. I then simulated the complete front-end path. The cursor readings below are evidence from design models, not measurements from a calibrated network analyzer, and the absolute dBV level in the complete-path plot depends on its microphone current-source stimulus. I therefore use absolute readings only to calculate differences within the same trace.

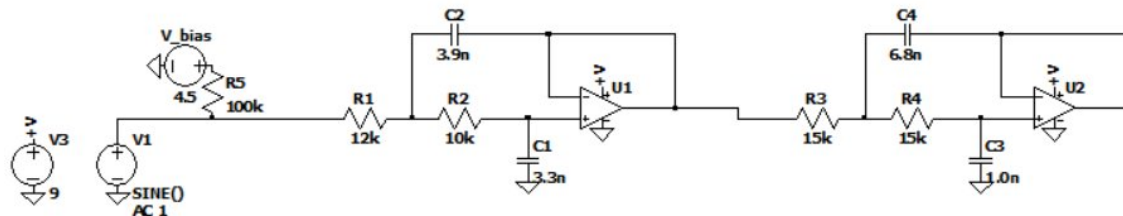


Figure 58. Isolated two-section unity-gain Sallen-Key SPICE bench.

Retained SPICE design evidence from the earlier report.

This bench contains the two filter sections, source, and 4.5 V bias; it does not contain the complete microphone amplifier. Isolating the cascade lets the two pole pairs be checked directly. The bias establishes a valid single-supply operating point but does not set the small-signal pole frequencies.

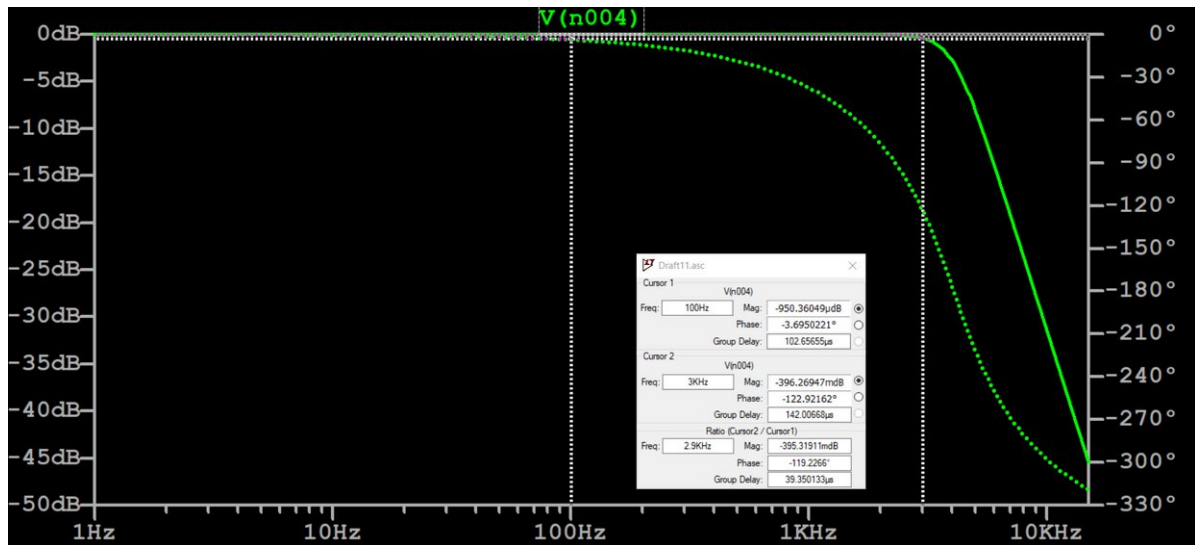


Figure 59. SPICE passband-flatness check from 100 Hz to 3 kHz.

Retained SPICE design evidence from the earlier report.

The cursor reads -0.000950 dB at 100 Hz and -0.396269 dB at 3 kHz. Their difference is -0.395319 dB, comfortably inside the assigned -1 to +1 dB band. This matters to MFCC recognition because the analog circuit should not impose a large artificial tilt across the speech spectrum before the MEL filters analyze it.

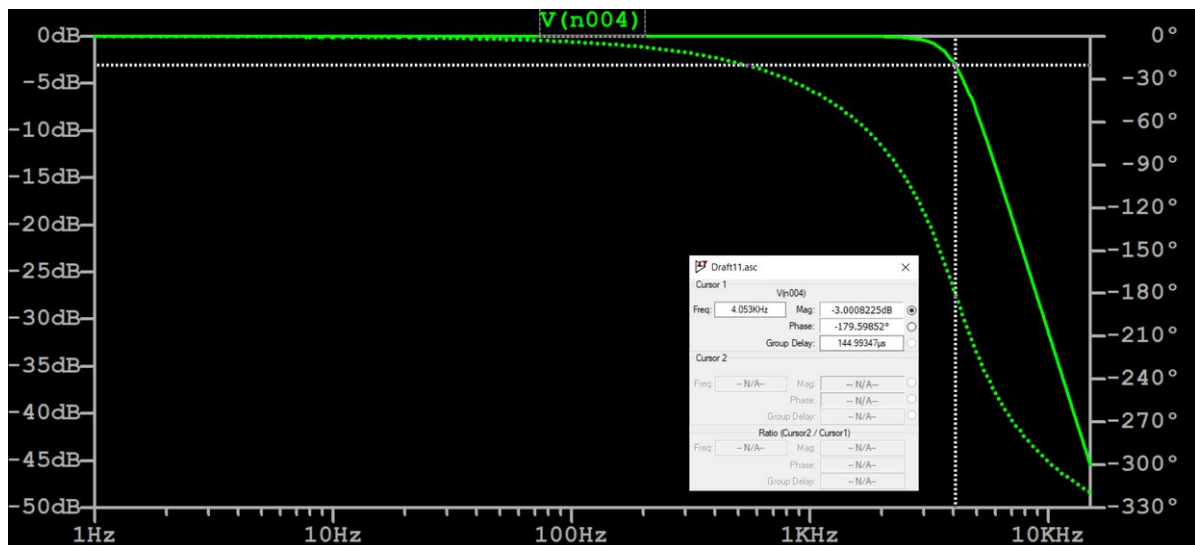


Figure 60. SPICE -3 dB corner check at 4.053 kHz.

Retained SPICE design evidence from the earlier report.

The cursor reads -3.0008225 dB at 4.053 kHz. The simulated -3 dB point is within the required 4 to 4.5 kHz range.

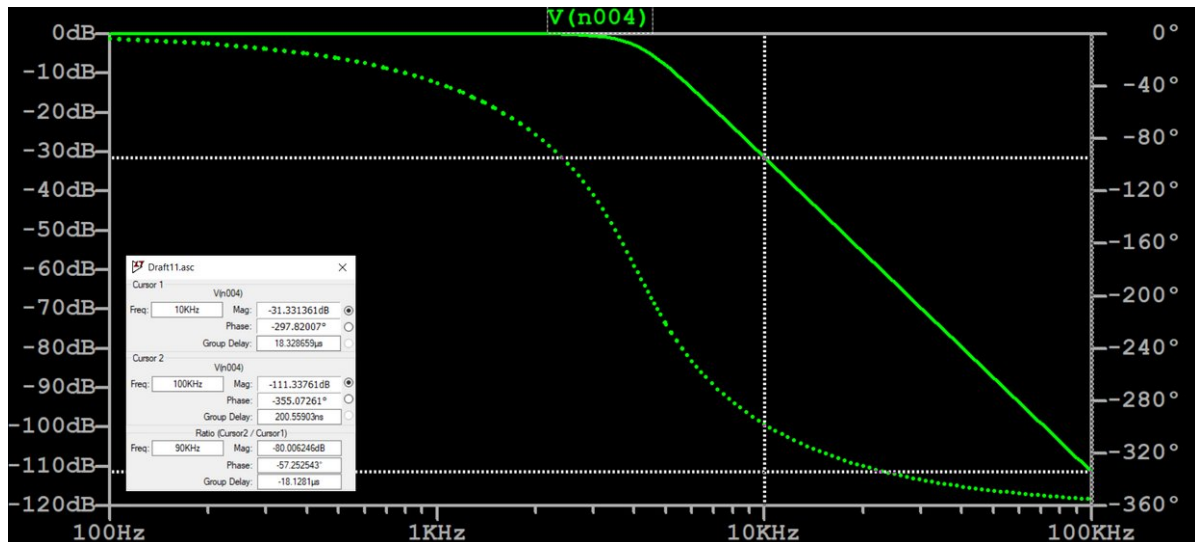


Figure 61. SPICE fourth-order roll-off check from 10 to 100 kHz.

Retained SPICE design evidence from the earlier report.

The response is -31.331361 dB at 10 kHz and -111.33761 dB at 100 kHz. The one-decade change is -80.006246 dB, demonstrating that both second-order sections contribute to the expected fourth-order asymptote. The purpose is anti-alias attenuation above the speech band, not precision analyzer calibration.

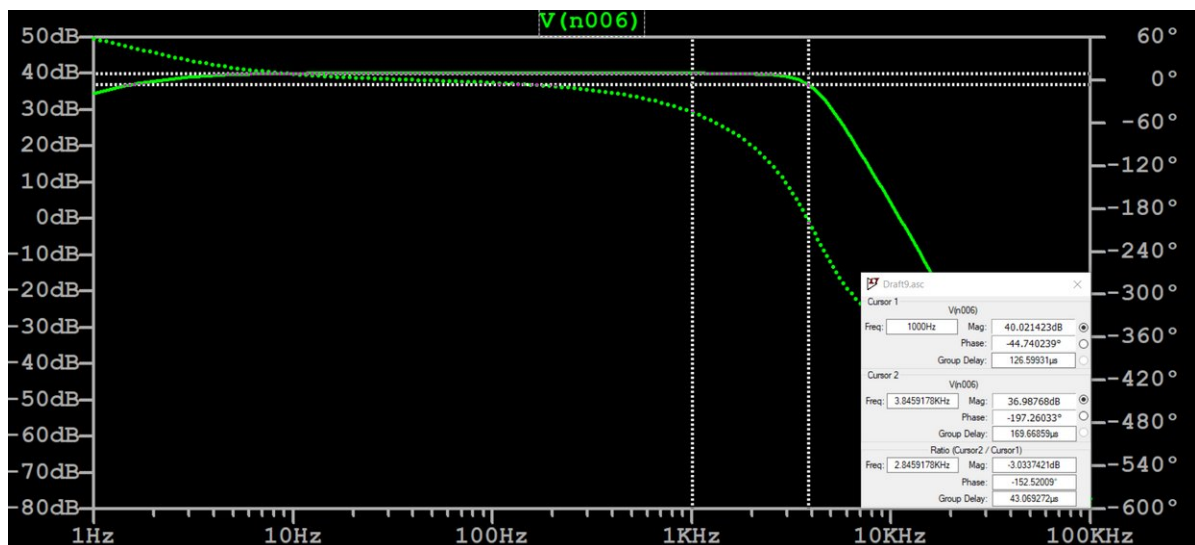


Figure 62. Complete microphone/front-end AC-response simulation.

Retained SPICE design evidence from the earlier report.

The same trace reads 40.021423 dBV at 1 kHz and 36.987680 dBV at 3.8459178 kHz, a relative change of -3.033742 dB. Because the retained microphone model is excited by an AC current source, 40.021 dBV is not safely interpreted as '40 dB amplifier gain.' I therefore use this plot only for its response shape and approximately 3.846 kHz relative corner.

19.3 Original schematic and first single-sided PCB layout

The final circuit and layout evidence answer different questions from SPICE. The schematic records the implemented bias, coupling, gain, and filter network; the layout records whether that circuit was converted into a fabricable through-hole board under a tight single-sided process. Figure 63 shows the final component values. The recalled DC readings guided bring-up but do not constitute a calibrated transfer-function sweep.

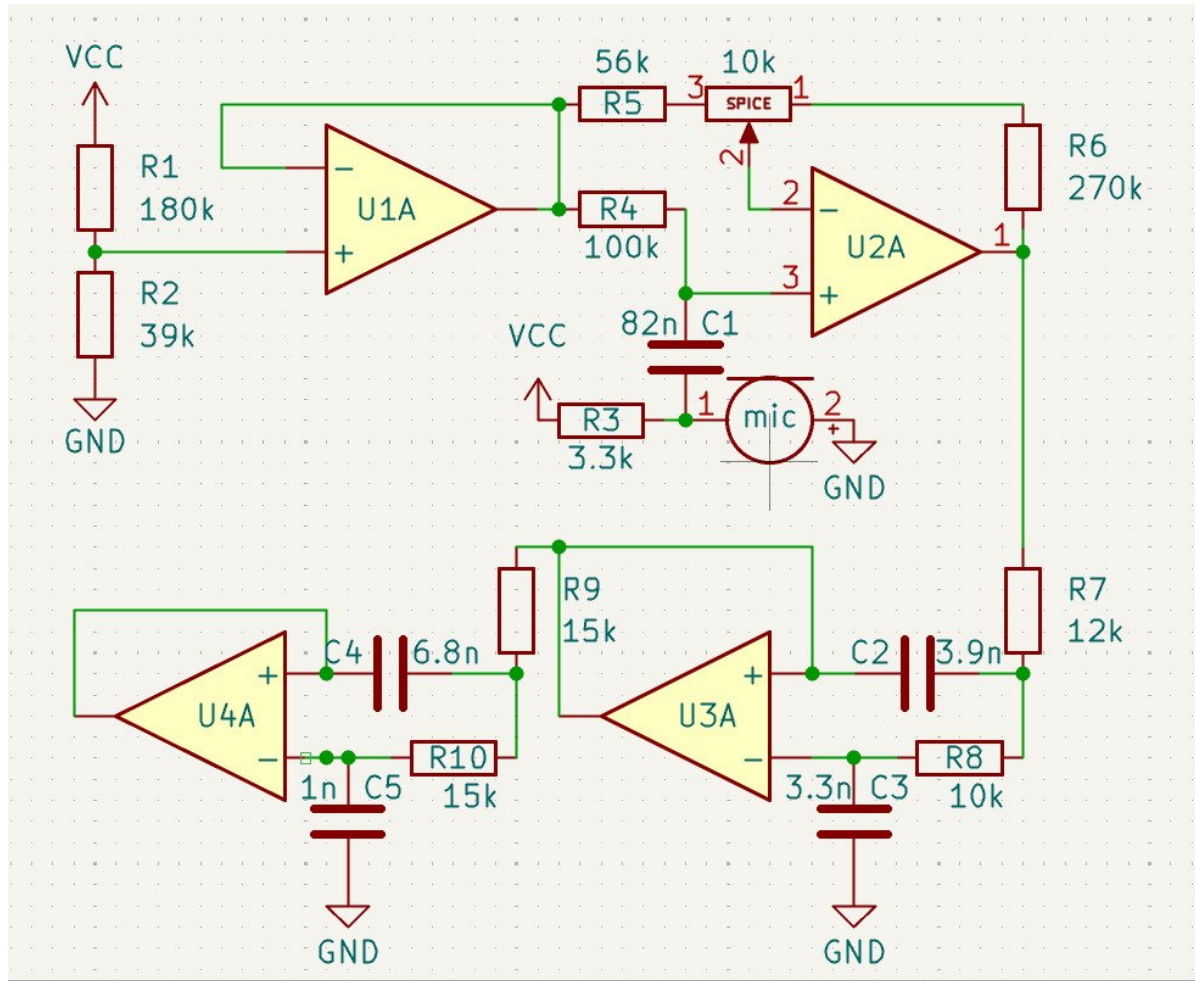


Figure 63. Original analog-front-end schematic with the implemented component values.

Confirmed final implemented schematic retained from the earlier report and project design files.

One LM324 package performs four roles: reference buffer, microphone amplifier, first Sallen-Key buffer, and second Sallen-Key buffer. I selected the final 180 kOhm/39 kOhm divider on a breadboard under the real supply instead of assuming an exact 9 V source. At the recalled 9.16 V supply, the nominal divider predicts 1.631 V, consistent with the recalled approximately 1.632 V DC operating point. That is only 18 mV, or about 0.55% of the 3.3 V full scale, below the ideal 1.65 V midpoint. The 82 nF coupling capacitor with its 100 kOhm return gives an approximate 19.4 Hz high-pass corner, and the 10 kOhm potentiometer provides an ideal gain range of roughly 5.09 to 6.0 V/V.

I set the 10 kOhm potentiometer from the sampled signal rather than from a nominal gain target. A temporary FPGA instrument read the ADC over SPI and streamed the centered samples over UART to a MATLAB live display. I increased the gain until spoken digits occupied a useful fraction of the ADC range, then backed it off until none of the tested utterances clipped at the 3.3 V limits. Under the same room conditions, ambient noise and background music remained below the spoken-digit excursions rather than dominating the capture, while the digits retained visibly different temporal information.

The retained diagnostic source corroborates this workflow: it averages 32 twelve-bit conversions, subtracts 0x800, and sends signed 16-bit samples at 921,600 baud. Its MATLAB wrapper shows a 0.600 s trace at 17,847.5 samples/s, summary statistics, and a live FFT. That 17.8475 kSa/s rate belongs to the temporary diagnostic design and is separate from the recognizer's approximately 16.526 kSa/s acquisition cadence. This exposed the actual FPGA input codes without repeated oscilloscope trigger, time-base, and range adjustments. Across the later 30-utterance dataset, none of 491,520 samples reaches a rail; the range is -1,888 to +2,009 versus -2,048/+2,047 limits.

A fresh repeat trial on 12 July 2026 reproduced the temporary-instrument workflow and supplied visible evidence that had not survived from the original bring-up [8]. The paired captures show one live utterance: the first records time-domain headroom and envelope, and the second shows the spectrum computed by the MATLAB monitor from those ADC samples.

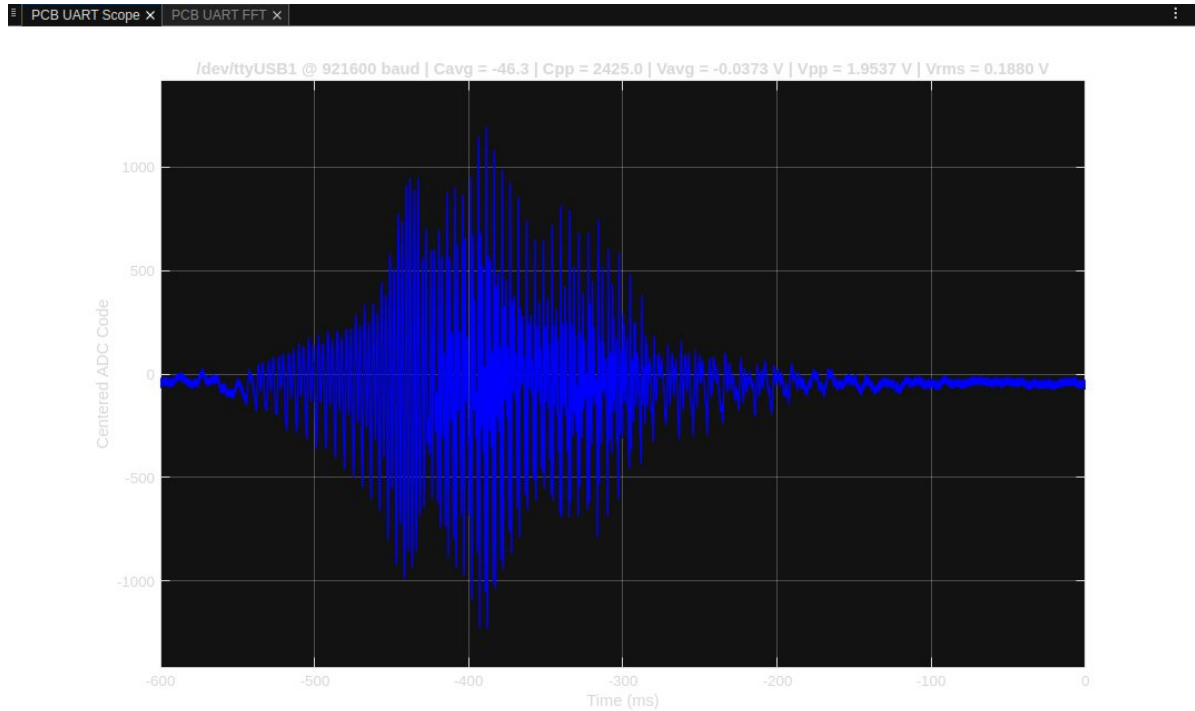


Figure 64. Fresh live ADC-to-UART spoken-utterance waveform used to verify gain and headroom.

MATLAB live capture from the retained temporary Basys 3 ADC-to-UART instrument, 12 July 2026.

The 600 ms record contains a quiet lead-in, rising speech onset, high-energy voiced interval, and decay rather than a stationary laboratory tone. The monitor reports a mean of -46.3 centered codes (-0.0373 V), 2,425 codes peak-to-peak (1.9537 Vpp), and 0.1880 Vrms. The trace remains far from the nominal -2,048/+2,047 rails while using a substantial portion of the converter range. This is the direct visual justification for the potentiometer setting: useful speech excursion is preserved without clipping.

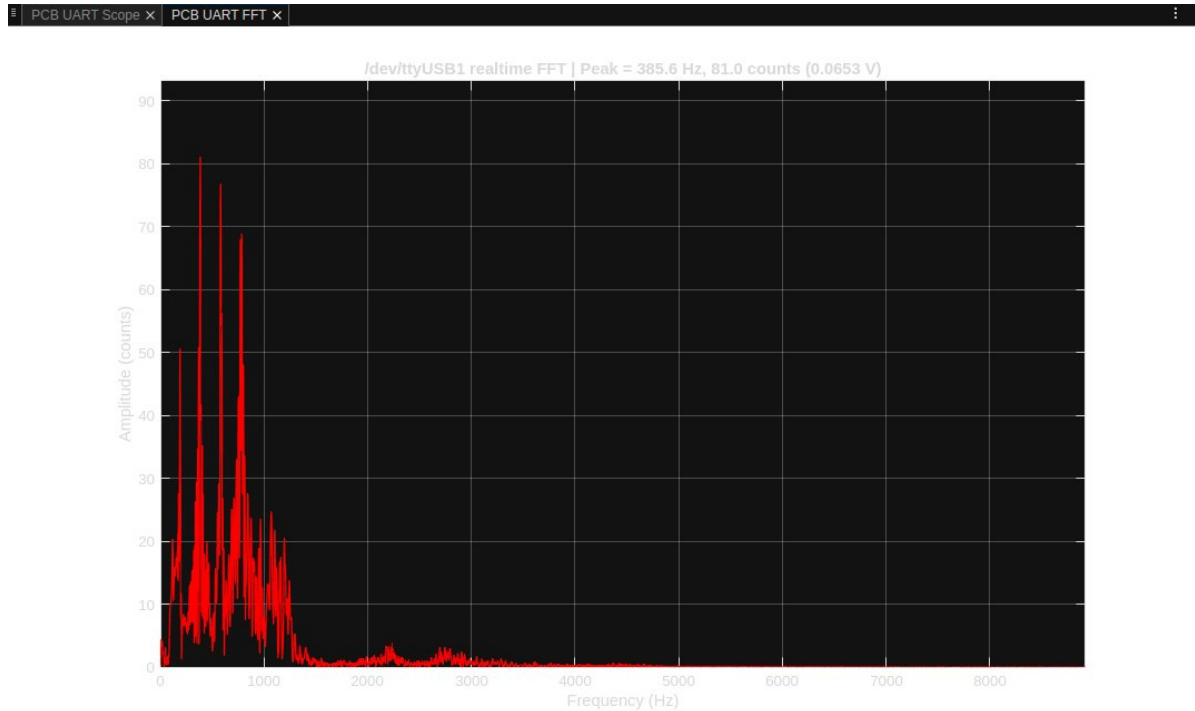


Figure 65. Live MATLAB FFT of the same streamed spoken utterance.

MATLAB live capture from the retained temporary Basys 3 ADC-to-UART instrument, 12 July 2026.

The diagnostic FFT reports its largest component at 385.6 Hz with 81.0 counts (0.0653 V) and shows multiple voiced-speech harmonics concentrated mainly below approximately 1.2 kHz. It confirms that the streamed waveform contains structured speech-band information rather than only DC offset or broadband noise. This is a MATLAB analysis of the temporary ADC stream, not the recognizer's Xilinx FFT output and not a calibrated analog transfer-function measurement.

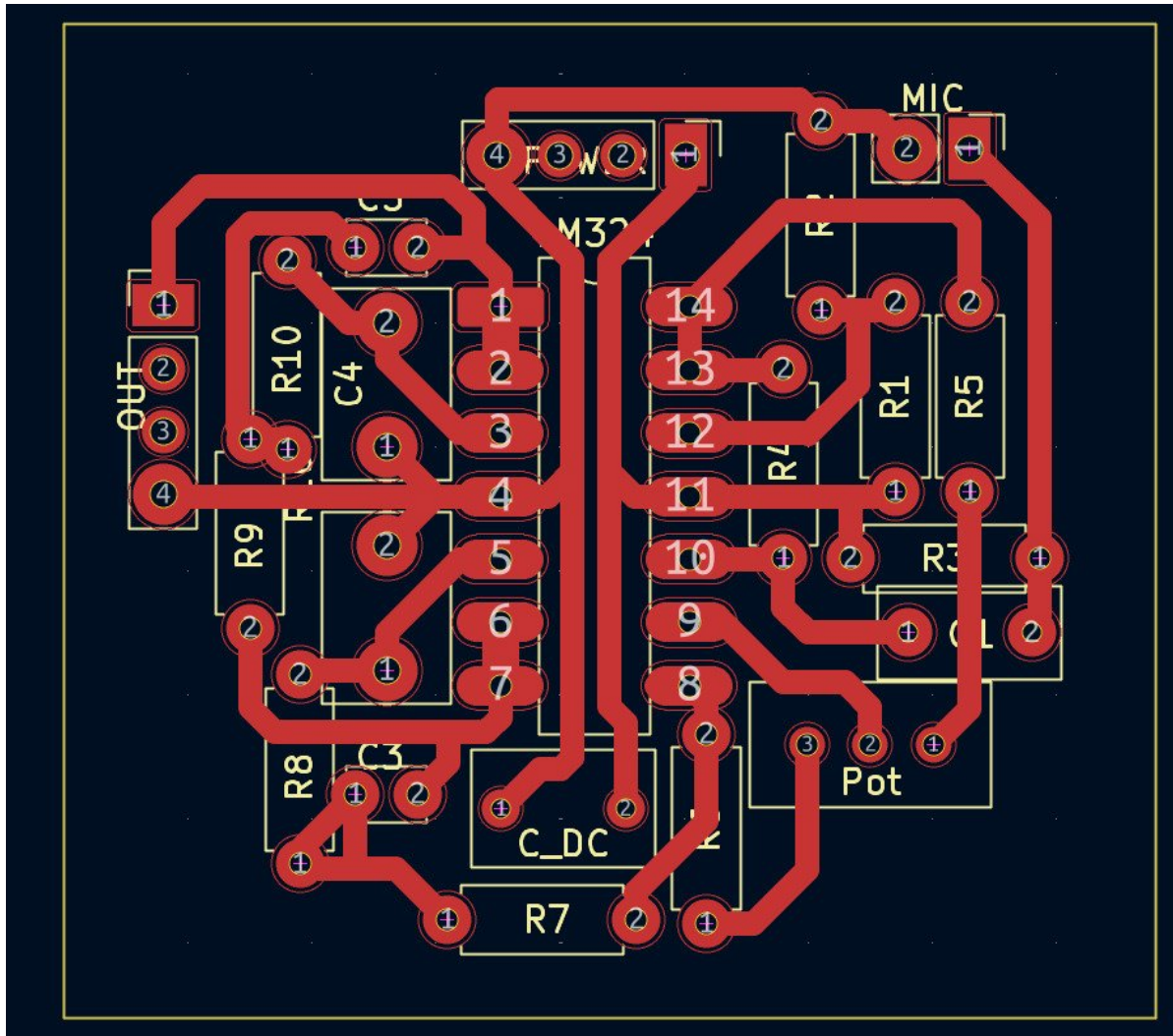


Figure 66. First 44 mm x 40 mm single-sided PCB layout.

Retained final layout evidence from the earlier report and project design files.

The artwork occupies 17.6 cm², or 70.4% of the 25 cm² assignment limit. It contains 21 through-hole footprints, 120 front-copper segments, no vias, and predominantly 1 mm traces with two 1.5 mm segments. Those wide traces and the no-via topology are deliberate home-etching choices: they increase tolerance to mask defects, copper undercut, drilling error, and hand soldering.

19.4 Exposure, etching, assembly, and first operation

I fabricated the board using a photoresist workflow. The mask had to be printed at the correct scale and orientation, held against the copper, exposed, developed, etched without breaking narrow traces, drilled, and assembled. The following sequence documents that process.

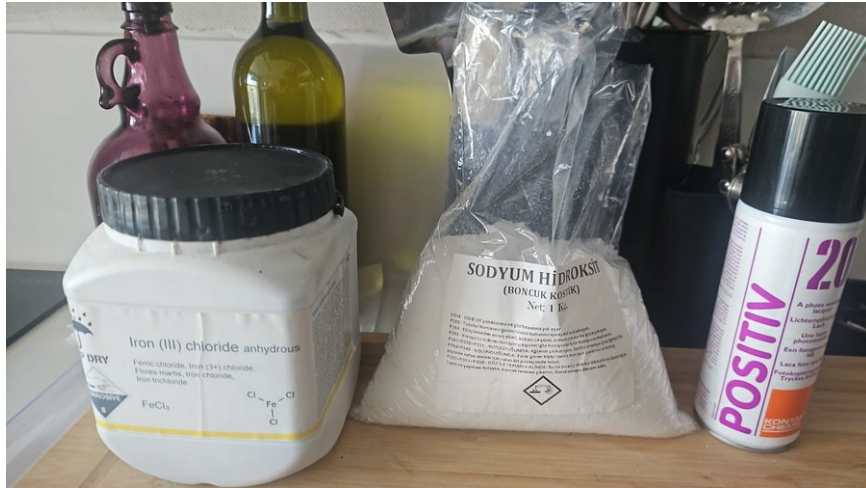


Figure 67. Ferric-chloride etchant, sodium-hydroxide developer, and photoresist materials.

Retained fabrication photograph from the project archive.

I used sodium hydroxide to develop the photoresist and ferric chloride to etch the exposed copper.



Figure 68. Fabrication tools and containers prepared for controlled handling.

Retained fabrication photograph from the project archive.

The development and etching containers were prepared before exposure.

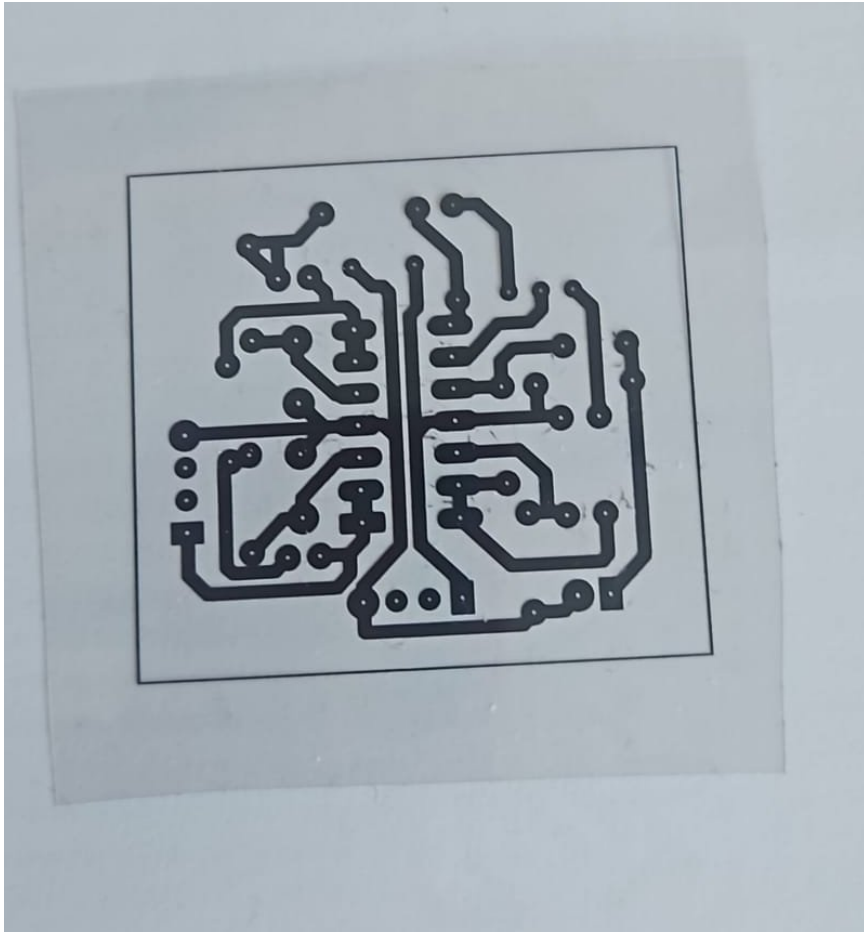


Figure 69. Printed transparency carrying the PCB artwork.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

I printed the copper-side artwork at 1:1 scale and checked its orientation.



Figure 70. UV exposure of the artwork and photosensitive board.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

The transparency was held against the photoresist board during UV exposure.

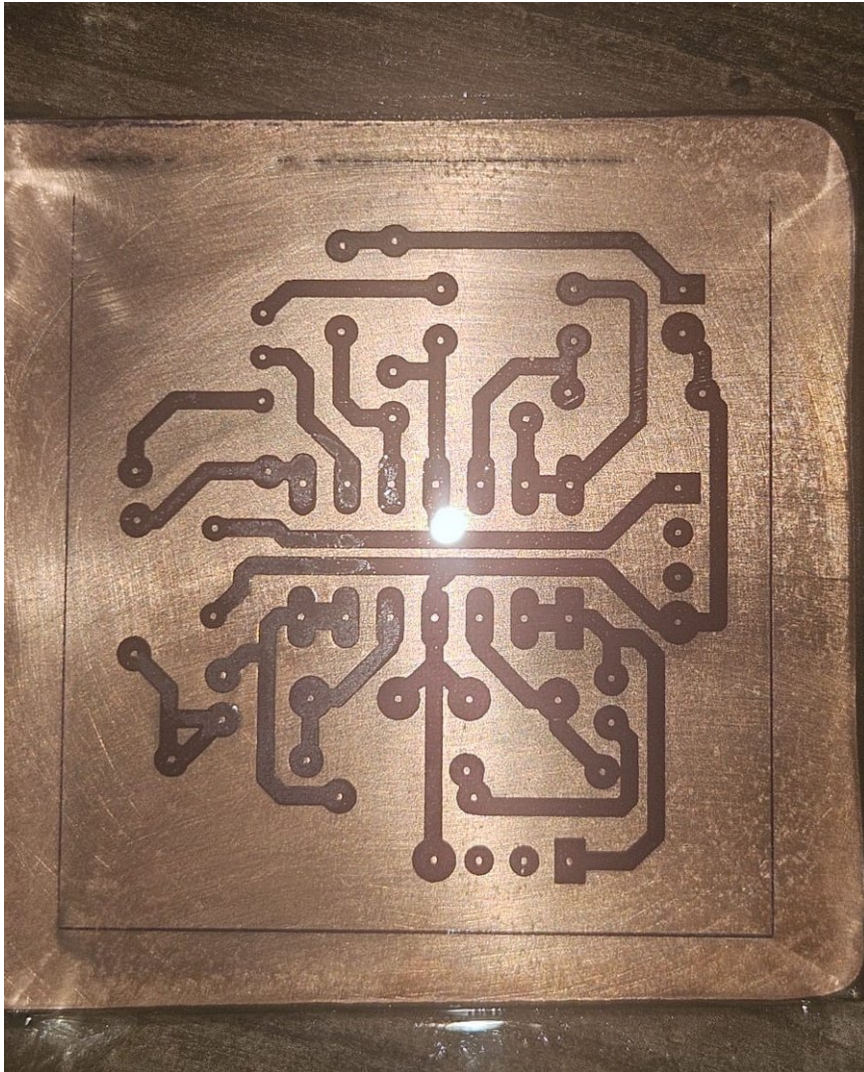


Figure 71. Protected copper pattern after development and etching.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

After development and etching, continuous traces and separated pads remained visible.

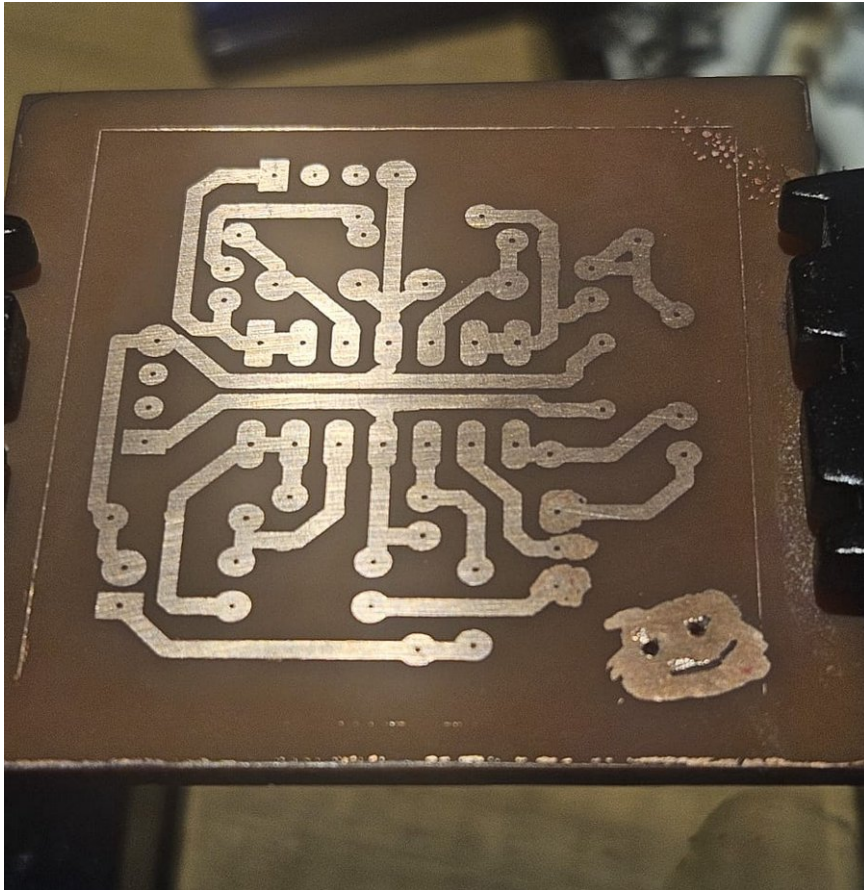


Figure 72. Etched and drilled single-sided copper board.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

The remaining copper forms the intended traces and pads with the surrounding copper removed. This is the first direct physical proof that the layout was manufacturable with the chosen process.

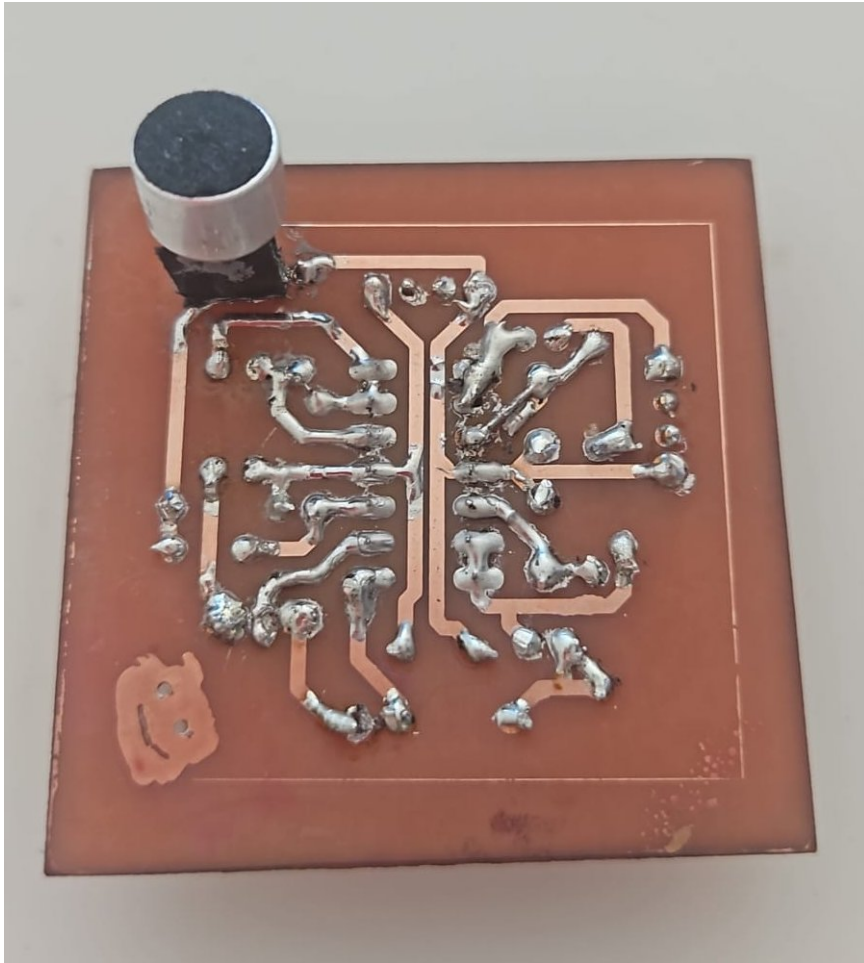


Figure 73. Solder-side view of the assembled first board.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

The photograph records the hand-soldered connections and microphone installation. It closes the path from pads in the CAD layout to real electrical joints.

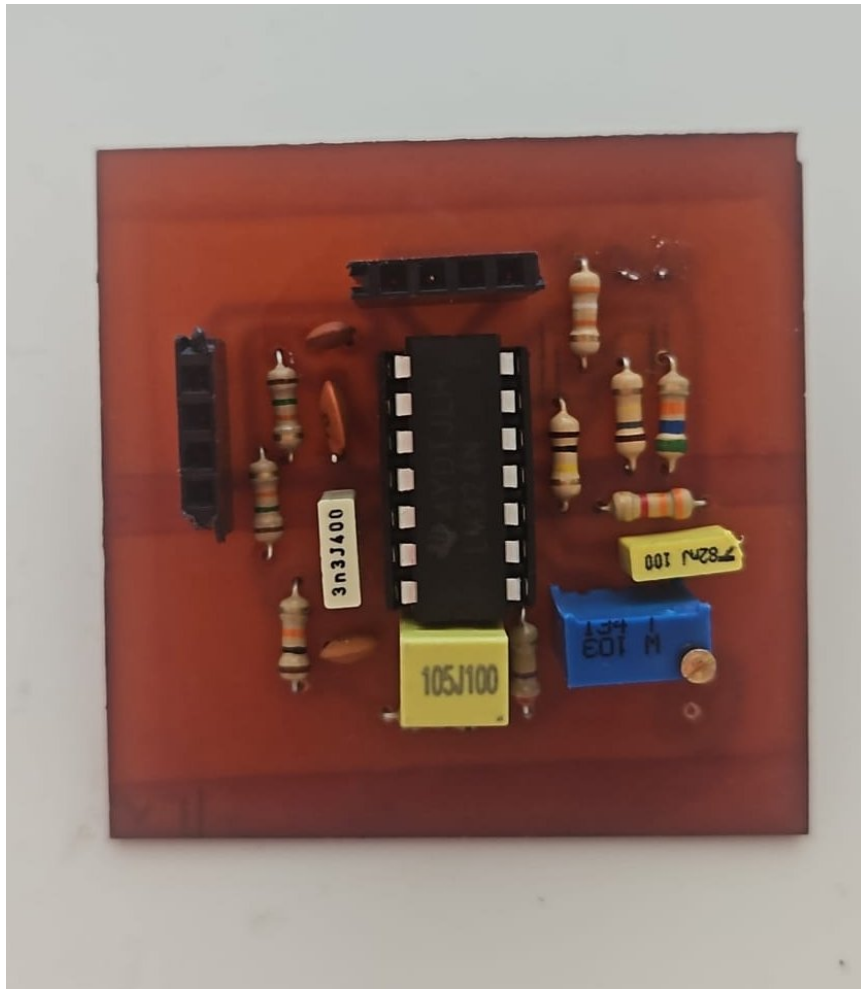


Figure 74. Component-side view of the completed first analog-front-end PCB.

Retained fabrication photograph from the project archive, cropped to the relevant process area.

The completed board includes the op-amp, passive filter components, adjustment element, and connectors. Its successful first operation is significant because the entire PCB workflow was new to me when I began the assignment.

My first-operation verdict followed a staged electrical check rather than appearance alone. I recall measuring the ADC-side quiescent output at approximately 1.632 V, close to the converter midpoint, and roughly 8 V at the microphone bias node. Only after those checks did I connect the board to the Basys 3, load the retained diagnostic bitstream, and inspect the changing microphone samples in MATLAB.

Table 19. Bench DC checks across the original and later PCB revisions.

Bring-up step	Observed result	Engineering decision
ADC-side DC operating point	Original board: recalled approximately 1.632 V; later double-sided SMD revision: 1.652 V current bench reading	Both readings are close to the 1.65 V midpoint; the later value is only 2 mV above it
Electret microphone bias	Original board: recalled approximately 8 V; later double-sided SMD revision: 8.21 V current bench reading	The 8.21 V reading confirms that the populated SMD microphone-bias path is energized
Original live UART/MATLAB observation	Changing, visibly unclipped samples were observed; the trace was not retained	Approve the board and retain the empirically selected potentiometer setting

Seeing live ADC codes was the final criterion in the original approval. The fresh 12 July 2026 trial above reproduces and quantifies that method; it is not presented as the original bring-up screenshot. The result came from checking the operating point, microphone bias, acquisition path, and sampled waveform in sequence.

The frequency-response figures remain simulations, and no calibrated measured transfer-function sweep, transient oscilloscope trace, or power measurement is retained. The approximately 1.632 V output and 8 V microphone-bias values are measurements I recall from the original bring-up; the meter photographs were not saved. The temporary acquisition source, bitstream, MATLAB wrapper, fresh live waveform/FFT captures, and later unclipped utterance dataset now survive together, so the ADC-to-UART verification method and its practical outcome are directly supported.

19.5 Later two-layer home-fabricated revision

The preceding subsections document my first single-sided through-hole board. I later rebuilt the same microphone and filter concept as a compact two-layer board that also carries the ADC121S101 interface and a direct Basys 3 Pmod connection. This revision uses a mixture of surface-mount and through-hole parts, aligned front- and back-copper artwork, and hand-finished inter-layer connections.

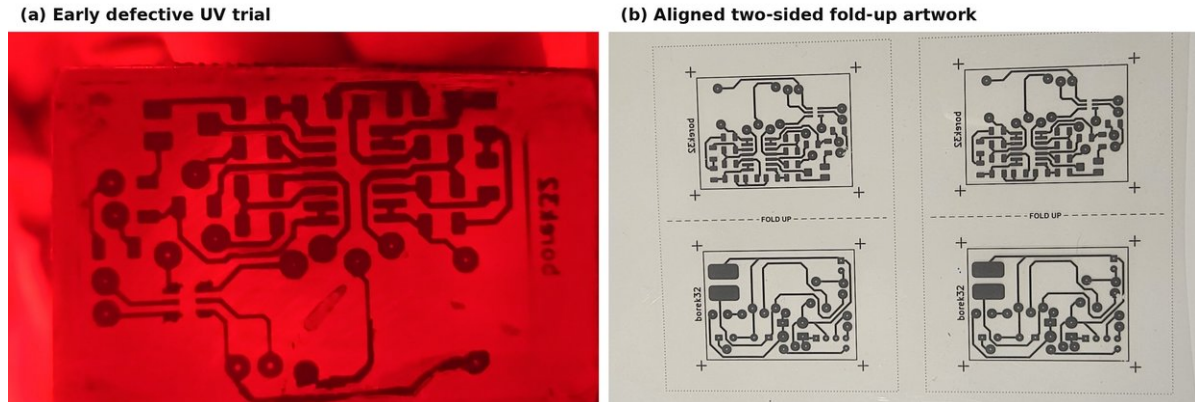


Figure 75. Development progression of the later two-layer board: (a) early defective UV-exposure trial retained as failure evidence; (b) aligned fold-up artwork used for front- and back-copper registration.

Metadata-clean fabrication photographs from July and August 2026; the defective trial was discarded and was not used for hardware measurements.

The failed transfer remains useful development evidence because it changed how I registered the two copper sides. The fold-up print keeps the front and back artwork on one sheet, reducing independent movement during exposure. I retain the failure explicitly instead of presenting every fabrication attempt as successful.

The later layout is a separate revision from Figure 66. It contains 27 footprints, 75 pads, 287 routed segments across F.Cu and B.Cu, and 18 vias, rather than the first board's 21 through-hole footprints, 120 F.Cu segments, and no vias. Keeping both sets of figures preserves the actual development path.

19.6 Schematic and PCB net reconciliation

The recovered KiCad schematic and PCB did not initially carry one complete common set of net names. I preserved the board geometry and used KiCad's native connectivity model to assign every physical copper island to the corresponding schematic net. The ADC ground, originally represented by a separate local /gnd label, was joined to the board-wide GND net.

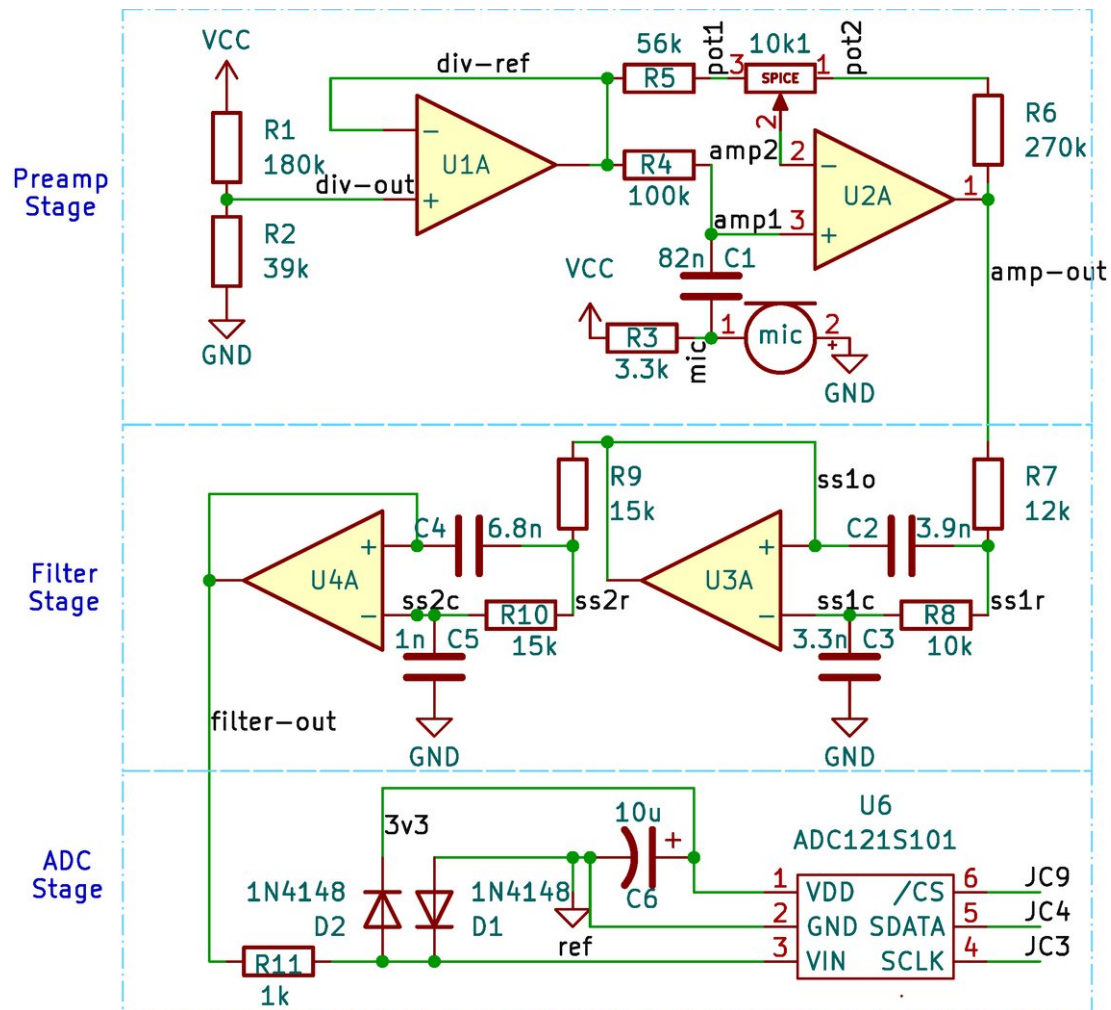


Figure 76. Reconciled schematic of the later microphone preamplifier, fourth-order filter, protection network, ADC121S101, and JC interface.

Reconciled KiCad schematic from the retained named-net project, 19 August 2026.

The schematic makes the analog-to-digital boundary explicit: the microphone and gain stage feed the two filter sections, the clamp/reference network protects the ADC input, and the serial pins leave as JC9 chip select, JC4 serial data, and JC3 serial clock. The schematic and PCB DRC/ERC findings have been resolved.

The successful copper is mirrored relative to the logical ADC121S101 and LM324 footprint numbering, so I reconciled the layout from the actual electrical functions rather than copying the printed geometric pad numbers. KiCad found 22 physical copper islands: 21 use the exact 21-name schematic set and one isolated unused Pmod pad intentionally remains no-net. All 75 pads, 287 routed segments, and 18 vias are accounted for, the geometry fingerprint is unchanged, and the native check reports zero unconnected items.

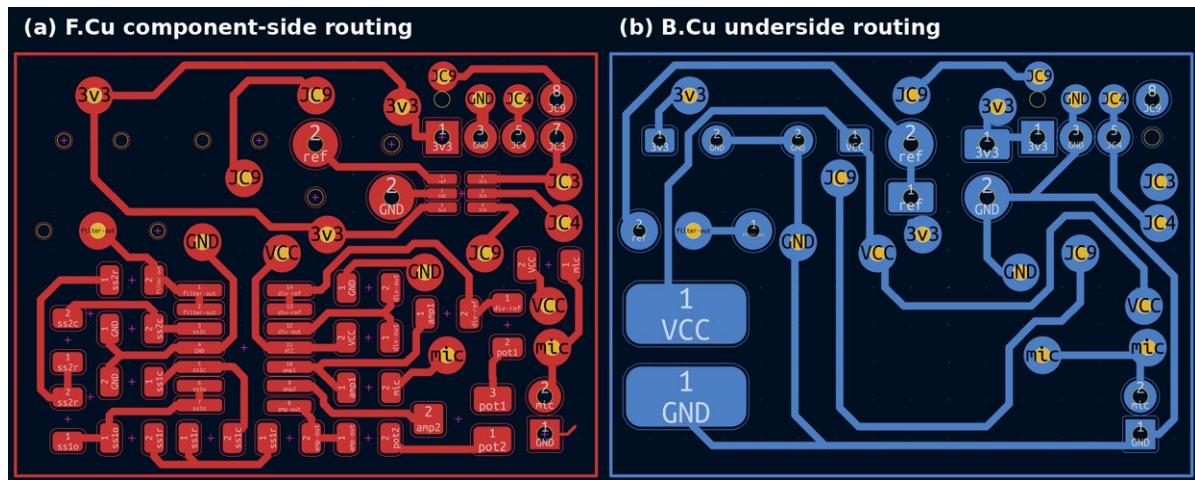


Figure 77. Isolated named-net routing views of the later PCB: (a) F.Cu component-side routing; (b) B.Cu underside routing.

Derived from the retained F.Cu and B.Cu KiCad working views after net assignment.

The separated layers make the routing and via relationships readable without footprint text obscuring the copper. Red identifies F.Cu and blue identifies B.Cu. The net labels provide traceability to the reconciled schematic. The PCB and schematic rule findings described during development have since been resolved.

Net reconciliation and resolution of the PCB DRC/ERC findings are complete. Earlier clearance, courtyard, board-outline, connector, and schematic-structure findings describe the development revision rather than outstanding work. The direct Pmod geometry requires JC10 to remain electrically isolated because JC10/R18 is a separate FPGA output.

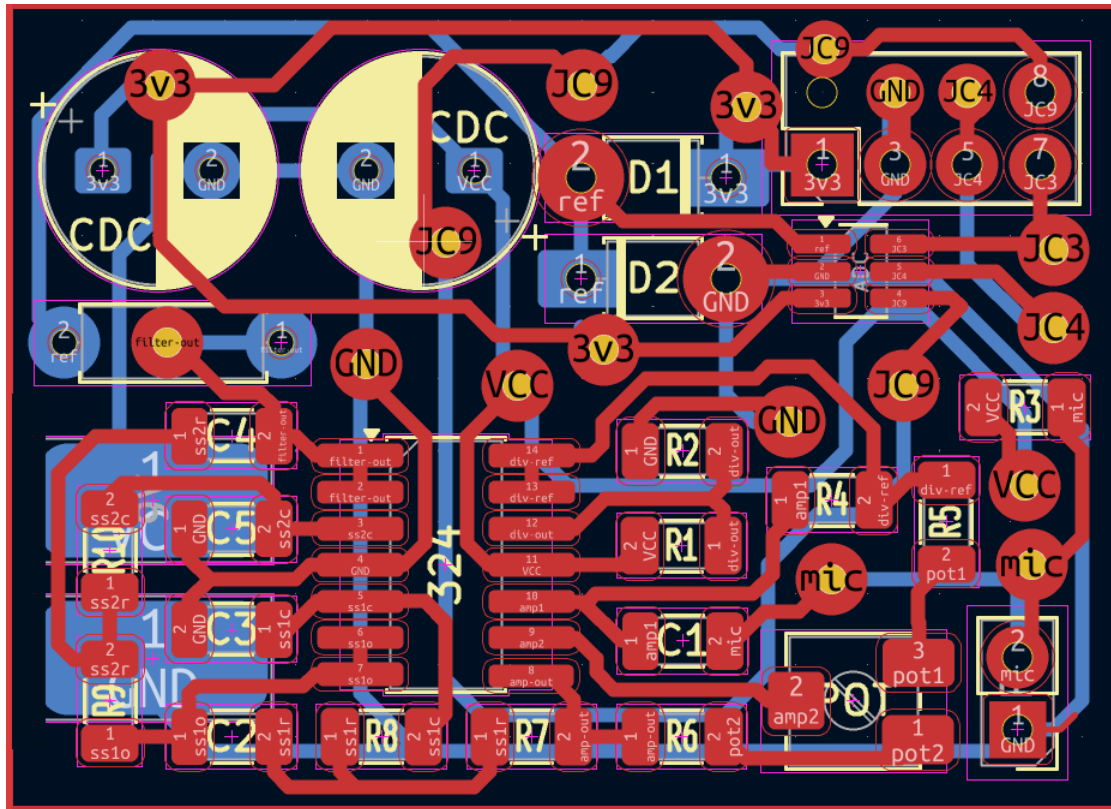


Figure 78. All-layers KiCad working view of the later PCB with footprints, pads, vias, and assigned net names visible.

Retained KiCad working screenshot; included for placement and traceability context rather than as the primary routing figure.

This denser view shows how the component placement, pad labels, and both copper layers fit together. Figures 76 and 77 are the clearer electrical views; this screenshot is retained because it connects them to the actual working layout used during recovery.

19.7 ADC121S101 serial-alignment trial

During standalone ADC-to-UART diagnosis, I saw approximately -2048 centered counts at 0 V but values near 0/-1/+1 with the ADC input near 3.3 V. That pattern pointed upstream of MATLAB's signed display conversion. I therefore preserved the receiver and built an isolated datasheet-derived candidate that asserts CS before the first clock transition, captures exactly 16 falling-edge positions, retains the complete DB11 through DB0 word, discards the first complete frame after start, averages 32 unsigned conversions, and subtracts 0x800 exactly once after averaging [9].

The behavioral model returned -2048, -2047, -1, 0, +1, and +2047 for raw codes 000, 001, 7FF, 800, 801, and FFF, with 16 falling edges in each checked frame. The standalone design routed with WNS +6.874 ns and WHS +0.106 ns, all 192 nets routed, and no DRC or methodology findings, and was programmed through JTAG. Known-voltage endpoint tests and raw receiver-word capture have been completed. The corrected receiver has also been integrated into the full MFCC design.

Corrected ADC receiver integration and the endpoint checks are complete. The standalone simulation, implementation, and programming results document that development step. The dated JC build figures below retain their original build identity; they are not new timing measurements of subsequent revisions.

19.8 Full MFCC integration on Pmod JC

For full-system testing I built an isolated top_module pin variant with spi_clk on JC3/N17, spi_miso on JC4/P18, and cs_out on JC9/P17. The implementation routed all 12,824 routable nets and achieved WNS +0.671 ns and WHS +0.020 ns. Routed DRC reported 11 advisories and no warnings or errors. It used 5,603 LUTs, 8,437 registers, 23.5 BRAM tiles, and 15 DSP blocks.

Retained bitstream SHA-256:

2f1a6697f5937033b59397478b328e61b7bad764fd204695cf9bb3adea60b3ac.

At 19:54:59 TRT on 19 August 2026, the programming script selected the single expected Digilent target, verified that exact bitstream hash, programmed the xc7a35t, and recorded startup HIGH, FPGA DONE = 1, and PROGRAM_RESULT = PASS. Programming also restored the configuration-time COE reference baseline [10].

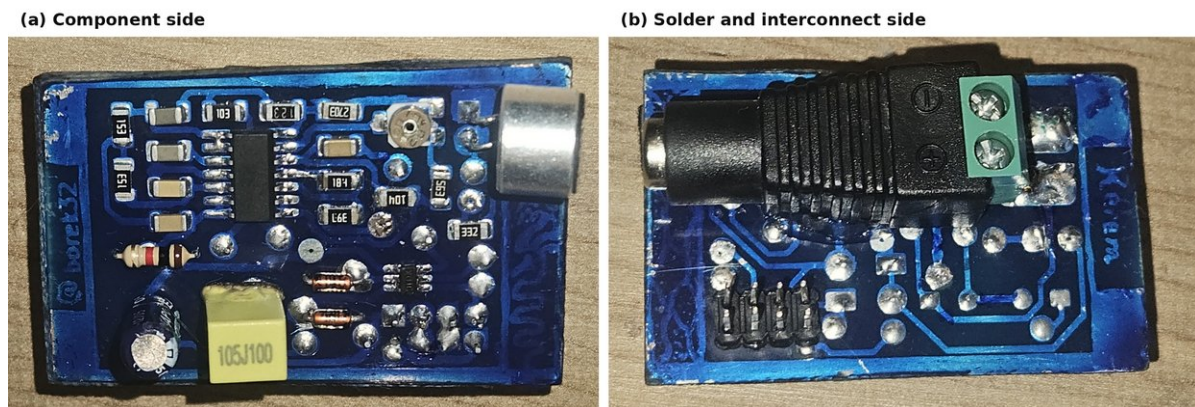


Figure 79. Final populated home-fabricated microphone and ADC PCB: (a) component side; (b) solder and interconnect side.

Metadata-clean final-board photographs from 19 August 2026.

The component side shows the LM324 analog chain, adjustment network, microphone, protection parts, and ADC section. The reverse view records the hand-soldered joints, power input, and direct interconnect construction. The images are presented as built, without cosmetic retouching. A current bench check of this later two-layer, SMD-populated revision measured 1.652 V at the ADC-side quiescent output, only 2 mV above the ideal 1.65 V midpoint, and 8.21 V at the microphone-bias node.

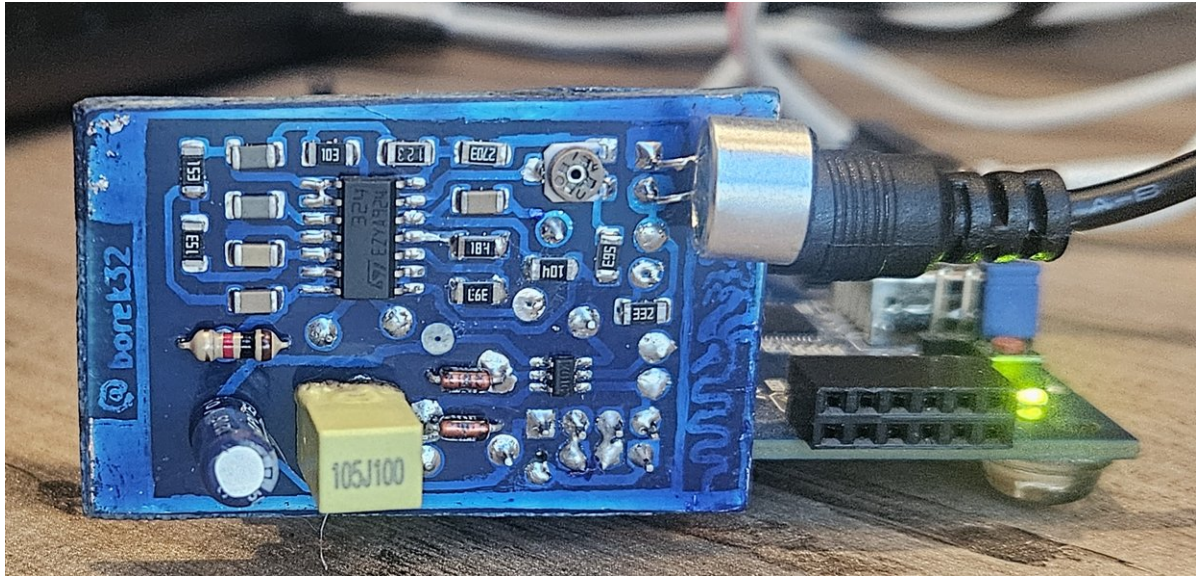


Figure 80. Final home-fabricated microphone and ADC PCB connected to the Basys 3 Pmod JC during full-design bench use on 19 August 2026.

Metadata-clean bench photograph retained with the home-fabrication evidence bundle.

Together with the programming log, this photograph documents the physical path from the schematic and copper layers to the programmed FPGA. Labeled live spoken-digit comparison, including the full frame sequence and final display/UART result, is complete. Runtime reference recording, selected-slot readback, and immediate recall have also been completed. These activities are no longer outstanding project tests.

The system combines net reconciliation, full-design implementation, programming, physical integration, and completed live recognition tests. The dated implementation report contains 36 TIMING-18 warnings because external I/O delays were absent. Its internal timing result does not establish external SPI or board-interface timing closure.

20 Discussion, Limitations, and Conclusion

20.1 Engineering decisions viewed as a system

The final architecture reflects several practical tradeoffs: one clock domain simplified acquisition, RAM-separated stages simplified latency handling, and sequential arithmetic kept resource use manageable. The FFT wrapper retains only the one-sided spectrum needed by the filter bank. COMP uses energy alignment and SAD/L1 instead of a heavier classifier, while DEBUG exposes each intermediate memory when integration fails.

Table 20. Major decisions, their rationale, and the evidence visible at system level.

Decision	Reasoning	Observed consequence
33.333 MHz ADC enable	Remove the generated-clock/CDC sensitivity while remaining close to the original 36 MHz target	Positive current internal timing margins; historical board captures remain version-labeled
Correct FFT sample and address contracts	Preserve synchronous BRAM responses through AXI backpressure and align power latency explicitly	Retained bins 0..255 at matching addresses; AMD parity 256/256
Sequential stage reuse	Spoken-digit latency permits resource sharing	Complete pipeline fits with 15 DSPs and 23.5 BRAM tiles
COE-initialized writable reference RAM	Provide a repeatable startup set while allowing speaker-specific slot replacement without resynthesis	Immediate compare operation plus volatile runtime enrollment
SAD/L1 templates	Transparent arithmetic without multipliers or square roots	30 retained pre-correction development utterances separate 30/30 in leave-one-out analysis
Optional full-memory UART	Preserve deep diagnosis without slowing normal mode	1,362 words per frame only when selected

20.2 Limitations and tests

Completed hardware work is reflected in the preceding sections. The remaining limitations and the two follow-up measurements are listed below.

Table 21. Remaining limitations and tests.

Category	Scope
Limitation	The internal timing result does not close external SPI and board-interface timing.
Limitation	The analog front end lacks a calibrated physical transfer-function sweep.
To test next	Capture reference memory before and after flush, including the 512-word clear and flush_done event.
To test next	Measure the physical front-end response with calibrated equipment.

20.3 Conclusion

The prototype implements the intended record/compare architecture on the Basys 3. It acquires a fixed-length recording, extracts eight features per frame, writes or reads 16-frame templates, and reports the nearest-template digit. Module regressions, the focused COMP record-and-recall simulation, and the current routed checkpoint support those claims. The routed result covers the constrained internal 100 MHz paths; external board-interface timing remains outside that claim.

At the module boundaries, WINDOW writes addresses 0 through 511 in order, the FFT stores the 256 retained bins at matching addresses and agrees with the AMD test model, MEL multiplies $P[k]$ by $H[m,k]$, and COMP writes and recalls runtime templates. The 30-record collection gives 30/30 in its offline leave-one-out analysis. Labeled live recognition and runtime reference recording, slot readback, and recall have also been completed. The offline and live tests describe distinct parts of the verification work.

The analog input path has two documented hardware stages: the first single-sided board and the later two-layer microphone/ADC revision. The later board has a reconciled 21-net KiCad representation and completed DRC/ERC corrections. The corrected ADC receiver is integrated into the full design, and endpoint tests are complete. The remaining work is the before/after reference-memory flush capture and a calibrated front-end sweep. Internal timing closure alone does not close the external interface.

21 References

- [1] AMD/Xilinx, Fast Fourier Transform LogiCORE IP Product Guide, PG109.
- [2] AMD/Xilinx, Vivado Design Suite User Guide: Designing with IP, UG896.
- [3] Digilent, Basys 3 FPGA Board Reference Manual, Revision C.
- [4] L. R. Rabiner and R. W. Schafer, Theory and Applications of Digital Speech Processing, Prentice Hall, 2011.
- [5] J. G. Proakis and D. G. Manolakis, Digital Signal Processing: Principles, Algorithms, and Applications, 4th ed., Pearson, 2007.
- [6] Current project RTL, constraints, initialized memories, VHDL testbenches, MATLAB checkers, and routed Vivado reports, reviewed July-August 2026.
- [7] Retained FPGA-UART spoken-digit and stationary-tone captures, acquired during project development and rechecked July 2026.
- [8] Retained temporary PCB ADC-to-UART source, bitstream, MATLAB live-capture wrapper, and fresh waveform/FFT captures, reviewed July 2026.
- [9] Texas Instruments, ADC121S101 12-Bit, 500-kSPS to 1-MSPS Analog-to-Digital Converter, Revision H.
- [10] Retained home-fabricated microphone/ADC PCB KiCad source, named-net verification, photographs, JC-mapped implementation reports, and programming logs, reviewed August 2026.

Appendix A Supplementary Testbench Views

Appendix A collects the retained verification contact sheet and six earlier numerical views. The current checker for each module remains in the main chapter; these figures preserve the earlier debugging and verification history.

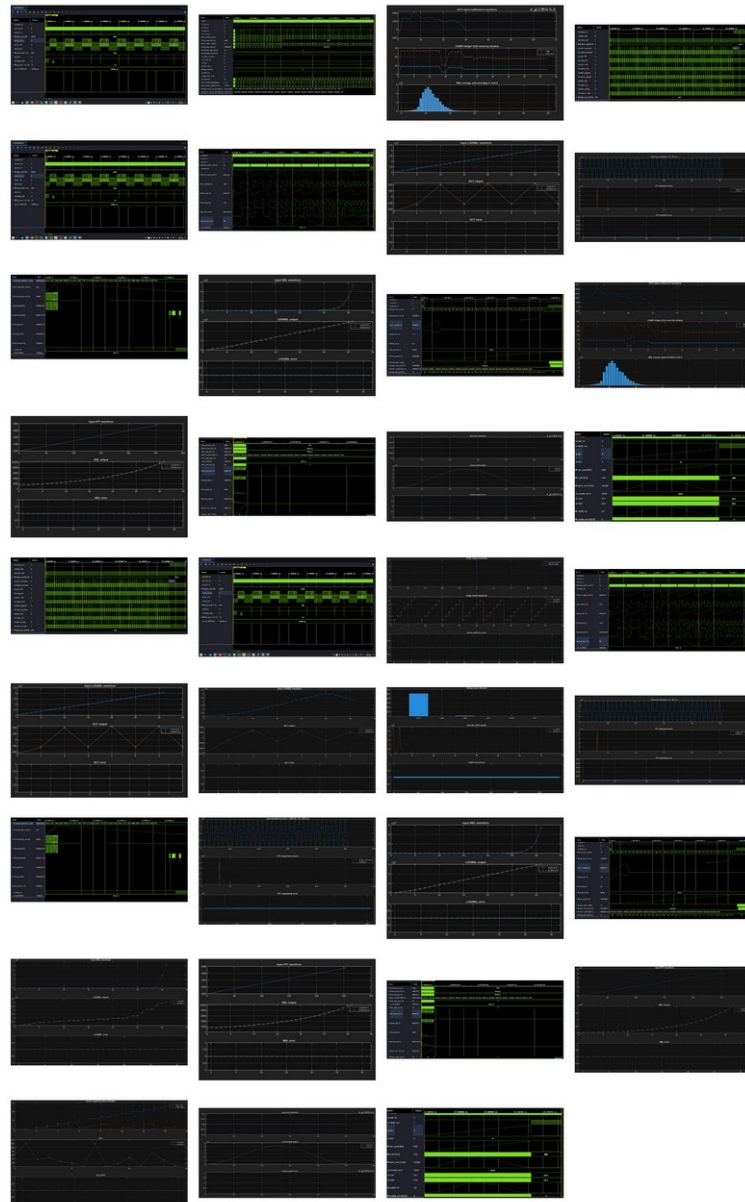


Figure 81. Contact sheet of the retained module-verification views.

Unique retained view from the indexed module-verification set.

This file is a contact sheet rather than a WINDOW-only checker despite its filename. It is retained here with the correct description because it records the full gathered-evidence overview. The WINDOW-specific clean checker appears in Section 7, and each constituent unique view is reproduced individually in the relevant chapter or this appendix.

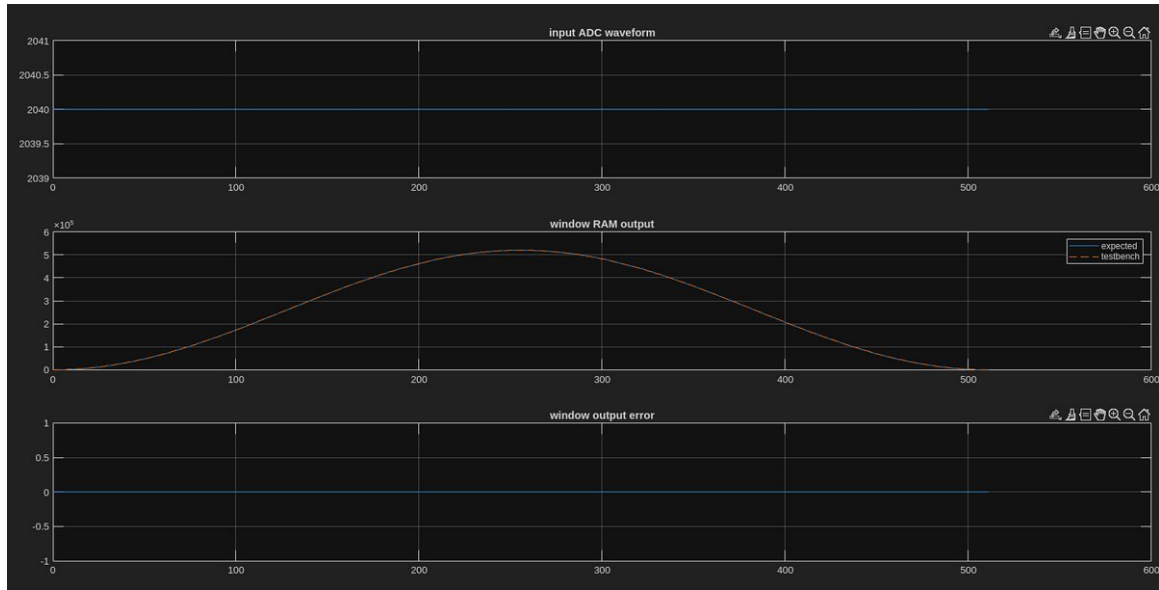


Figure 82. Earlier WINDOW numerical comparison.

Unique retained view from the indexed module-verification set.

The original export already shows the expected and observed Hann products overlapping with a zero error trace. The cleaner main-chapter export confirms the same 512-word result and is easier to read; retaining both records the verification history without changing the conclusion.

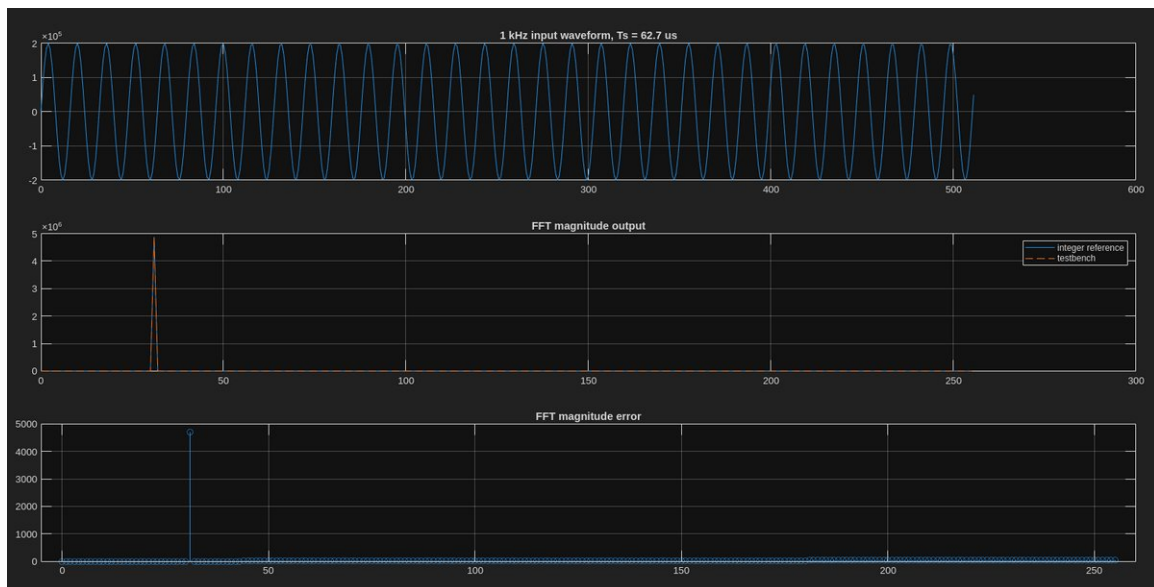


Figure 83. Pre-fix FFT address-defect evidence.

Unique retained view from the indexed module-verification set.

This historical view records the original one-address shift. It is retained as defect-discovery evidence only; the current post-fix result is the exact-address, skid-protected regression in Section 8.

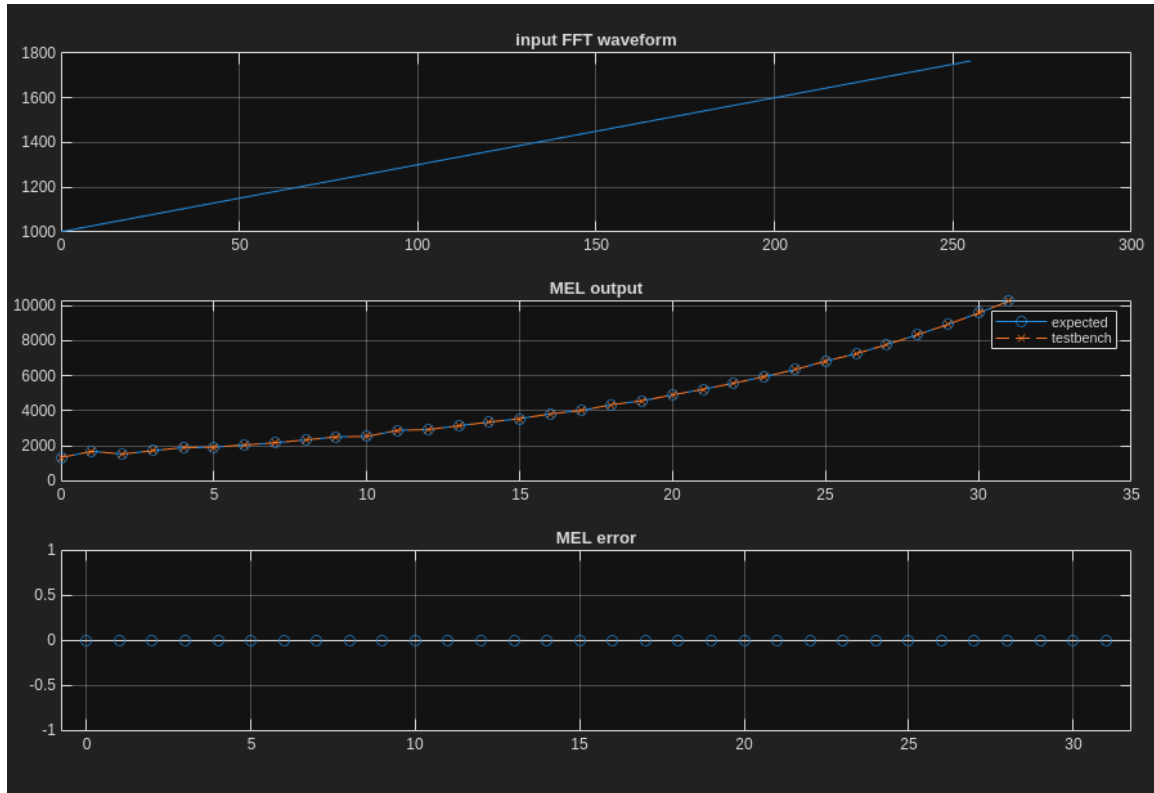


Figure 84. Earlier MEL ramp comparison before final bin-address identity.

Unique retained view from the indexed module-verification set.

The expected and observed traces overlap because both the historical RTL and its checker modeled the same residual $k+1$ FFT request. It is retained as verification-history evidence only; Section 9 contains the corrected $P[k]$ with $H[m,k]$ regression.

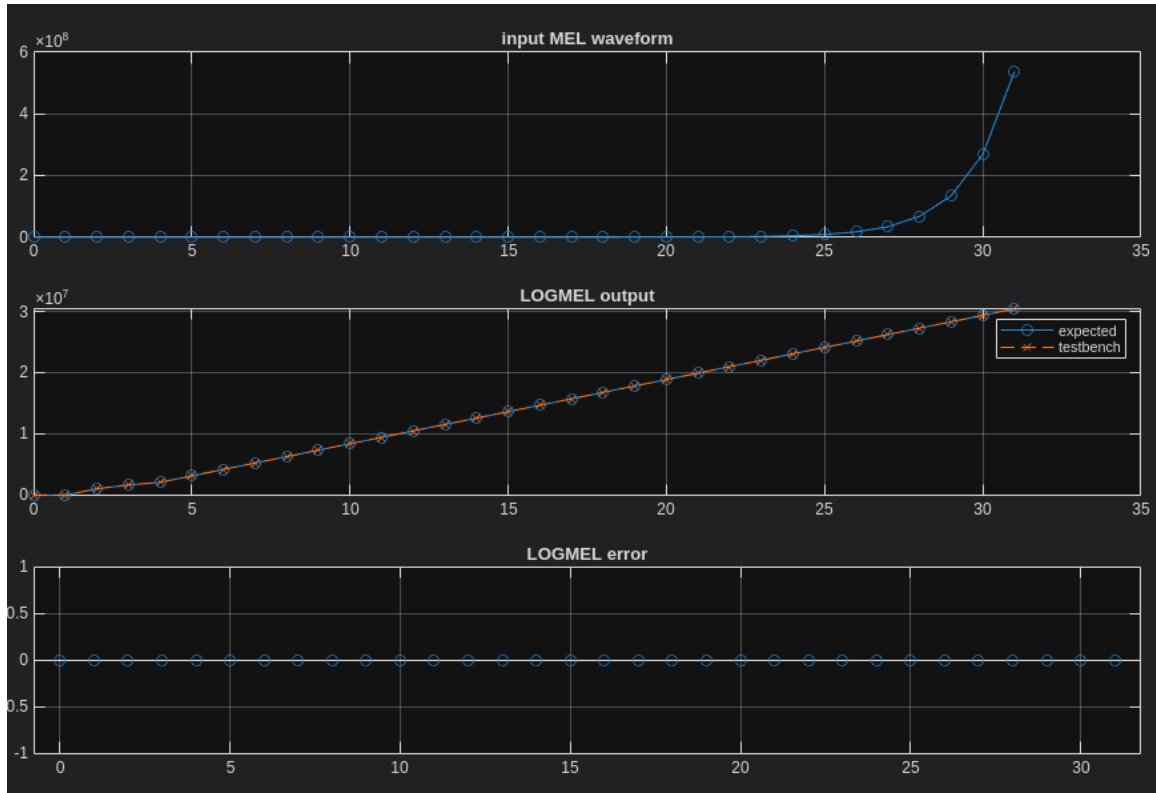


Figure 85. Earlier LOGMEL dynamic-range comparison.

Unique retained view from the indexed module-verification set.

The input grows over a very wide range while the output becomes approximately linear in log space. Expected and simulated words overlap; the current export adds the boundary-focused interpretation used in the main chapter.

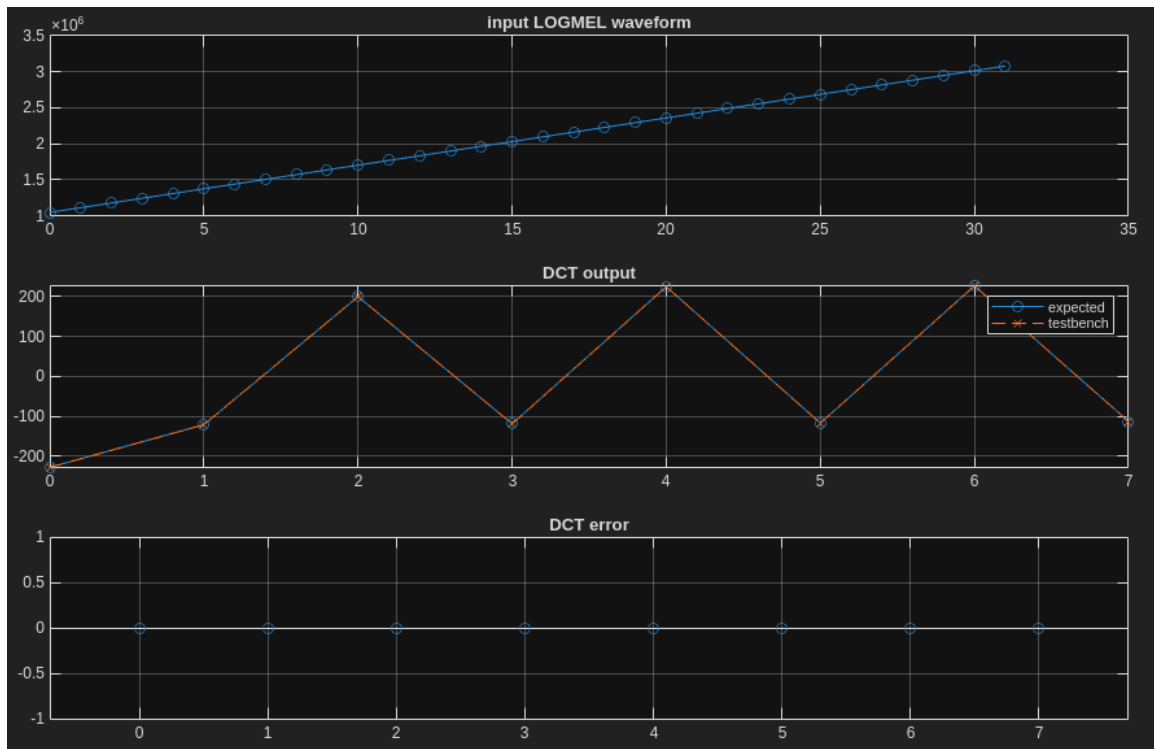


Figure 86. Earlier DCT coefficient comparison.

Unique retained view from the indexed module-verification set.

All eight signed outputs coincide with the expected vector and the error remains zero. This is the original numerical companion to the raw repeated-cosine XSim trace.

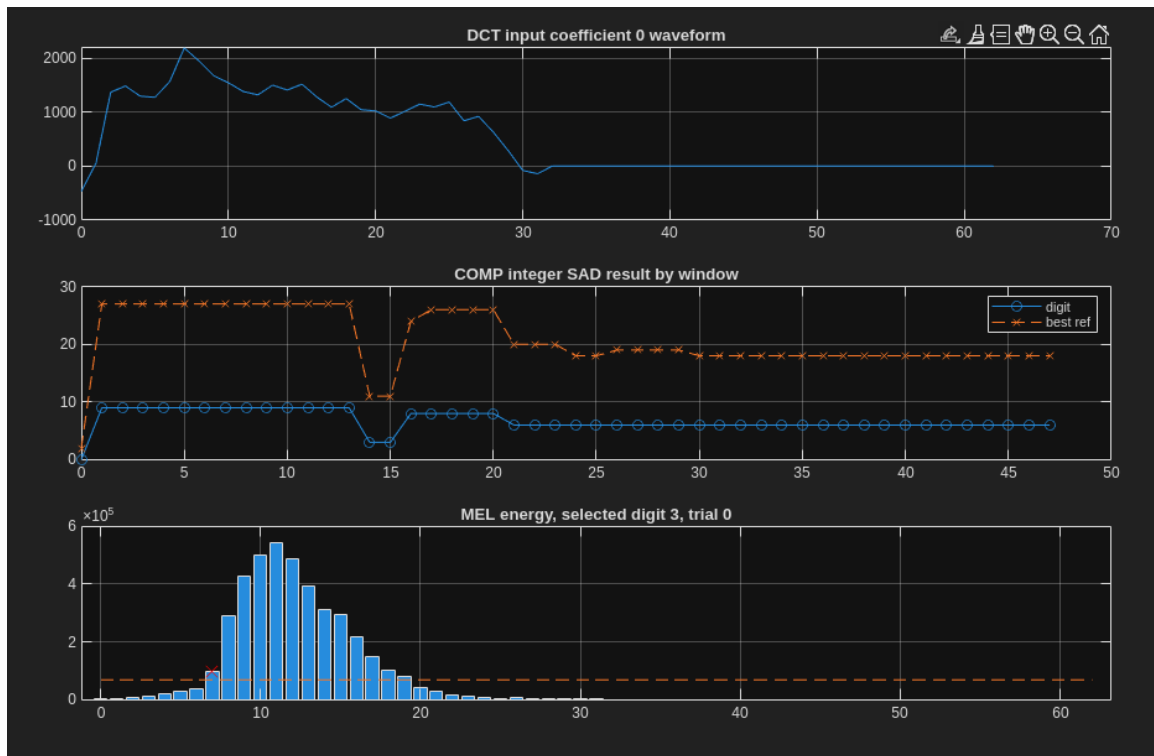


Figure 87. Earlier COMP onset and reference-search comparison.

Unique retained view from the indexed module-verification set.

The plotted MEL energy selects the active interval and the stored best-reference trace identifies the digit-9 result. The current checker in Section 12 supplies the exact threshold, reference, and SAD values behind this view.

Appendix B Complete Programmed-Board Tone Evidence

Section 16 shows the complete 1 kHz set; Appendix B collects the remaining 26 retained tone plots. No DCT image was saved for 2 kHz, and no LOGMEL image was saved for 4 kHz.

B.1 100 Hz

At the low end of the sweep, FFT energy remains near the lower stored addresses and MEL energy remains in the earliest bands. This provides a reference against which the higher-frequency movement can be judged.

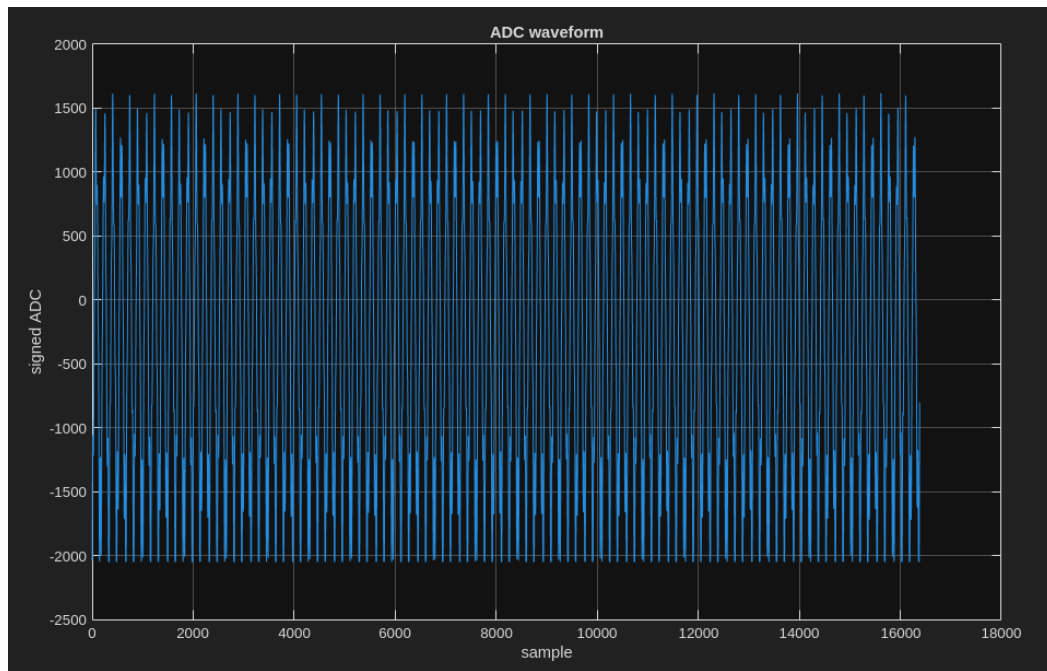


Figure 88. Programmed-board 100 Hz experiment: ADC.

UART capture from the programmed Basys 3, 3 June 2026.

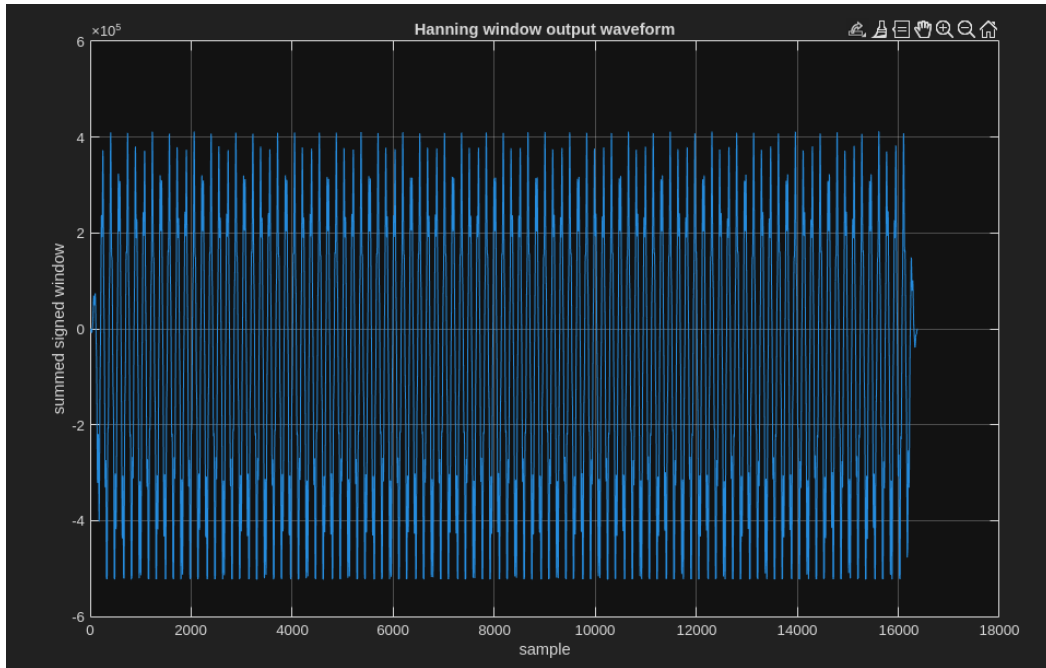


Figure 89. Programmed-board 100 Hz experiment: Hann overlap-add.

UART capture from the programmed Basys 3, 3 June 2026.

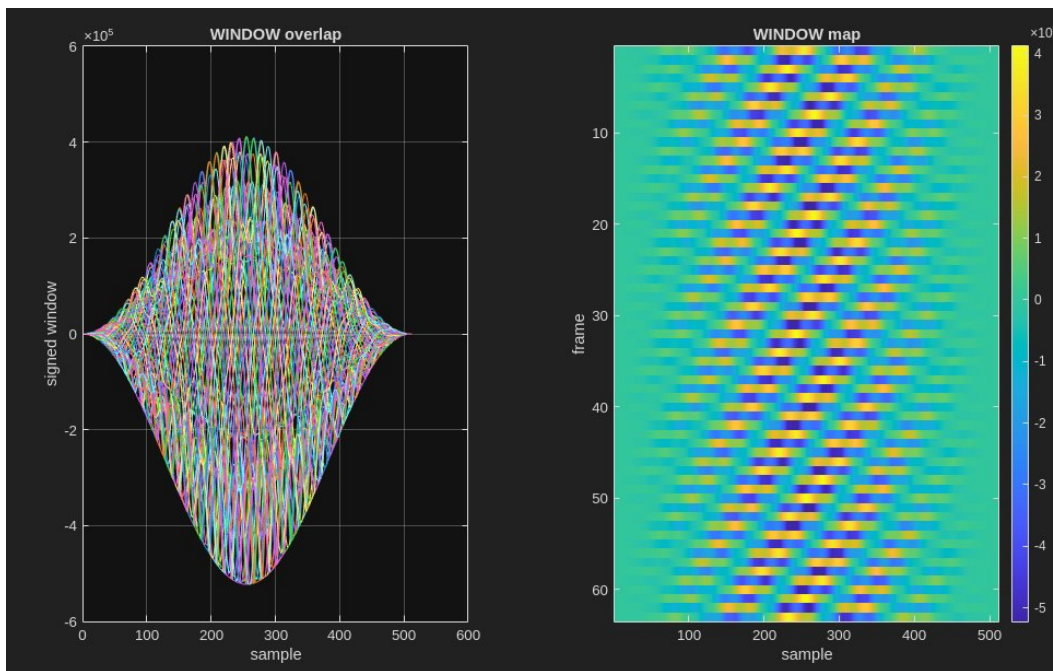


Figure 90. Programmed-board 100 Hz experiment: WINDOW.

UART capture from the programmed Basys 3, 3 June 2026.

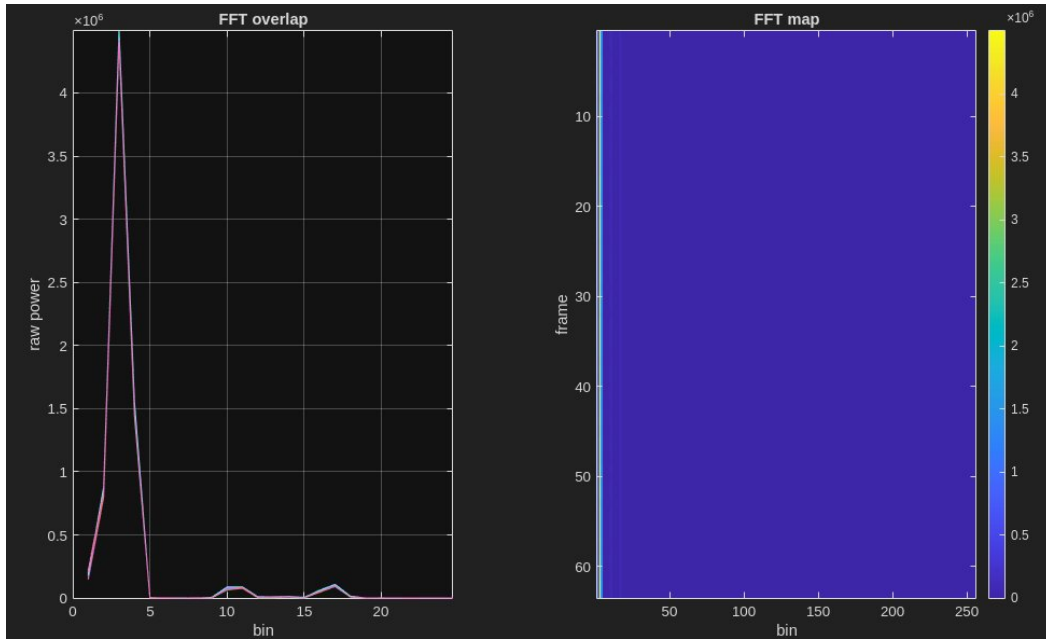


Figure 91. Programmed-board 100 Hz experiment: FFT.

UART capture from the programmed Basys 3, 3 June 2026.

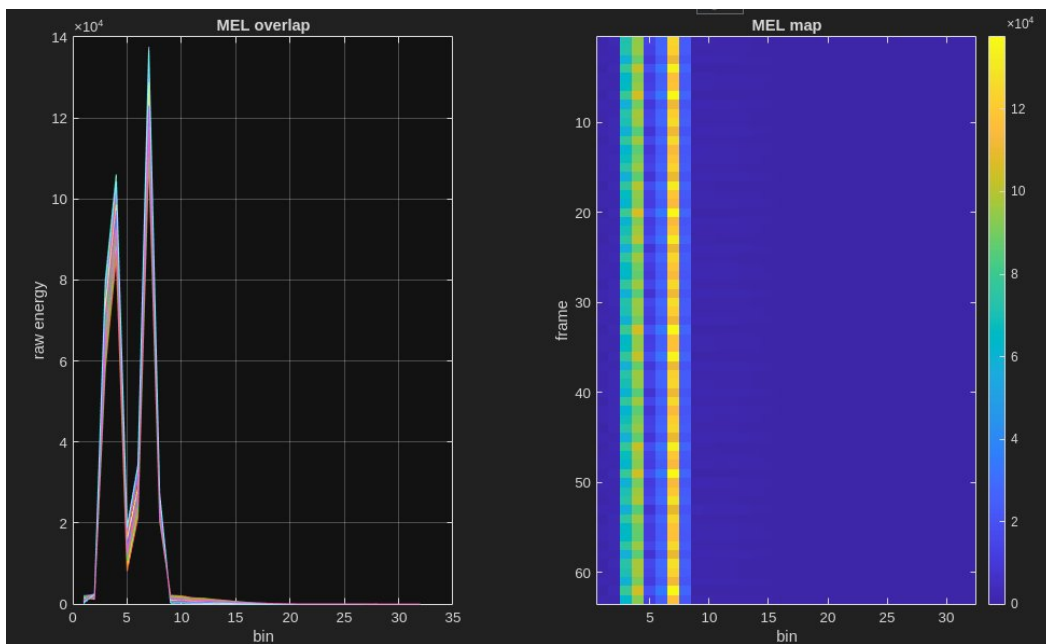


Figure 92. Programmed-board 100 Hz experiment: MEL.

UART capture from the programmed Basys 3, 3 June 2026.

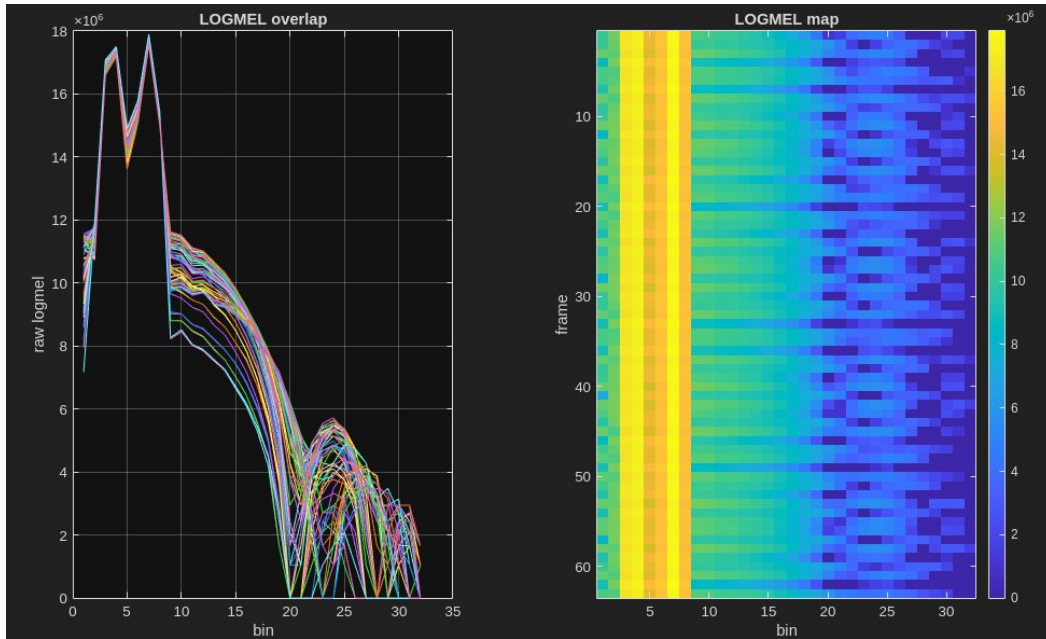


Figure 93. Programmed-board 100 Hz experiment: LOGMEL.

UART capture from the programmed Basys 3, 3 June 2026.

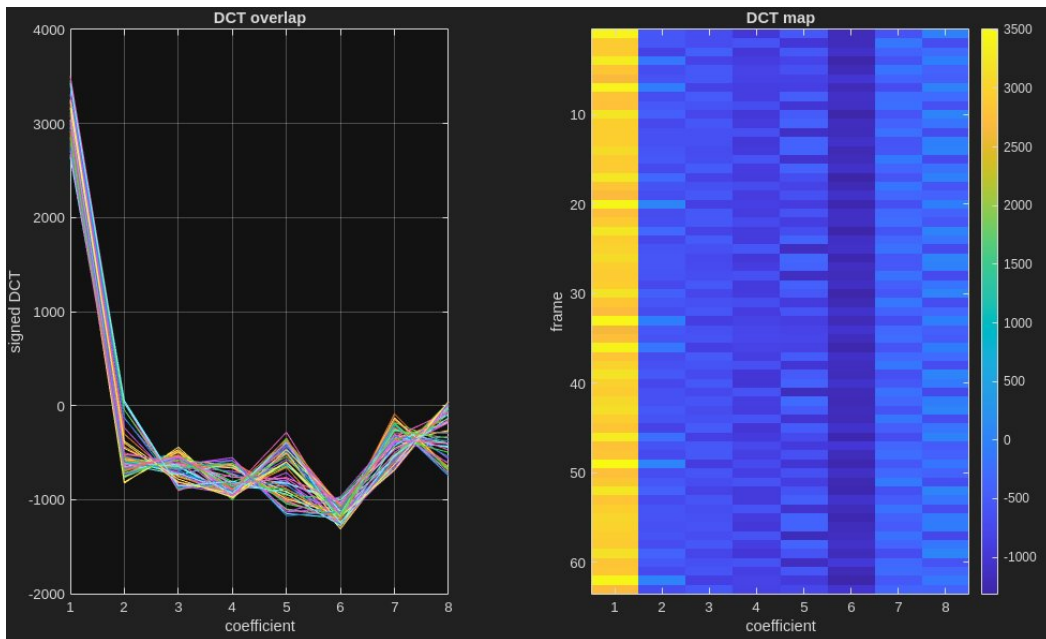


Figure 94. Programmed-board 100 Hz experiment: DCT.

UART capture from the programmed Basys 3, 3 June 2026.

B.2 500 Hz

The 500 Hz run moves the dominant FFT/MEL regions above the 100 Hz case while retaining stable frame-to-frame structure through DCT.

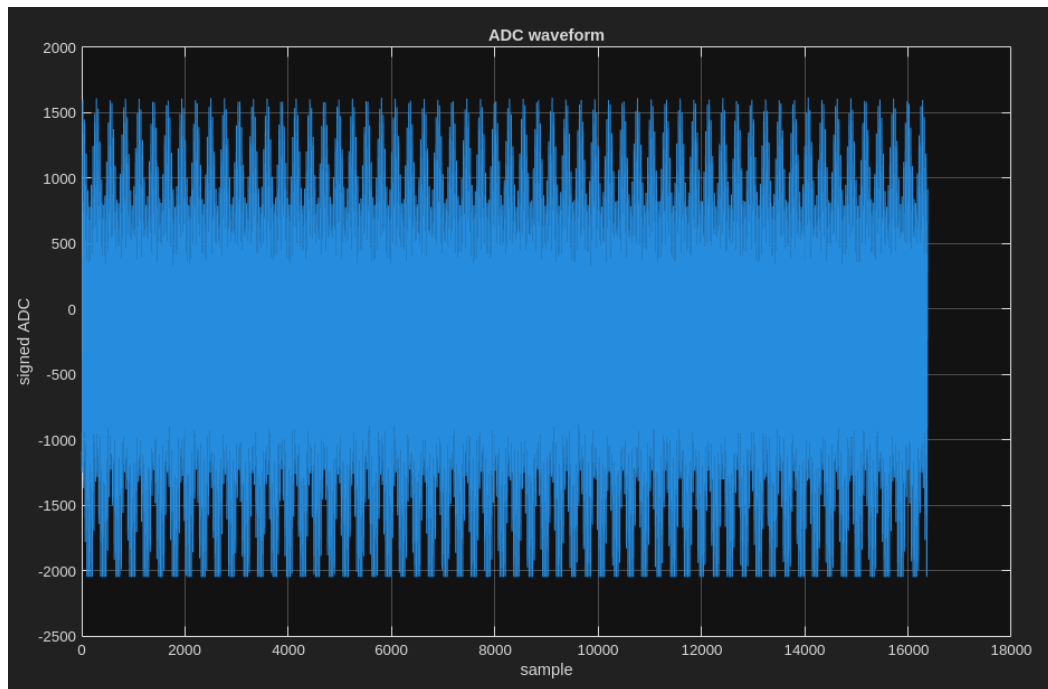


Figure 95. Programmed-board 500 Hz experiment: ADC.

UART capture from the programmed Basys 3, 3 June 2026.

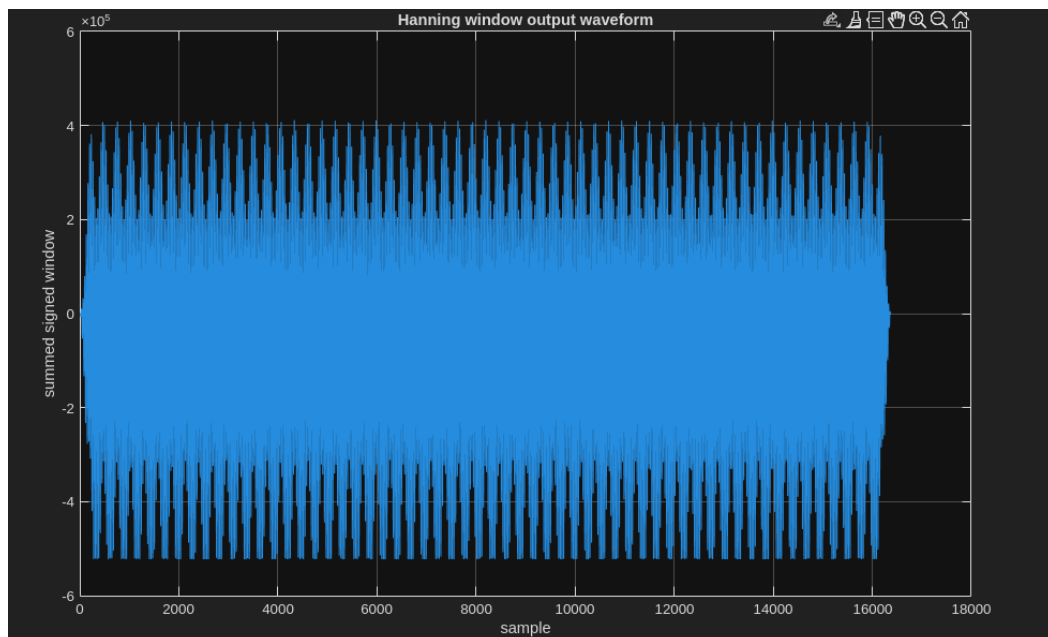


Figure 96. Programmed-board 500 Hz experiment: Hann overlap-add.

UART capture from the programmed Basys 3, 3 June 2026.

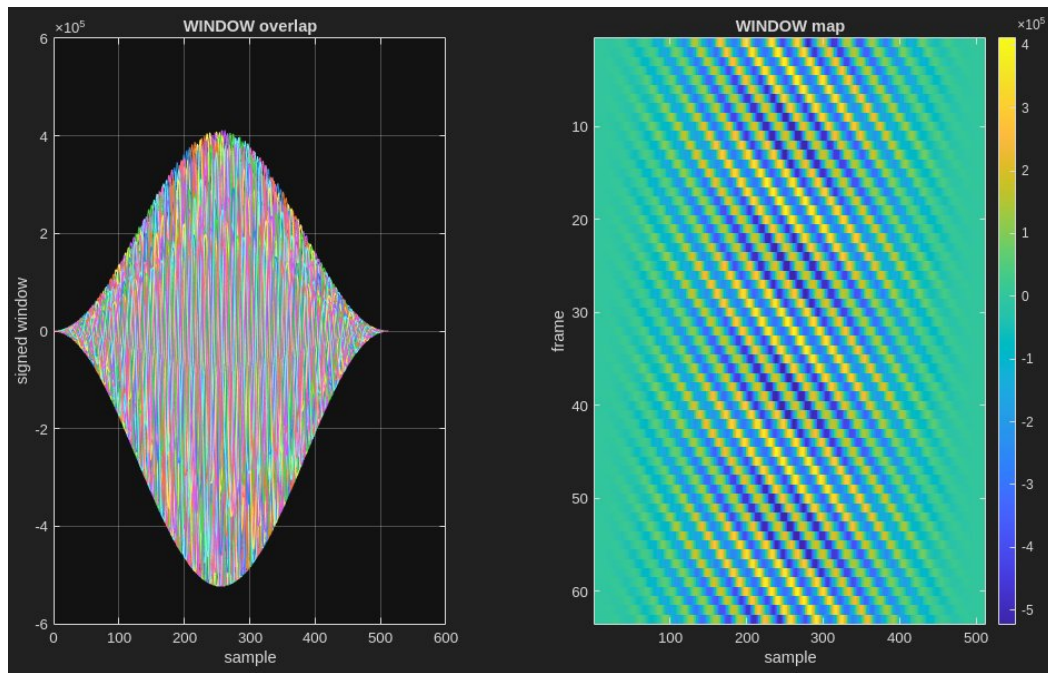


Figure 97. Programmed-board 500 Hz experiment: WINDOW.

UART capture from the programmed Basys 3, 3 June 2026.

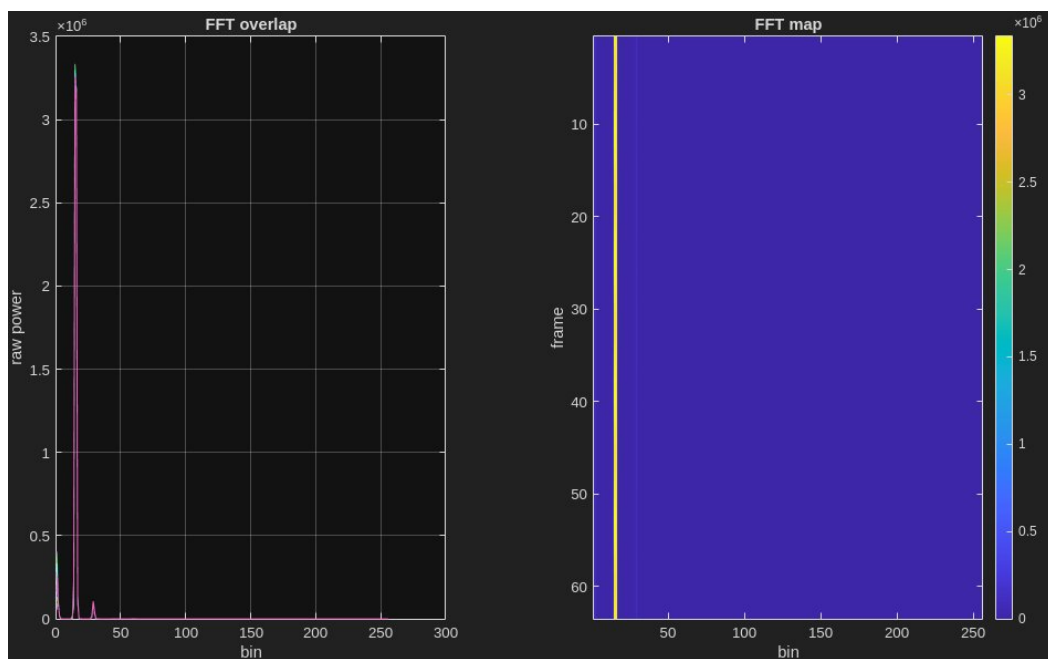


Figure 98. Programmed-board 500 Hz experiment: FFT.

UART capture from the programmed Basys 3, 3 June 2026.

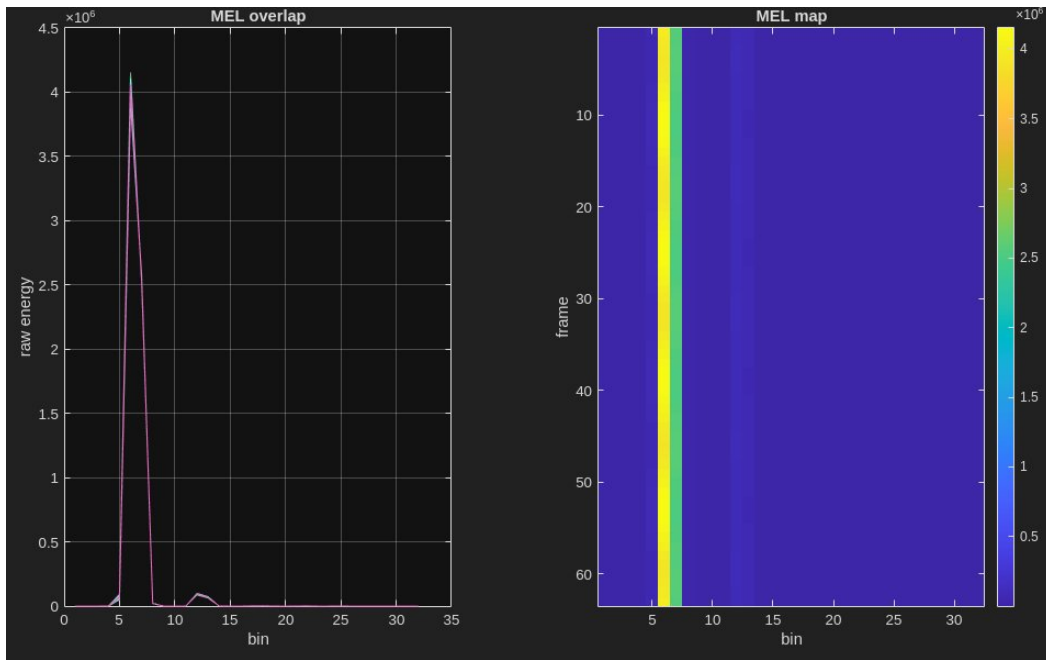


Figure 99. Programmed-board 500 Hz experiment: MEL.

UART capture from the programmed Basys 3, 3 June 2026.

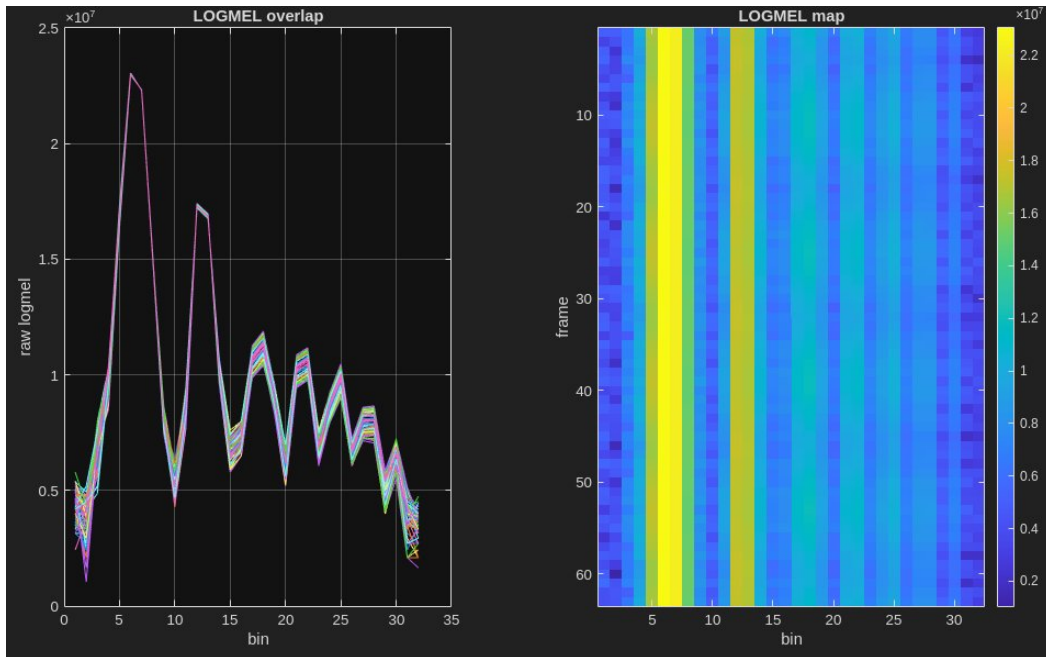


Figure 100. Programmed-board 500 Hz experiment: LOGMEL.

UART capture from the programmed Basys 3, 3 June 2026.

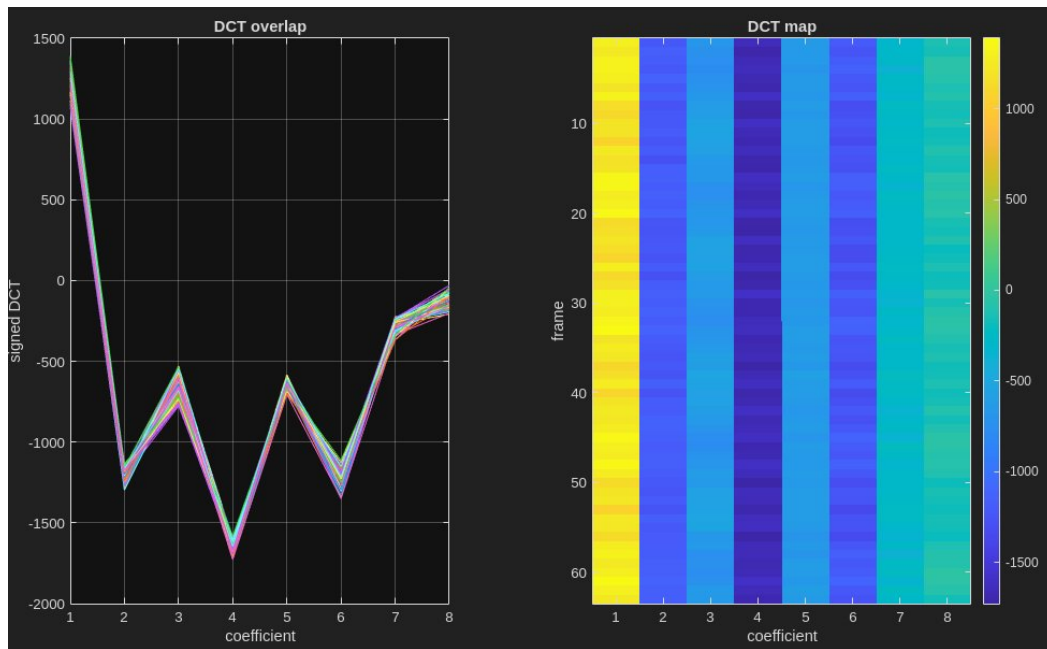


Figure 101. Programmed-board 500 Hz experiment: DCT.

UART capture from the programmed Basys 3, 3 June 2026.

B.3 2 kHz

The 2 kHz run moves the spectral concentration well above the 1 kHz case. Six stage views are retained; no DCT image exists for this specific run.

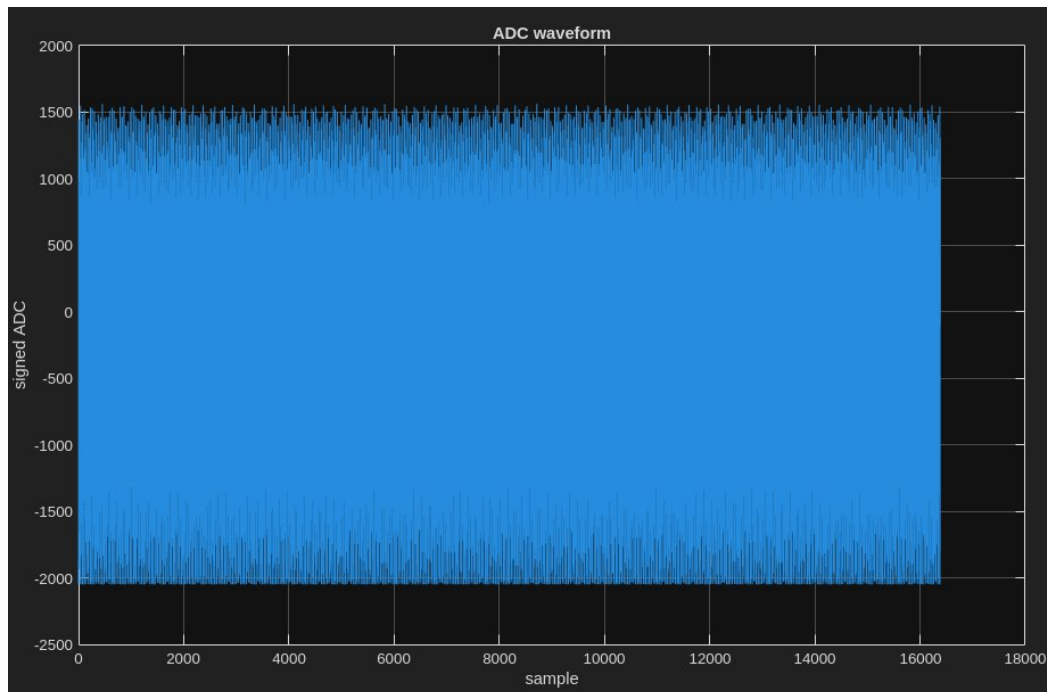


Figure 102. Programmed-board 2 kHz experiment: ADC.

UART capture from the programmed Basys 3, 3 June 2026.

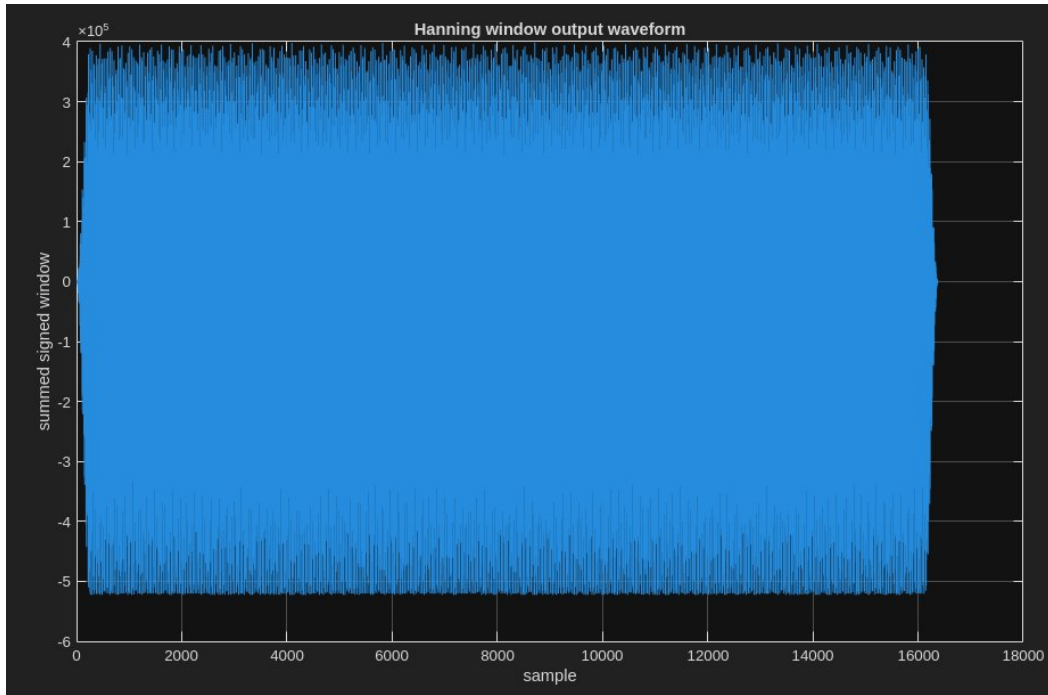


Figure 103. Programmed-board 2 kHz experiment: Hann overlap-add.

UART capture from the programmed Basys 3, 3 June 2026.

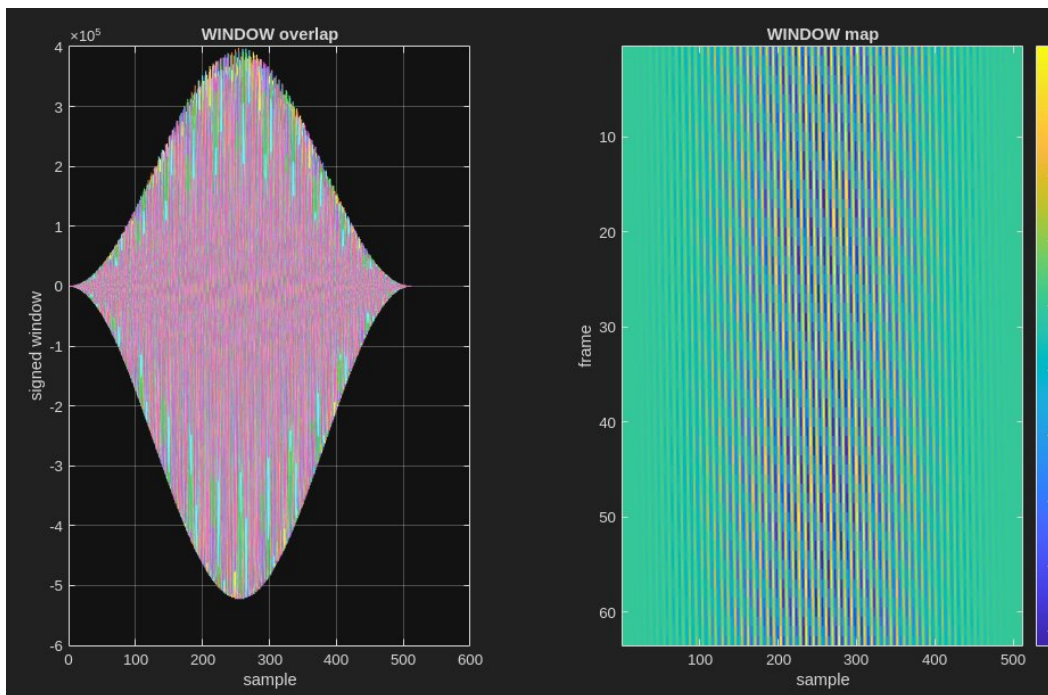


Figure 104. Programmed-board 2 kHz experiment: WINDOW.

UART capture from the programmed Basys 3, 3 June 2026.

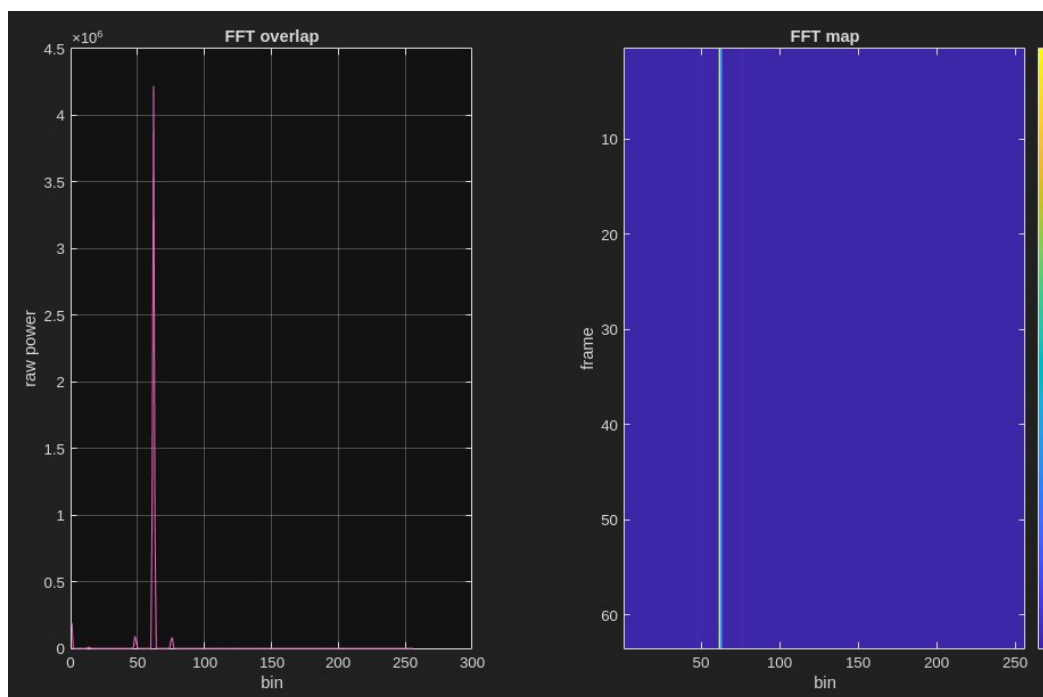


Figure 105. Programmed-board 2 kHz experiment: FFT.

UART capture from the programmed Basys 3, 3 June 2026.

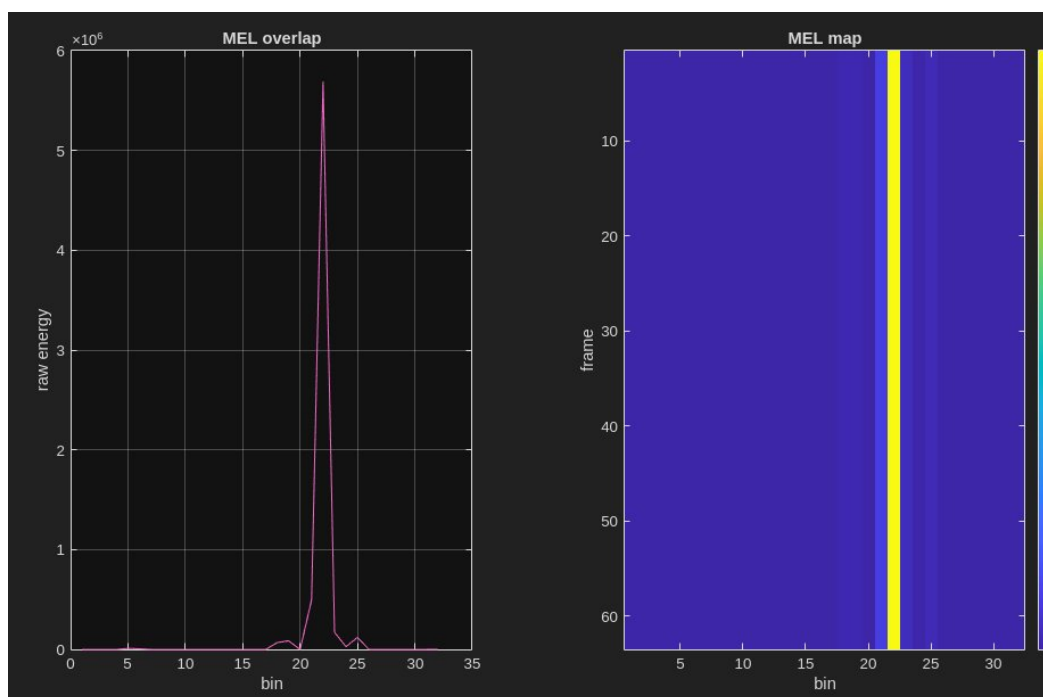


Figure 106. Programmed-board 2 kHz experiment: MEL.

UART capture from the programmed Basys 3, 3 June 2026.

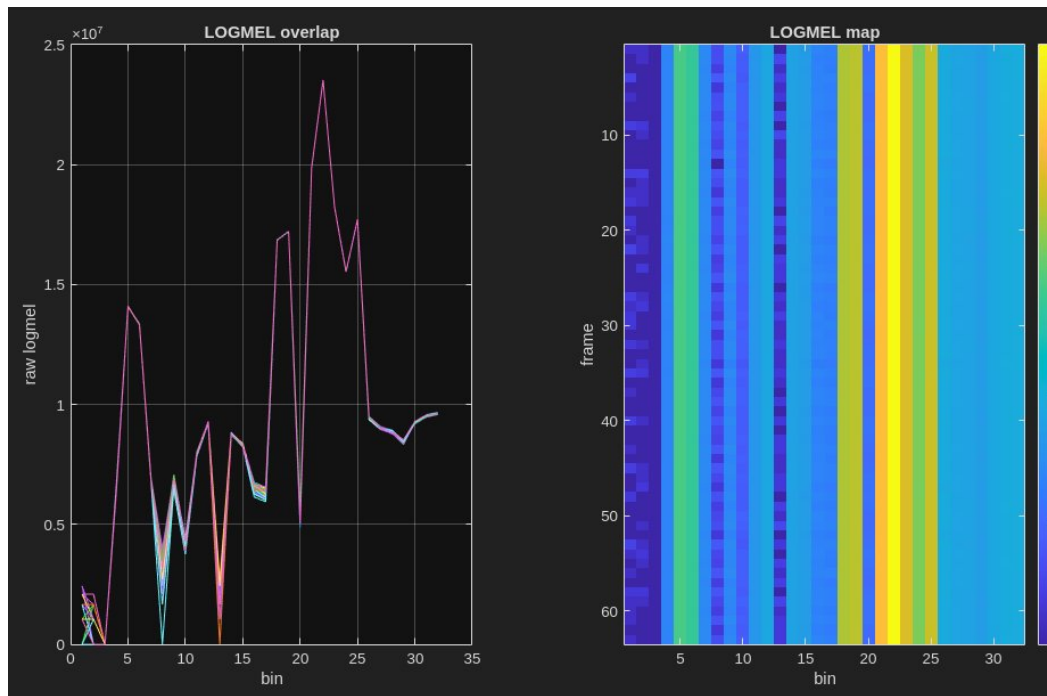


Figure 107. Programmed-board 2 kHz experiment: LOGMEL.

UART capture from the programmed Basys 3, 3 June 2026.

B.4 4 kHz

The 4 kHz run occupies the high end of the stored one-sided spectrum and the highest MEL region. Six stage views are retained; the saved set omits LOGMEL but includes the final DCT map.

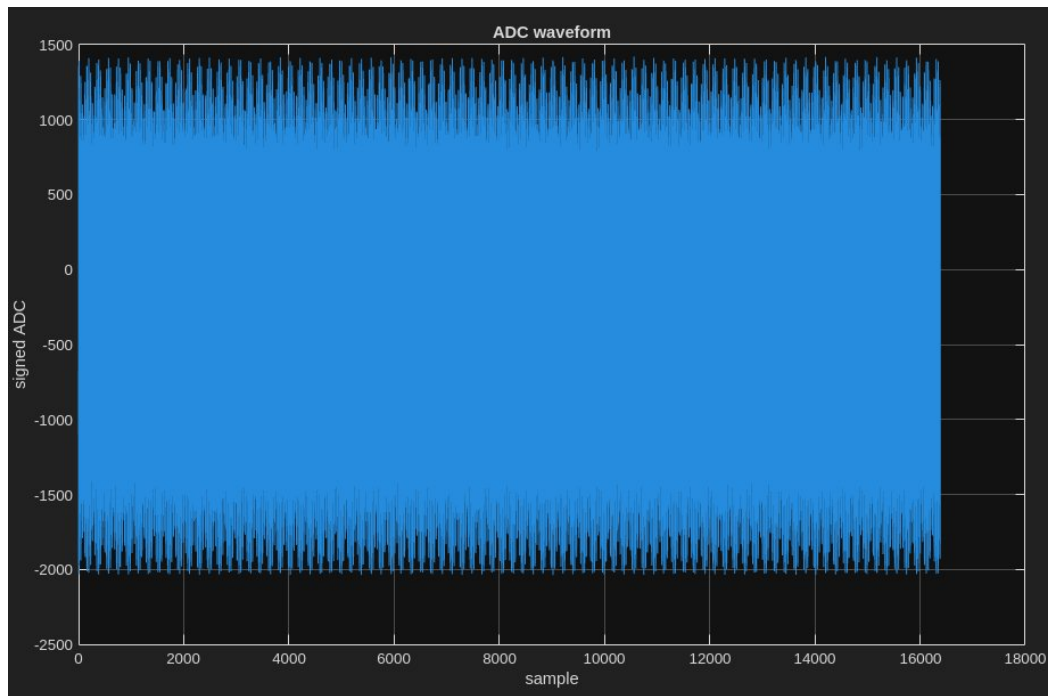


Figure 108. Programmed-board 4 kHz experiment: ADC.

UART capture from the programmed Basys 3, 3 June 2026.

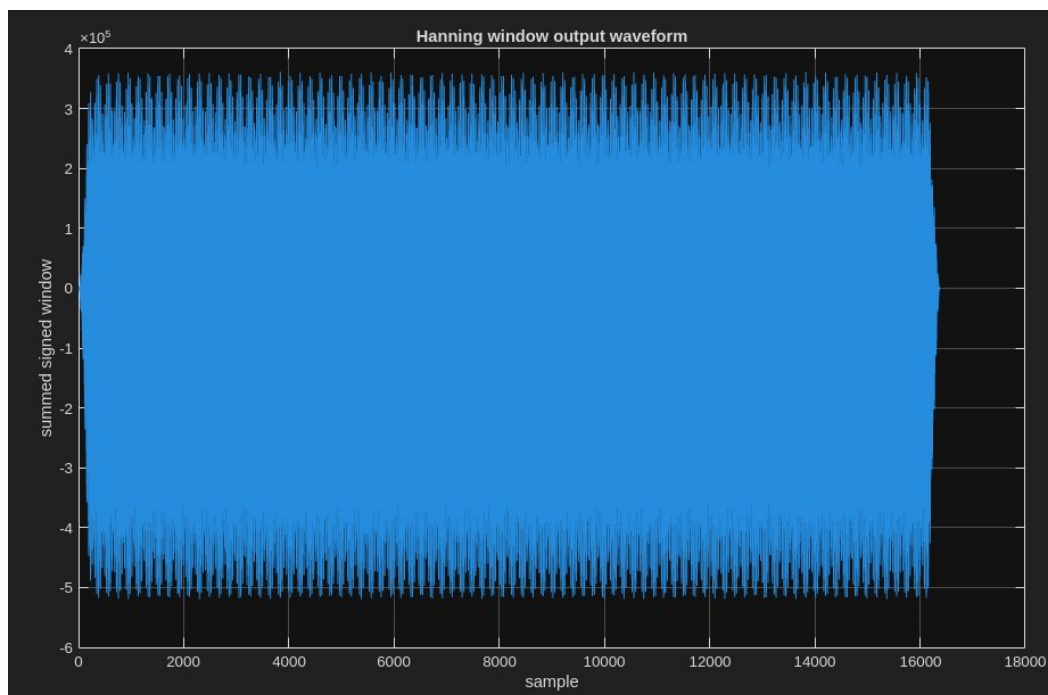


Figure 109. Programmed-board 4 kHz experiment: Hann overlap-add.

UART capture from the programmed Basys 3, 3 June 2026.

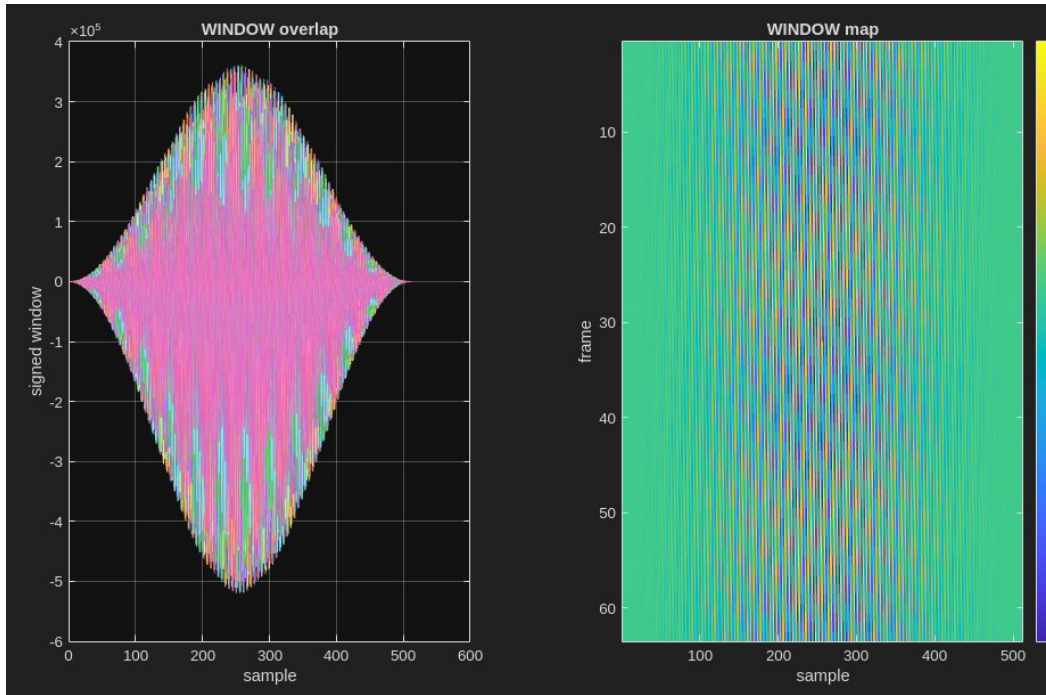


Figure 110. Programmed-board 4 kHz experiment: WINDOW.

UART capture from the programmed Basys 3, 3 June 2026.

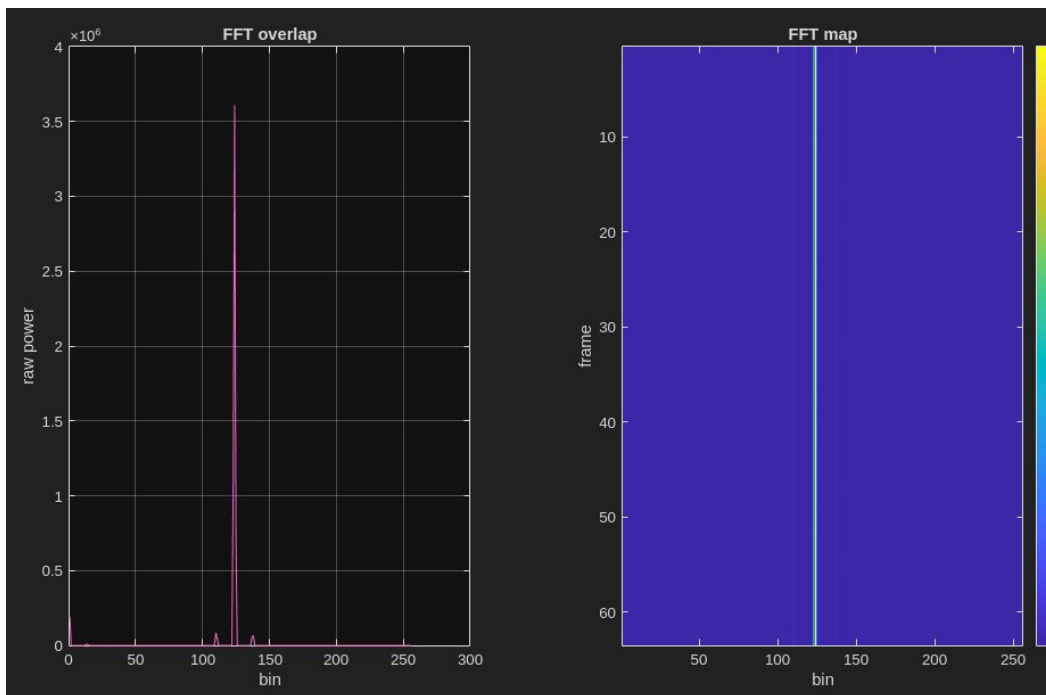


Figure 111. Programmed-board 4 kHz experiment: FFT.

UART capture from the programmed Basys 3, 3 June 2026.

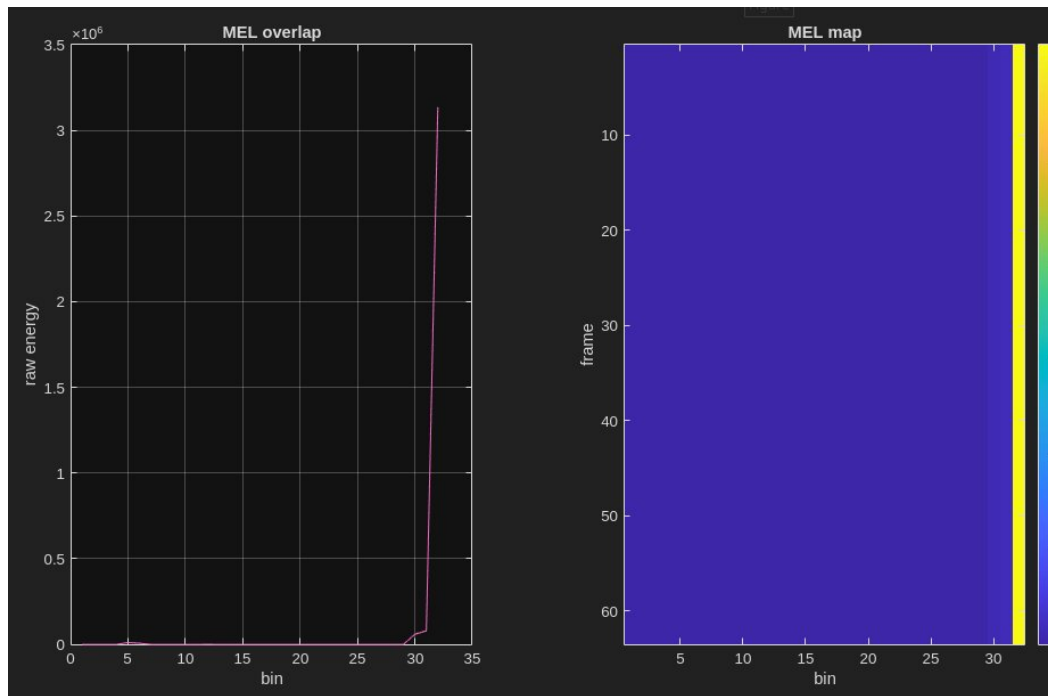


Figure 112. Programmed-board 4 kHz experiment: MEL.

UART capture from the programmed Basys 3, 3 June 2026.

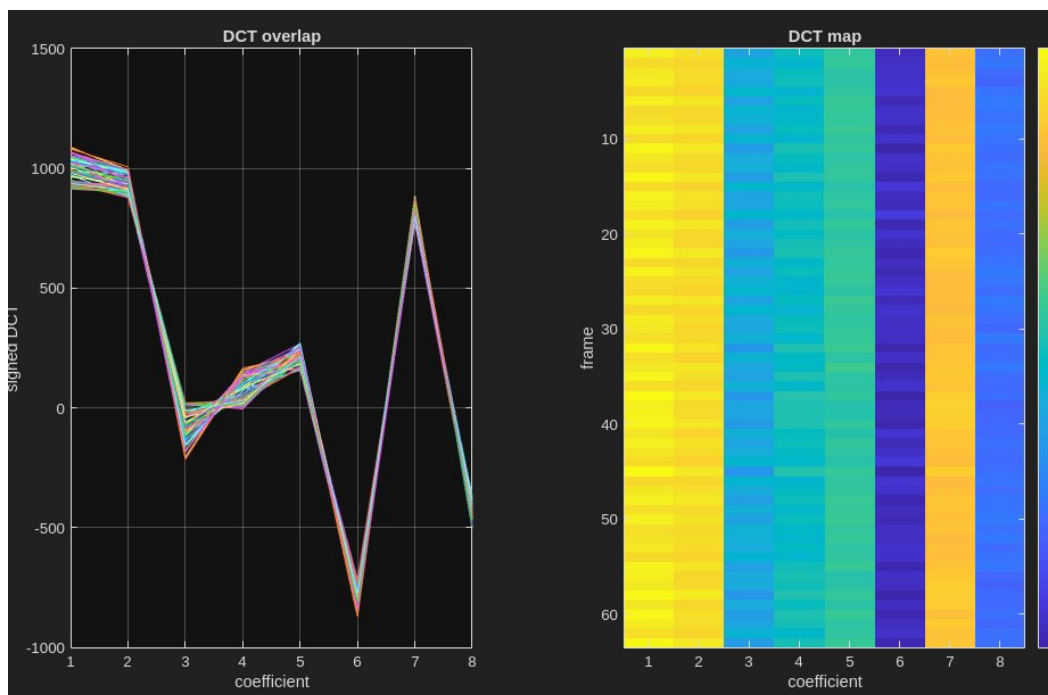


Figure 113. Programmed-board 4 kHz experiment: DCT.

UART capture from the programmed Basys 3, 3 June 2026.

Appendix C Evidence Labels and Reproduction Guide

Figures are labeled as simulation, programmed-board capture, derived plot, or historical artifact, with acquisition dates where available.

C.1 Reproduction guide

A technical reader can reproduce the reasoning in the same order as the architecture. Begin with `top_module.vhd` and `CTRL.vhd` to establish ownership and sequencing, then inspect `ADC.vhd`, `WINDOW.vhd`, `FFT.vhd`, `MEL.vhd`, `LOGMEL.vhd`, `DCT.vhd`, `COMP.vhd`, `debug.vhd`, and `seven-segment.vhd`. The initialized coefficient/reference memories define the fixed-point convention used by both the FPGA and the checkers.

1. Select the relevant VHDL testbench and run the XSim behavioral simulation.
2. Retain the raw waveform view that shows start/ready, address, latency, and write behavior.
3. Confirm that the testbench writes its complete address/value text output.
4. Run the matching MATLAB integer checker and inspect both expected/actual values and the error trace.
5. For board work, enable DEBUG, capture complete marker-framed packets at 1 Mbaud, and retain the original serial capture before plotting.
6. Treat any unexplained difference as a failure at the producing module boundary first. Only after that boundary passes should the next consumed representation be used as integration evidence.

C.2 To test next

The following two measurements remain:

1. Capture reference memory before and after flush, showing the 512-word clear and `flush_done` event.
2. Measure the physical front-end response with calibrated equipment.

Known-digit recording, selected-slot readback and recall, labeled live comparison with frame and result capture, ADC receiver integration and endpoint checks, and PCB DRC/ERC resolution are complete. Status corrected on 17 September 2026.